PRINCIPLES OF ECONOMIC...

COURSE CONTENTS

Introduction to Economic 01

Markets and Welfare 02

Economics of the Public Sector 03

Firm Behavior and the Organization of Industry 04

The Theory of Consumer Choice 05

The Real Economy in the Long Run 06

INTRODUCTION TO ECONOMICS

Two Tech Cities, One Tough Choice

Once there were two neighbouring technology cities, Techville and Innovia. Both received ₹500 crore, 500 engineers, limited land, electricity, and water to build a world-class technology industry.

Techville decided to use most of its resources to produce semiconductors. At first, production increased rapidly and everyone celebrated. But soon the city faced electricity shortages, water scarcity, pollution, and a shortage of engineers in other sectors.

Innovia made a different choice. It divided its resources between semiconductors, AI, healthcare, and infrastructure. It also encouraged factories to save electricity and water. Its chip production was lower than Techville's, but its overall productivity and quality of life were higher.

One day, a Techville engineer asked: “Why didn't we simply use all our resources to produce more chips?” The economist smiled and replied: “Because resources are limited. Every choice has a cost. When you choose one use, you give up another.”



If you had ₹500 crore, 500 engineers, and limited resources, what would you produce and what would you be willing to give up?

DEFINITION OF ECONOMICS

Social science that studies how individuals, businesses, governments, and societies allocate scarce resources to satisfy their needs through the production, distribution, and consumption of goods and services.

PRINCIPLES OF ECONOMICS

How people make decisions:

Principle 1: People Face Trade-offs

Principle 2: The Cost of Something Is What You Give Up to Get It

Principle 3: Rational People Think at the Margin

Principle 4: People Respond to Incentives

PRINCIPLES OF ECONOMICS

How People Interact:

Principle 5: Trade Can Make Everyone Better Off

Principle 6: Markets Are Usually a Good Way to Organize Economic Activity

Principle 7: Governments Can Sometimes Improve Market Outcomes

PRINCIPLES OF ECONOMICS

How the Economy as a Whole Works:

Principle 8: A Country's Standard of Living Depends on Its Ability to Produce Goods and Services

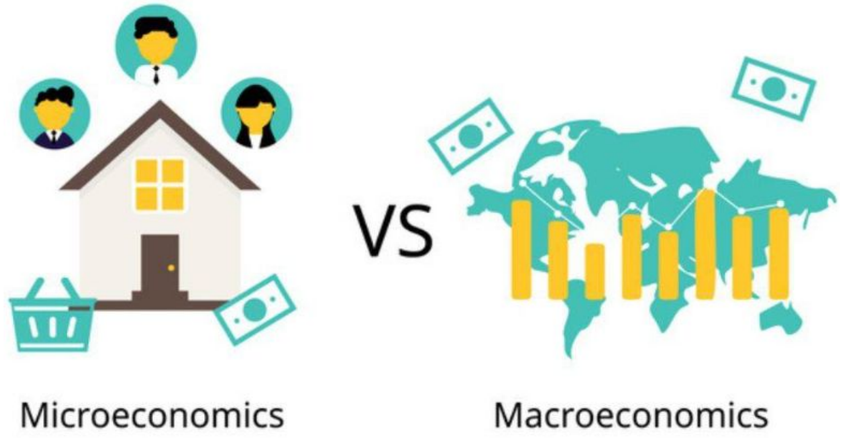
Principle 9: Prices Rise When the Government Prints Too Much Money

Principle 10: Society Faces a Short-Run Trade-off between Inflation and Unemployment

TYPES OF ECONOMICS

Microeconomics: the study of how households and firms make decisions and how they interact in markets.

Macroeconomics: the study of economy-wide phenomena, including inflation, unemployment, and economic growth.



EXAMPLES

Example	Microeconomics Connection	Macroeconomics Connection
Solar Power Plant	How a company decides how much solar electricity to produce based on cost, technology, and electricity demand.	How expansion of solar energy affects India's energy security, imports, investment, and economic growth.
Electricity Consumption	How a household or company decides how much electricity to consume when electricity prices increase.	How national electricity demand, power shortages, and energy policies affect the overall economy.
Battery Storage	How a company decides the optimal size of a battery based on battery cost and expected electricity savings.	How large-scale battery storage can affect India's energy security, renewable-energy integration, and investment.
Electric Vehicles (EVs)	How a consumer decides between an EV and a petrol car by comparing purchase price, fuel/electricity cost, and maintenance cost.	How EV adoption affects oil imports, employment, government revenue, and economic growth.
Industrial Energy Use	How a factory decides whether to use electricity, natural gas, or solar power based on relative costs.	How changes in national energy prices affect industrial production, inflation, and economic growth.

1. You win $100 in a basketball pool. You have a choice between spending the money now or putting it away for a year in a bank account that pays 5 percent interest. What is the opportunity cost of spending the $100 now?

2. Explain whether each of the following government activities is motivated by a concern about equality or concern about efficiency. In the case of efficiency, discuss the type of market failure involved.

a) regulating cable TV prices.

b) providing some poor people with vouchers that can be used to buy food

c) prohibiting smoking in public places

d) breaking up Standard Oil (which once owned 90 percent of all oil refineries) into several smaller companies.

e) imposing higher personal income tax rates on people with higher incomes

f) instituting laws against driving while intoxicated

4. Classify the following topics as relating to microeconomics or macroeconomics.

a. family's decision about how much income to save
b. the impact of higher national savings on economic growth
c. a firm's decision about how many workers to hire
d. the relationship between the inflation rate and changes in the quantity of money

5. The company that you manage has invested $5 million in developing a new product, but the development is not quite finished. At a recent meeting, your sales-people reported that the introduction of competing products had reduced the expected sales of your new product to $3 million. If it would cost $1 million to finish development and make the product, should you go ahead and do so? What is the most that you should pay to complete development?

6. Classify each of the following statements as positive or normative. Explain.

a) Society faces a short-run trade-off between inflation and unemployment.
b) A reduction in the rate of money growth will reduce the rate of inflation.
c) The Federal Reserve should reduce the rate of money growth.
d) Society ought to require welfare recipients to look for jobs.
e) Lower tax rates encourage more work and more saving.

DEMAND AND SUPPLY


₹1,50,000

NEED vs WANT vs DEMAND

Checklist Criteria	Need	Want	Demand
Essential for survival/functioning	✔	✘	✘
Basic and universal	✔	✘	✘
Created by society or culture	✘	✔	✔
Can be influenced by marketing	✘	✔	✔
Unlimited in nature	✘	✔	✔
Specific form of satisfaction	✘	✔	✔
Includes purchasing power	✘	✘	✔
Results in economic transaction	✘	✘	✔

India’s Renewable Energy Market — 2019–2025

India’s renewable energy sector has undergone significant changes in recent years because of developments in technology, government policy, electricity consumption, production costs, infrastructure, and global energy markets. In 2019, India continued expanding solar and wind power as electricity consumption increased and renewable-energy projects received greater policy support. In 2020, the COVID-19 pandemic disrupted construction activities, transportation, equipment deliveries, and supply chains for renewable-energy projects. In 2021, global shortages of solar modules and other essential components, together with disruptions in international supply chains, created challenges for renewable-energy project developers. In 2022, international energy-market disruptions increased attention toward energy security and alternative energy sources, while India continued promoting renewable-energy investment and capacity expansion. In 2023, the rapid adoption of solar power, technological improvements, and increasing investment in renewable-energy infrastructure contributed to strong growth in the sector, while electricity consumption also increased because of economic activity and growing use of electrical equipment and digital technologies. In 2024, India continued adding substantial renewable-energy capacity, particularly through solar power, supported by policy initiatives, technological developments, and private investment; at the same time, constraints related to transmission networks, land, equipment availability, and grid integration remained important. In 2025, the renewable-energy sector continued to expand, with increasing attention not only to solar and wind generation but also to battery storage, green hydrogen, smart grids, and other technologies needed to manage the changing electricity system.



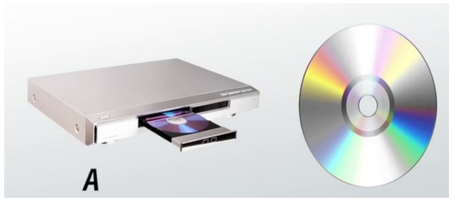
Year	Main factor	Curve affected	Equilibrium Price	Equilibrium Quantity
2019	Growing electricity demand + policy support for renewable energy	Demand	↑	↑
2020	COVID-19 disrupted construction, transportation and equipment supply	Supply	↑	↓
2021	Shortages and higher costs of solar modules/components	Supply	↑	↓
2022	Energy-market disruptions increased interest in alternative energy; renewable investment continued	Demand	↑	↑
2023	Rapid solar adoption, technological improvements and investment	Supply	↓	↑
2024	Expansion of renewable capacity supported by technology, policy and investment	Supply	↓	↑
2025	Growing demand for renewable electricity plus expansion of storage, green hydrogen and grid technologies	Demand	↑	↑

DEMAND AND QUANTITY DEMAND

The amount of a good that buyers are willing and able to purchase.

DEMAND OF DEMAND

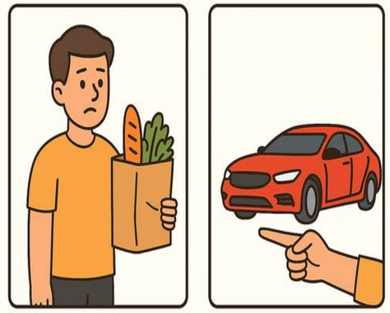
Direct and Derived Demand	Consumers buy wheat flour (direct demand), which creates demand for wheat from farmers (derived demand).
Recurring and Replacement Demand	A farmer buys fertilizers every cropping season (recurring demand) and replaces a broken tractor after years of use (replacement demand).
Complementary and Competing Demand	When farmers grow more rice, they also need more water for irrigation (complementary demand). At the same time, rice and wheat compete in the market as staple food choices (competing demand).
Individual and Market Demand	One farmer buys 5 bags of urea (individual demand), while all farmers in a region demand 10,000 bags (market demand).


A





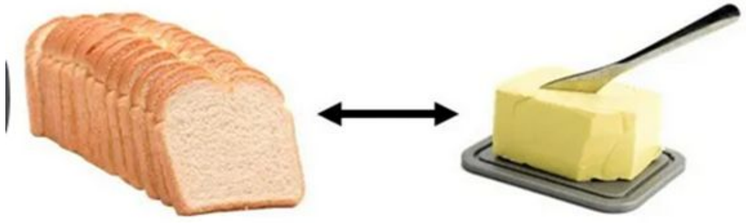


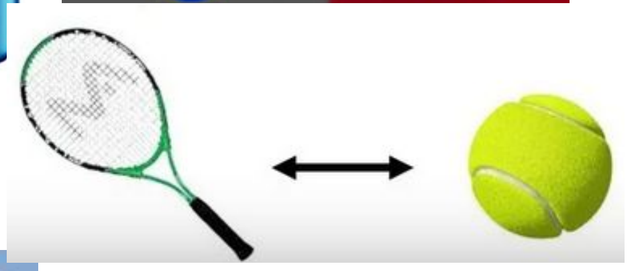



Coca-Cola
classic
vs.
pepsi


BURGER KING
McDonald's


chicken
mint leaves
coriander leaves
yogurt
ginger garlic paste
lemon juice
salt
green chillies
turmeric powder












Every 8 years
Every planting season
SEEDS
FERTILIZER


AMBULANCE


3号投料口
2号投料口
1号投料口

DETERMINATION OF DEMAND

- Price of the Product (P_x)
- Income (Y)
  Normal good: a good for which, other things equal, an increase in income leads to an increase in demand.
  Inferior good: a good for which other things equal; an increase in income leads to a decrease in demand.
- Prices of Related Goods (P_o)
  Substitutes: two goods for which an increase in the price of one leads to an increase in the demand for the other.
  Complements: two goods for which an increase in the price of one leads to a decrease in the demand for the other.
- Tastes (T)
- Expectations (E_f)
- Population (N)

Product	Cause	Effect on Demand
Normal Good	Income increases	Demand increases
Inferior Good	Income increases	Demand decreases
Substitute Good	Price of the related substitute increases	Demand for the good increases
Complementary Good	Price of the related complementary good increases	Demand for the good decreases

DEMAND FUNCTION

$$D_x = f\left(P_x, Y, P_o, T, E_f, N\right)$$

Linear form:

LAW OF DEMAND

The claim that other things equal, the quantity demanded of a good falls when the price of the good rises.

Reasons behind the Law of demand

1. Price effect
2. Income effect
3. Substitution effect

EXCEPTION TO THE LAW OF DEMAND

- Giffen Goods
- Snob Appeal
- Demonstration effect
- Future Expectation of Prices
- Goods with No Substitute

EXCEPTION TO THE LAW OF DEMAND

- Giffen Goods
- Snob Appeal
- Demonstration effect
- Future Expectation of Prices
- Goods with No Substitute

SUPPLY AND QUANTITY SUPPLIED

The amount of a good that sellers are willing and able to sell.

DETERMINATION OF SUPPLY

- Price of the Product (P_x)
- Cost of production (C)
- State of Technology (T)
- Government Policy (G)
- Population (N)

SUPPLY FUNCTION

$$S_x = f\ (P_x, C, T, G, N)$$

Linear form:

LAW OF SUPPLY

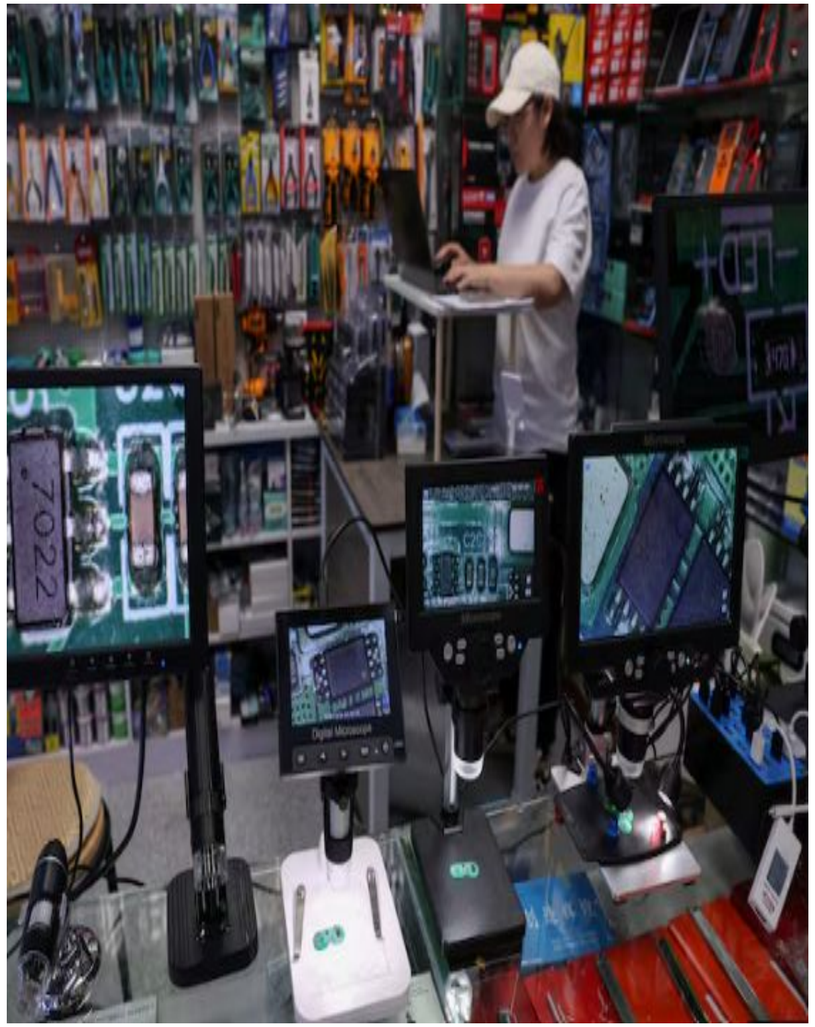
The claim that other things equal, the quantity supplied of a good rise when the price of the good rises.

EQULIBRIUM POINT

A situation in which the market price has reached the level at which quantity supplied equals quantity demanded.

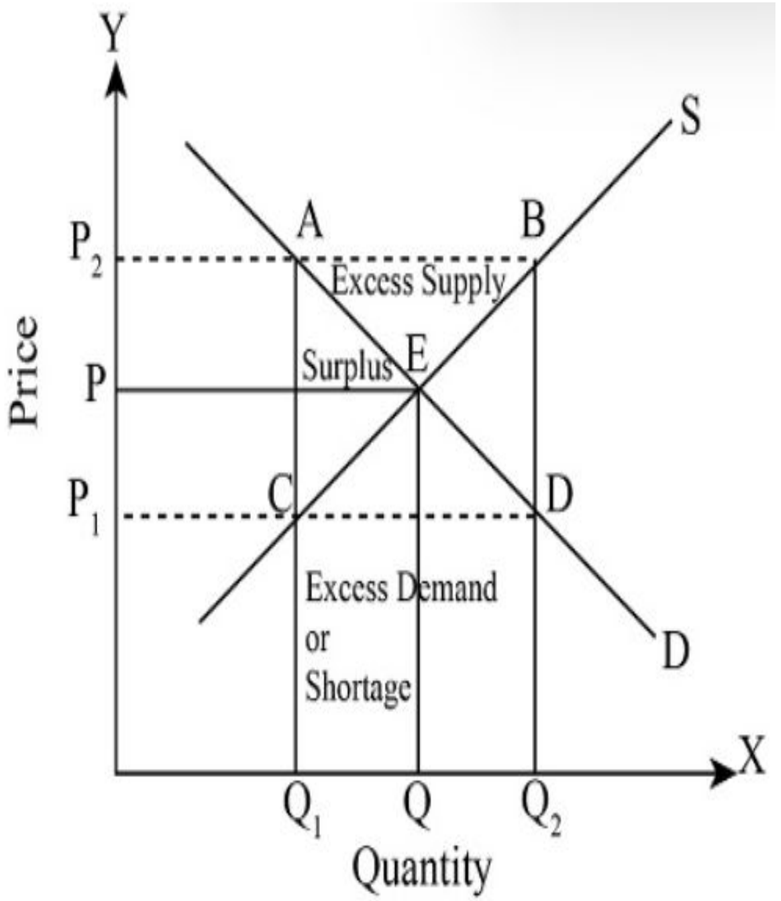
The Smartphone Chip Market

In **Bengaluru**, **Arjun**, the manager of a semiconductor company, watched his factory produce thousands of computer chips used in smartphones, laptops, and electric vehicles. One year, production was excellent, and the company produced a large quantity of chips. Confident about strong demand, Arjun set the price at **₹1,000 per chip**. But **Priya**, the procurement manager of a smartphone company, shook her head at the high price. “I cannot afford to buy chips at this price. Our production cost will become too high,” she complained. Soon, other electronics manufacturers also reduced their orders, and Arjun was left with a large stock of unsold chips. The very next year, a shortage of raw materials and strong demand for smartphones and EVs reduced chip production significantly. Arjun initially offered the chips at **₹600 per chip**, and manufacturers rushed to buy as many as they could. Soon, the available chips were insufficient to meet demand, and manufacturers began competing for the limited supply. Seeing the situation, **Mr. Rao**, a policymaker, explained, “We need a market price where the quantity of chips supplied equals the quantity demanded.” Finally, when the price settled at **₹800 per chip**, Arjun could sell his chips, Priya could obtain the required components, and manufacturers could continue producing smartphones and EVs



Excess demand: a situation in which the quantity demanded at a given price exceeds the quantity supplied

Excess supply: a situation in which the quantity supplied at a given price exceeds the quantity demanded.



THREE STEPS TO ANALYSING CHANGES IN EQUILIBRIUM

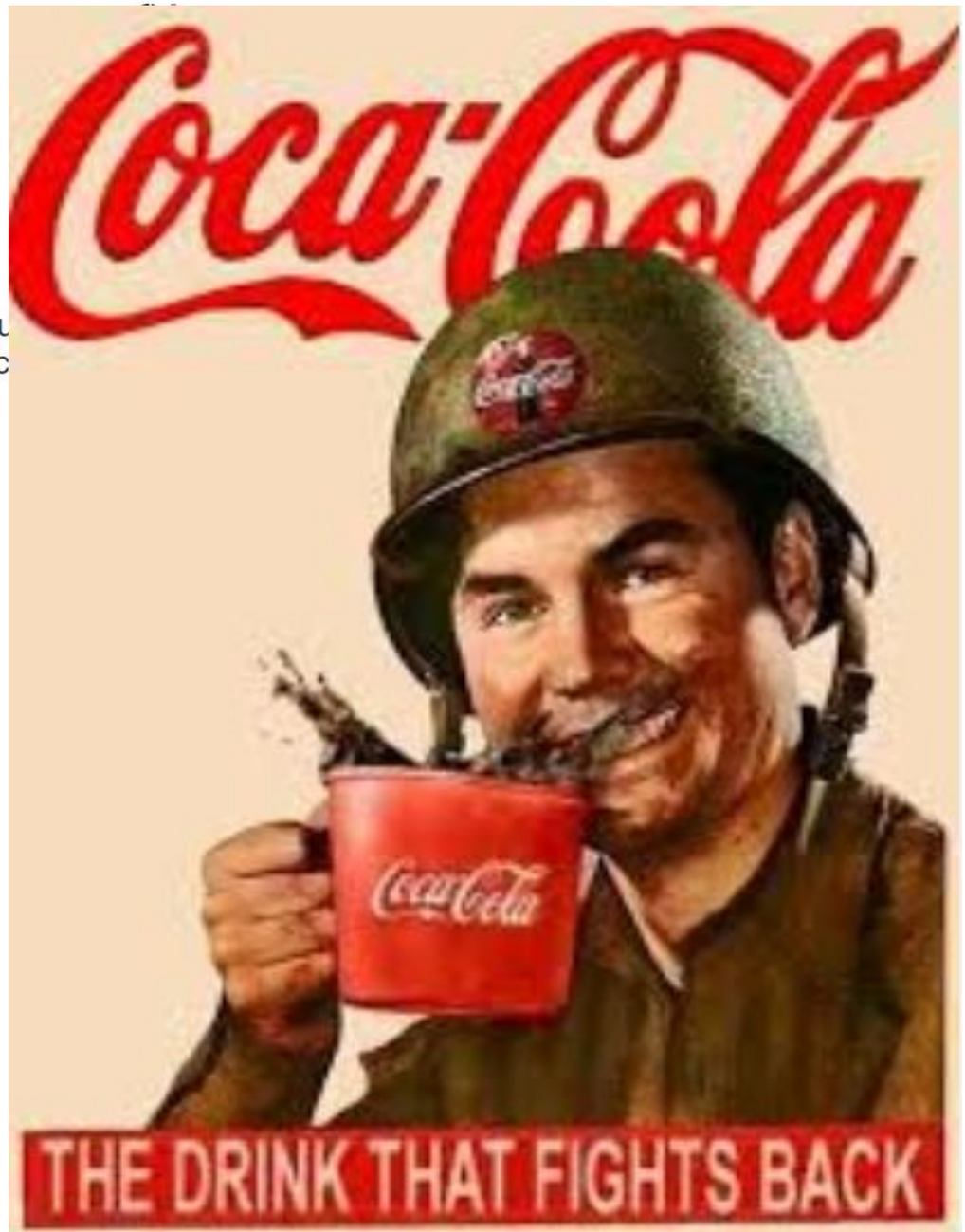
- A Change in Market Equilibrium Due to a Shift in Demand
- A Change in Market Equilibrium Due to a Shift in Supply
- Shifts in Both Supply and Demand

A Change in Market Equilibrium Due to a Shift in Demand

Coca-Cola and World War II

During World War II, the Coca-Cola Company set up over 60 bottling plants in Europe and the Pacific to serve American soldiers. More than 5 billion bottles were delivered between 1943 and 1945. As soldiers introduced locals to the drink, demand increased in areas where Coca-Cola had never been sold before. However, the production capacity remained fixed due to wartime resource constraints. The increased popularity created long lines and high local demand, but supply couldn't immediately increase.

Equ
pric
up



A Change in Market Equilibrium Due to a Shift in Supply

The 1972 U.S.–Soviet Grain Deal

In 1972, the Soviet Union quietly bought around 10 million tons of wheat and corn from the U.S. due to a poor harvest. These deals were made without notifying U.S. grain markets, leading to an unexpected fall in U.S. stockpiles. As a result, the price of wheat jumped from $1.73 to over $4.89 per bushel in just a few months. This sudden export led to shortages and price spikes in other countries, especially those that depended on U.S. grain imports.

Equilibr
price g
up



Shifts in Both Supply and Demand

The 2007–2008 Global Corn Price Surge

During 2007–2008, the corn market went through a major shift due to changes on both the supply and demand sides. A severe drought in the U.S. Midwest reduced corn production, while rising input costs made it harder for farmers to grow more, pushing the available quantity down. At the same time, new government policies promoted ethanol, increasing the use of corn for fuel, and growing meat consumption in countries like China and India raised the demand for corn-based animal feed. This demand shift, along with the supply drop, moved the market, even though less corn was sold.



Supply Change	Demand Change	Magnitude Considered	Price (P)	Quantity (Q)
↑	↑	ΔSupply > ΔDemand	Decreases	Increases
↑	↑	ΔDemand > ΔSupply	Increases	Increases
↓	↓	ΔSupply > ΔDemand	Increases	Decreases
↓	↓	ΔDemand > ΔSupply	Decreases	Decreases
↑	↓	ΔSupply > ΔDemand	Decreases	Increases
↑	↓	ΔDemand > ΔSupply	Decreases	Decreases
↓	↑	ΔSupply > ΔDemand	Increases	Decreases
↓	↑	ΔDemand > ΔSupply	Increases	Increases

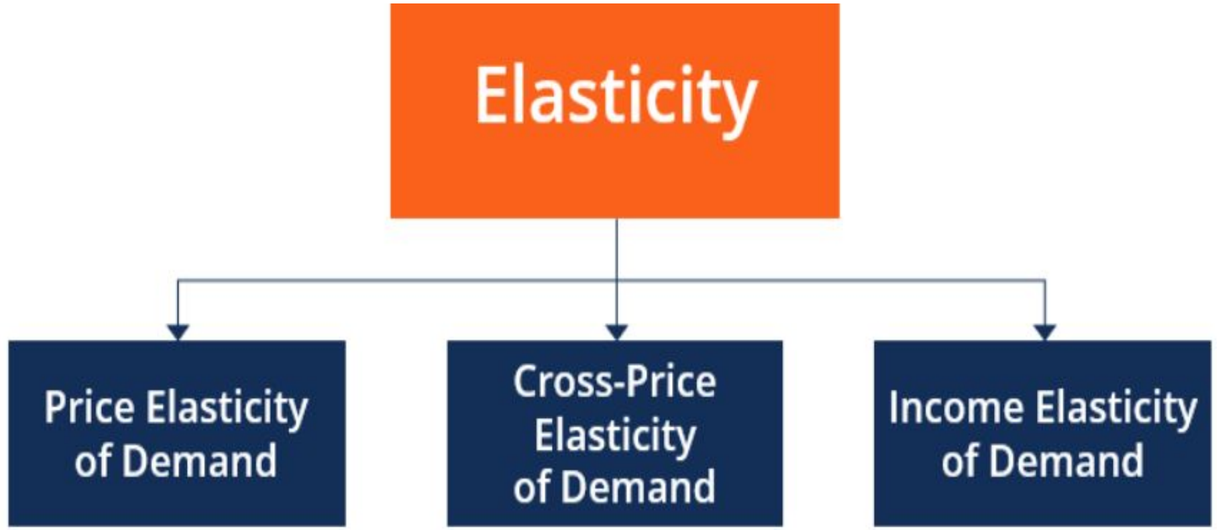
Same direction (↑↑ or ↓↓) → Quantity always follows that direction.

Opposite direction (↑↓ or ↓↑) → Quantity follows the stronger shift.

ELASTICITY
DEMAND

A measure of the responsiveness of quantity demanded or quantity supplied to a change in one of its determinants.

ELASTICITY DEMAND

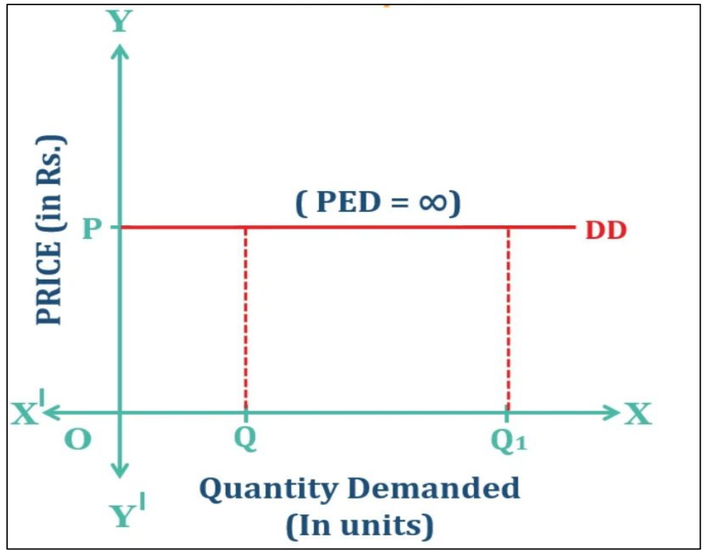


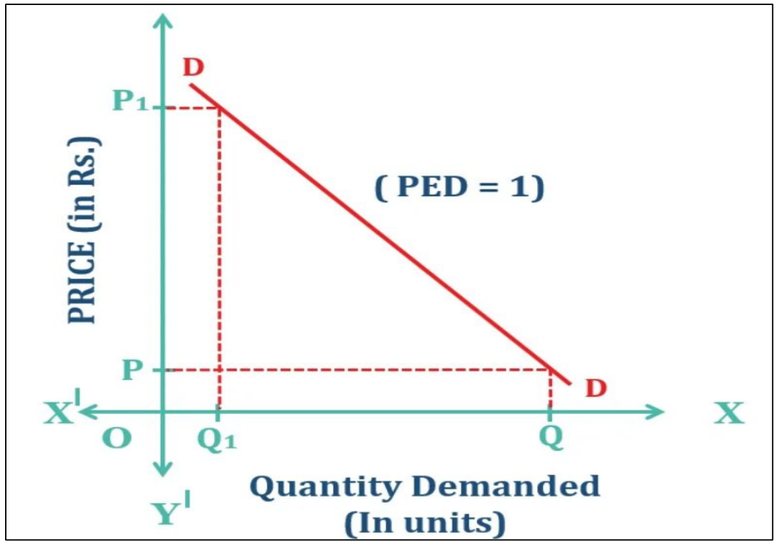
ELASTICITY DEMAND

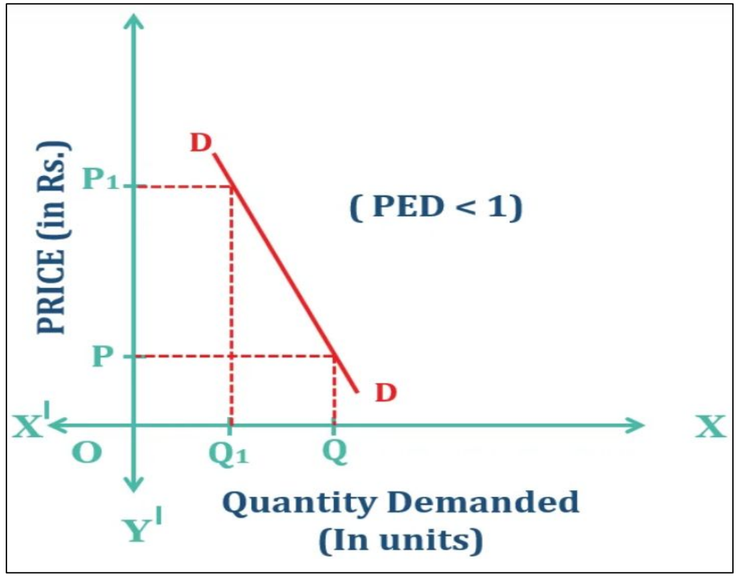
$$\text{PED} = \frac{\%\ \text{change in Quantity Demanded}}{\%\ \text{change in Price}} = \frac{\%\Delta \text{in QD}}{\%\Delta \text{in P}}$$

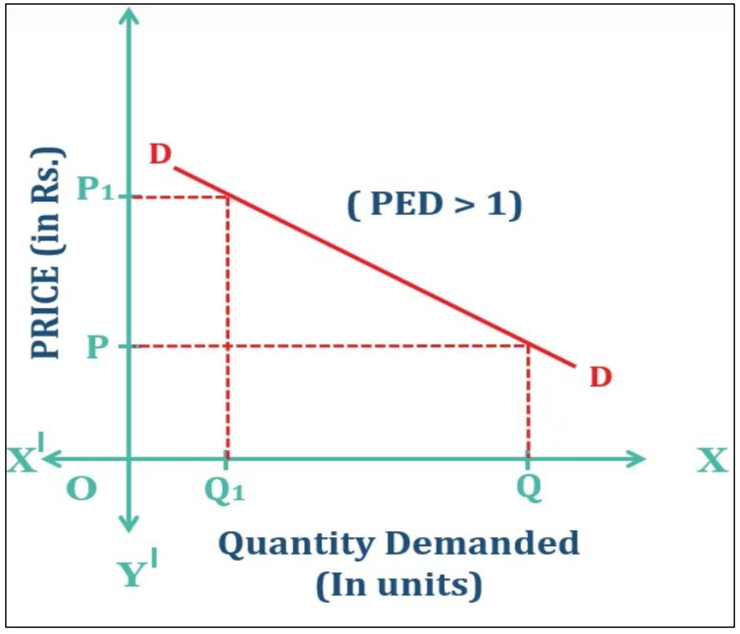
$$\text{YED} = \frac{\%\ \text{change in Quantity Demanded}}{\%\ \text{change in Income}} = \frac{\%\Delta \text{in QD}}{\%\Delta \text{in Y}}$$

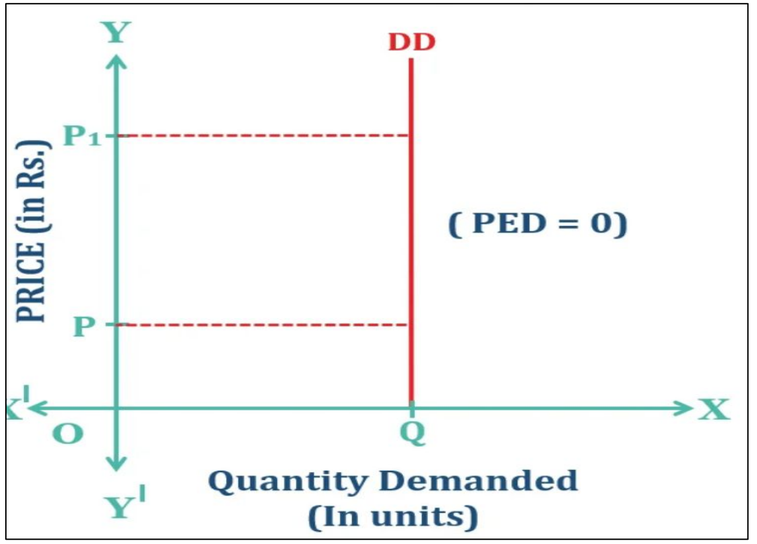
$$\text{XED} = \frac{\%\ \text{change in Quantity Demanded of good x}}{\%\ \text{change in Price of good y}} = \frac{\%\Delta \text{in QDx}}{\%\Delta \text{in Py}}$$


Y
PRICE (in Rs.)
(PED = ∞)
P
DD
X'
O
Q
Q1
X
Y'
Quantity Demanded
(In units)


D
P1
PRICE (in Rs.)
(PED = 1)
P
D
X'
O
Q1
Q
X
Y'
Quantity Demanded
(In units)


D
P1
PRICE (in Rs.)
(PED < 1)
P
D
X'
O
Q1
Q
X
Y'
Quantity Demanded
(In units)

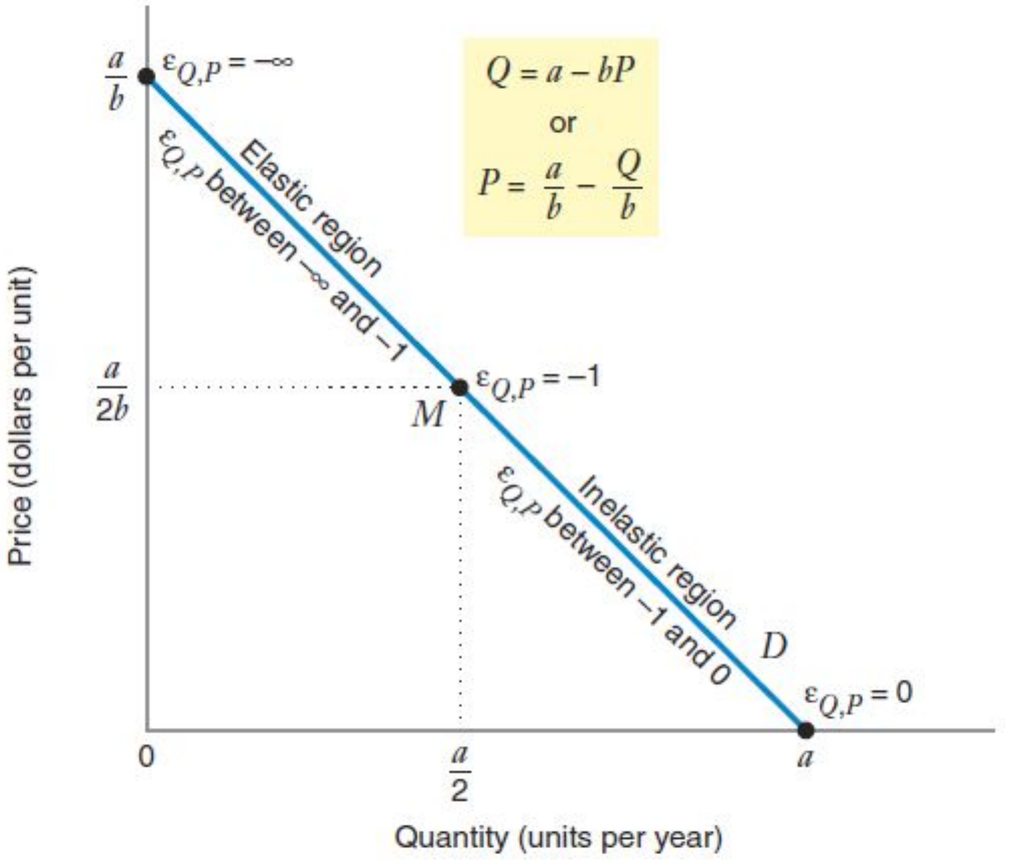

D
P1
PRICE (in Rs.)
(PED > 1)
P
D
X'
O
Q1
Q
X
Y'
Quantity Demanded
(In units)


Y
DD
P1
PRICE (in Rs.)
(PED = 0)
P
X'
O
Q
X
Y'
Quantity Demanded
(In units)

Elasticity in a linear Demand Curve

- $$\varepsilon_{Q,P} = \left({}^{\Delta Q}/_{\Delta P}\right)\left({}^{P}/_{Q}\right) = -b\left({}^{P}/_{Q}\right)$$

Elasticity falls from 0 to -∞ along the linear demand curve, but slope is constant.



Example: calculate elasticity when P = 30 and $Q_d = 400 - 10P$

Answer: $\varepsilon_{Q,P} = -3$ "elastic"

5 EXAMPLES OF ELASTIC GOODS

1. Soft drinks



2. Cereal



3. Clothing



4. Electronics



5. Cars



5 EXAMPLES OF INELASTIC GOODS

1. Life-Saving Medication



2. Gas



3. Electricity



4. Cigarettes



5. Post-Secondary Education



Revenue



Total Revenue (*TR*)

$$TR = P \times Q$$ P = price and Q = quantity

Average Revenue (*AR*)

$$AR = \frac{TR}{Q} = \frac{P \times Q}{Q} = P$$

Marginal Revenue (*MR*)

$$MR = \frac{dTR}{dQ}$$ The rate of change of total revenue with respect to quantity

Price Elasticity & Revenue

$TR = Q \bullet P$ *and*

$$MR = \frac{dTR}{dQ} = \frac{dQ}{dQ}P + Q\frac{dP}{dQ} = P + Q\frac{dP}{dQ}$$

Therefore:

$$MR = P\left(1 + \frac{Q}{P}\frac{dP}{dQ}\right)$$

But we know that:

$$E_P = \frac{P}{Q}\frac{dQ}{dP}$$

So that:

$$MR = P\left(1 + \frac{1}{E_P}\right)$$

Controls on Prices

Price Ceiling

A legal maximum on the price at which a good can be sold.

Price Floor

A legal minimum on the price at which a good can be sold.



The American Association of Ice-Cream Eaters complains that the $3 price is too high for everyone to enjoy a cone a day (their recommended daily allowance).

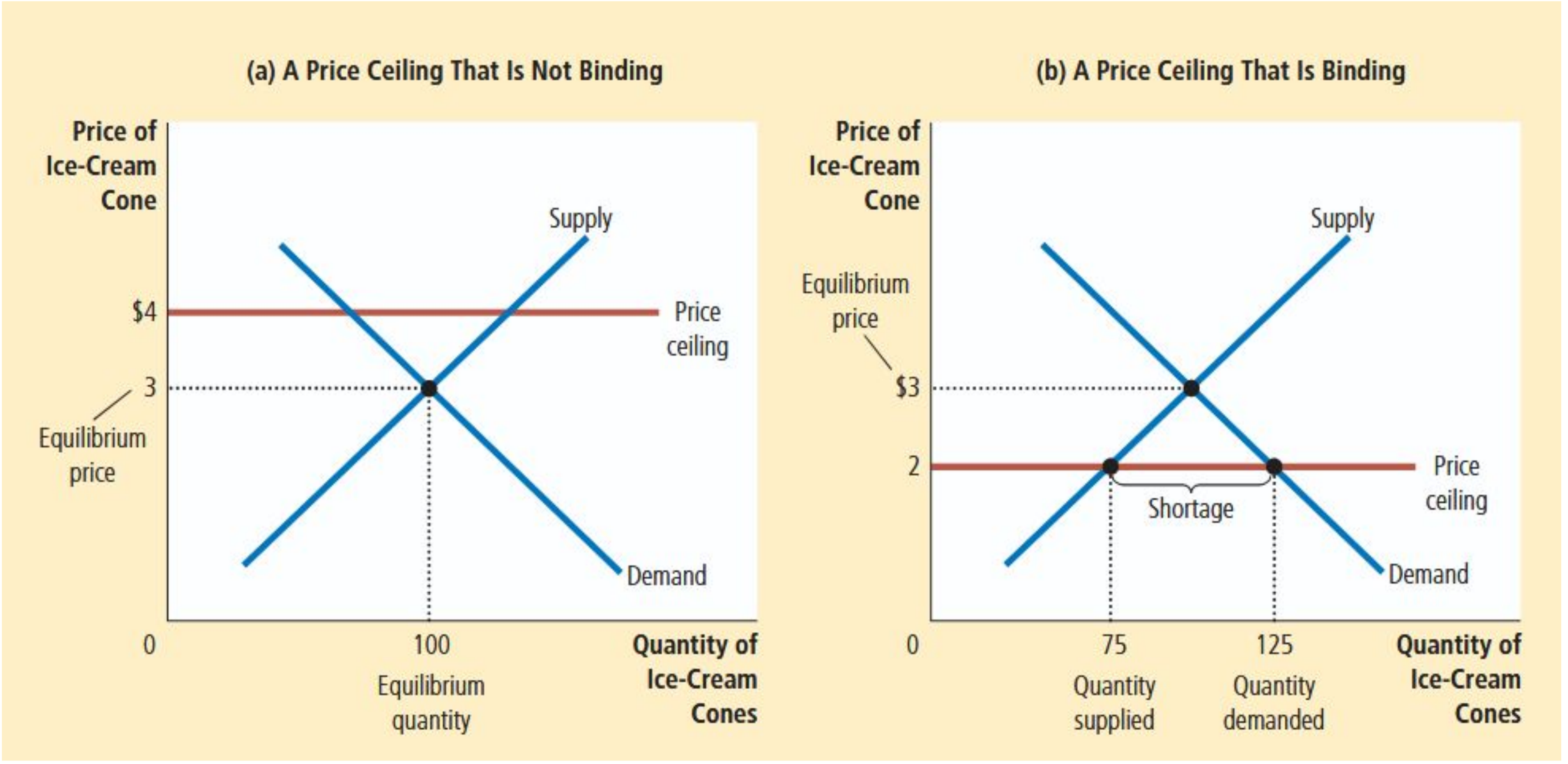
Meanwhile, the National Organization of Ice-Cream Makers complains that the $3 price—the result of "cutthroat competition"—is too low and is depressing the incomes of its members.

Each of these groups lobbies the government to pass laws that alter the market outcome by directly controlling the price of an ice- cream cone.

When the government, moved by the complaints and campaign contributions of the **Ice-Cream Eaters**, imposes a price ceiling on the market for ice cream, two outcomes are possible.

A. The government imposes a price ceiling of $4 per cone.
B. The government imposes a price ceiling of $2 per cone.




(a) A Price Ceiling That Is Not Binding
Price of Ice-Cream Cone
Supply
$4
Price ceiling
3
Equilibrium price
Demand
0
100
Equilibrium quantity
Quantity of Ice-Cream Cones
(b) A Price Ceiling That Is Binding
Price of Ice-Cream Cone
Supply
Equilibrium price
$3
2
Price ceiling
Shortage
Demand
0
75
Quantity supplied
125
Quantity demanded
Quantity of Ice-Cream Cones

Price ceiling

In the 1940s, as World War II created a housing crunch, New York City introduced rent control laws to keep housing affordable for returning soldiers and low-income families. The rent ceiling meant that landlords could not raise rents, even as demand for apartments surged. In the short run, this provided temporary relief. However, over the decades, the policy led to long-term shortages of rental housing. Landlords, receiving below-market rents, often neglect maintenance, leading to deteriorating buildings. At the same time, new construction slowed down, as developers were discouraged by the low profit potential. By the 1970s and 1980s, it was clear that the policy, though well-intentioned, had led to severe mismatches in housing demand and supply, just like a binding price ceiling. The result was long waiting lists, informal payments ("key money"), and discrimination in apartment allocation, as landlords informally selected tenants they preferred.



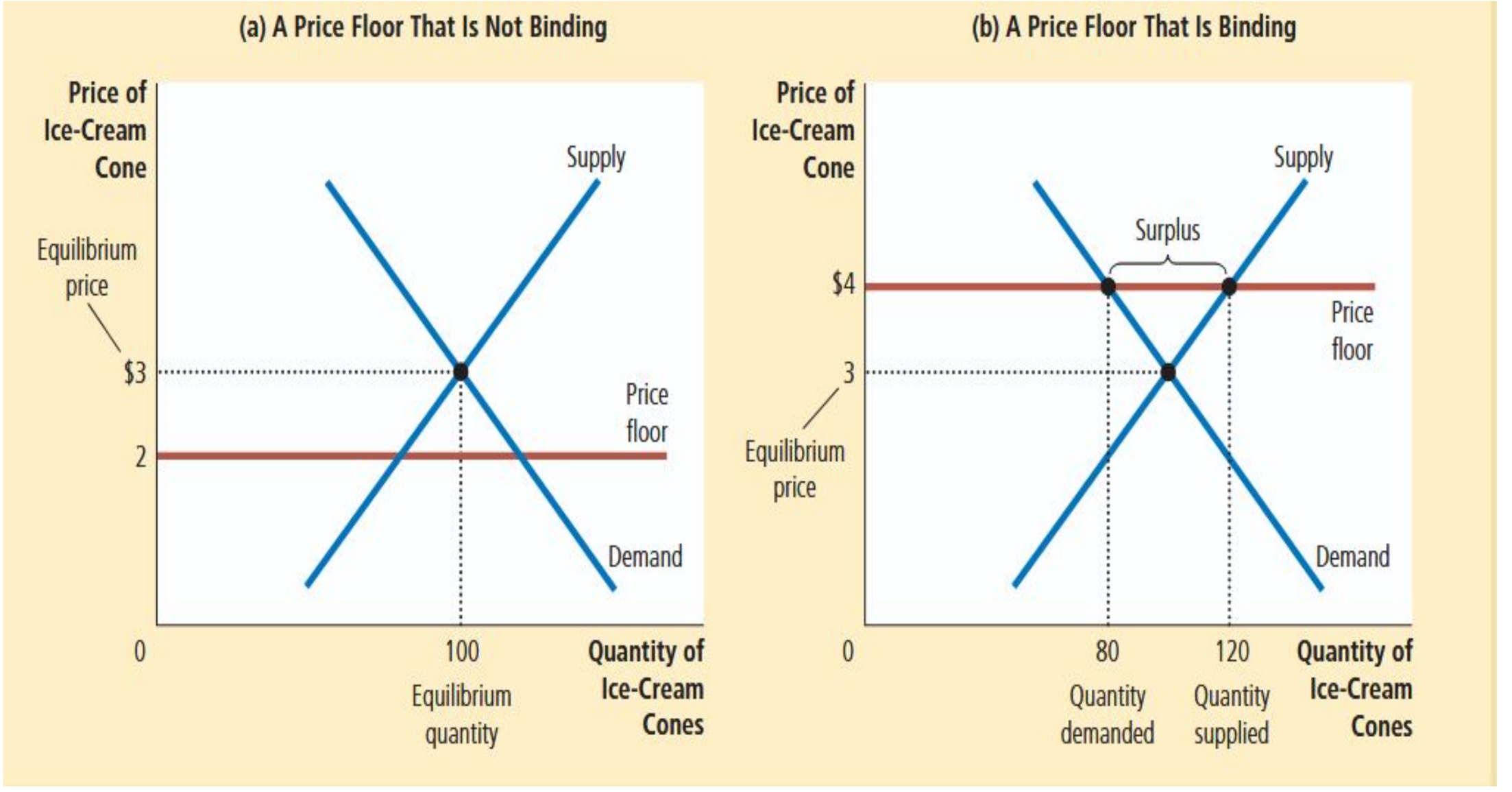
Imagine now that the government is persuaded by the pleas of the National Organization of Ice-Cream Makers whose members feel the $3 equilibrium price is too low. In this case, the government might institute a price floor.

When the government imposes a price floor on the ice-cream market, two outcomes are possible.

A. The government imposes a price floor of $2 per cone when the equilibrium price is $3.

B. Th government imposes a price floor of $4 per cone.




(a) A Price Floor That Is Not Binding
Price of Ice-Cream Cone
Supply
Equilibrium price
$3
2
Price floor
Demand
0
100
Equilibrium quantity
Quantity of Ice-Cream Cones
(b) A Price Floor That Is Binding
Price of Ice-Cream Cone
Supply
Surplus
$4
3
Price floor
Equilibrium price
Demand
0
80
120
Quantity demanded
Quantity supplied
Quantity of Ice-Cream Cones

Price floor

In the United States during the 1930s, the government introduced the first federal minimum wage through the Fair Labor Standards Act of 1938. At that time, the law mandated a minimum wage of $0.25 per hour, aimed at improving the living standards of low-income workers during the Great Depression. While the intention was to ensure a basic level of income, the policy had unintended effects in certain sectors. In regions where the market wage was naturally below the new legal minimum, employers found it too expensive to retain the same number of workers. As a result, youth and unskilled workers faced layoffs or reduced working hours. This mimicked the effect of a binding price floor: when the wage (price of labor) was set above the equilibrium, the quantity of labor supplied (workers willing to work) exceeded the quantity demanded (jobs offered)—creating unemployment in the affected segments. Over time, adjustments in skills, productivity, and inflation reduced these frictions, but the early years showed clear signs of surplus labor, particularly among minority and low-skilled youth populations.





CONSUMER SURPLUS

The extra benefit that consumers receive when they are willing to pay more for a good or service than the actual market price they pay.

In other words: Willingness to pay – Actual price paid.

The Jio Launch and Consumer Surplus in the Indian Telecom Market

In September 2016, Reliance Jio Infocomm Limited (Jio) disrupted the Indian telecom market by launching free voice calls and extremely low-cost 4G data plans. Before Jio's entry, leading telecom players such as Airtel, Vodafone, and Idea were charging around ₹250–₹300 per GB of mobile data.

Jio offered 4G data at ₹50 per GB or less, and for the first few months, it offered completely free voice and data services. This radically changed consumer behavior and created massive consumer surplus across the country.

The four typical mobile data users in 2016, reflecting their willingness to pay per GB of data:

User	Willingness to Pay per GB (₹)	Actual Price Paid with Jio (₹)	Consumer Surplus (₹)
Ramesh	300	50	250
Sneha	250	50	200
Imran	200	50	150
Kavita	100	50	50

Before Jio, only Ramesh and Sneha may have bought mobile data at ₹250–₹300 per GB. After Jio, even users like Imran and Kavita, who were previously excluded due to high prices, entered the market, reflecting increased consumer access and welfare.

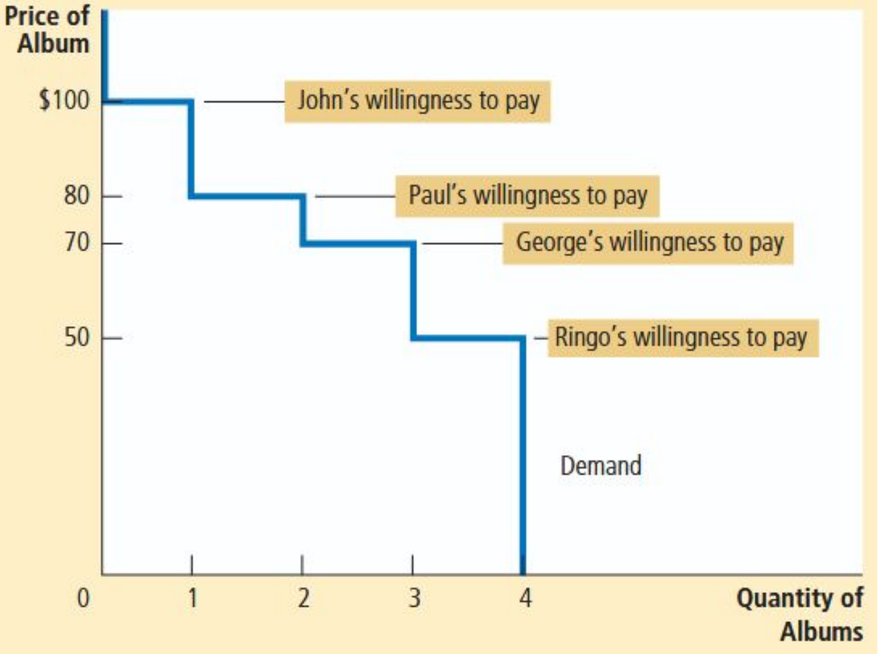
You own a premium Boba Tea DIY Kit, featuring organic Assam tea pearls, flavored syrups, reusable cups, and eco-friendly straws. It's a limited-edition kit curated for Boba lovers — especially popular among Gen Z.

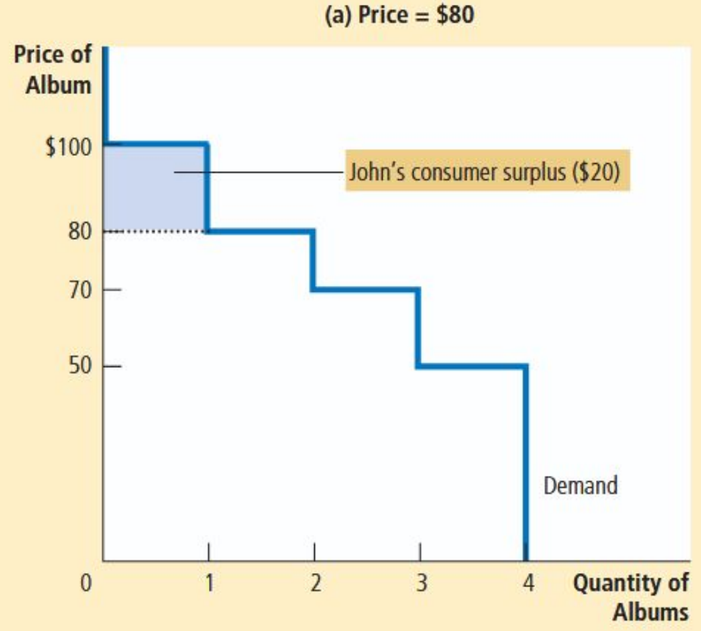
You're not a big fan of Boba tea yourself, so you decide to auction it off.

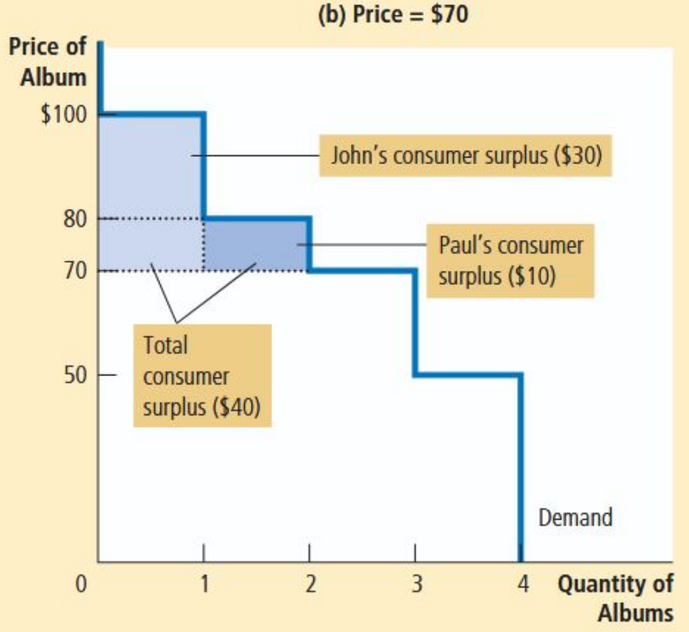


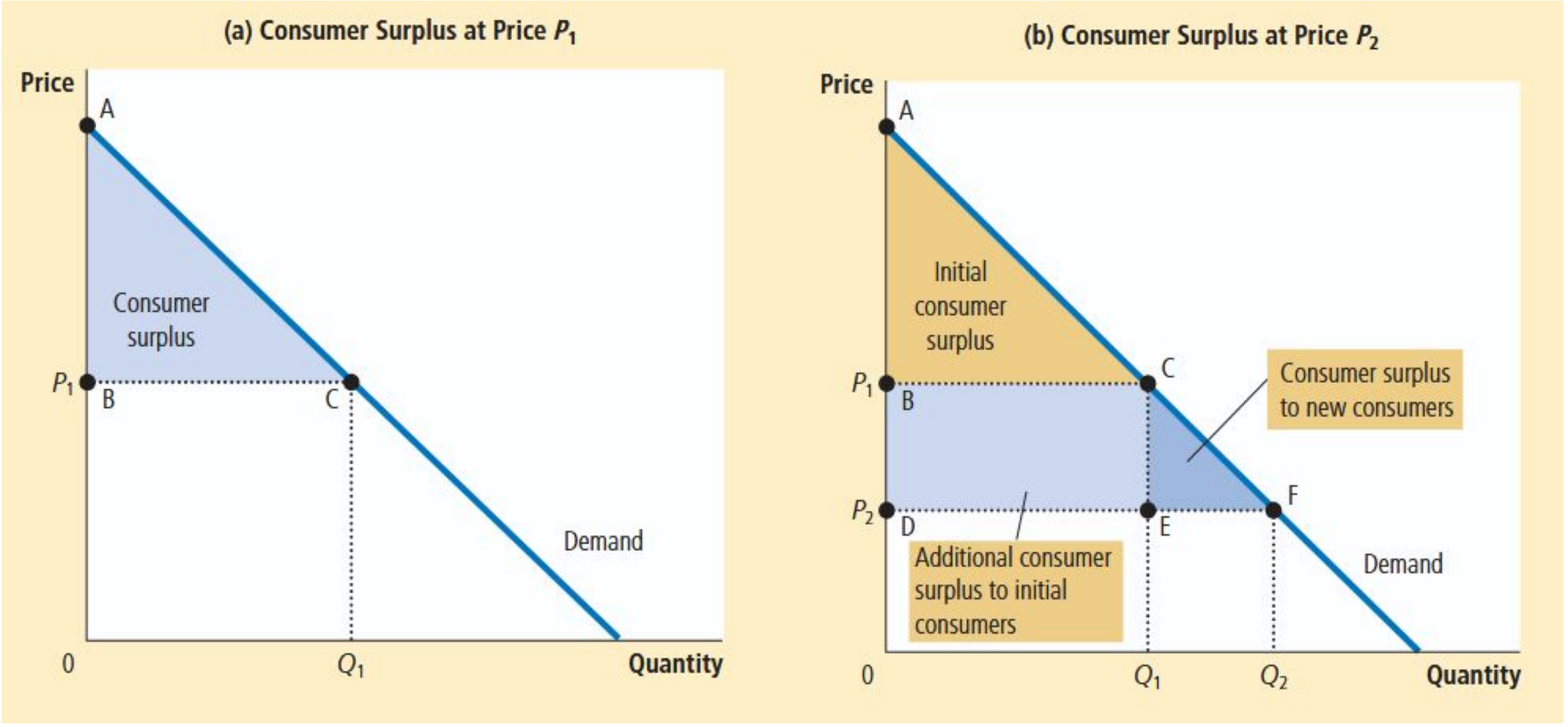
Buyer	Willingness to Pay
John	$100
Paul	80
George	70
Ringo	50

Price	Buyers	Quantity Demanded
More than $100	None	0
$80 to $100	John	1
$70 to $80	John, Paul	2
$50 to $70	John, Paul, George	3
$50 or less	John, Paul, George, Ringo	4








(a) Consumer Surplus at Price P_1
Price
A
Consumer surplus
P_1
B
C
Demand
0
Q_1
Quantity
(b) Consumer Surplus at Price P_2
Price
A
Initial consumer surplus
P_1
B
C
Consumer surplus to new consumers
P_2
D
E
F
Additional consumer surplus to initial consumers
Demand
0
Q_1
Q_2
Quantity



PRODUCER SURPLUS

The extra benefit that producers receive when they sell a good at a market price higher than the minimum price, they are willing to accept.

In other words: Actual price received – Minimum acceptable price

Wheat Procurement under MSP

In 2022, the Government of India set the Minimum Support Price (MSP) for wheat at ₹2,015 per quintal to ensure a fair return to farmers. The actual cost of production, as estimated by various state agricultural departments and farmers' cooperatives, varied between ₹1,200 to ₹1,800 per quintal, depending on the region, irrigation facilities, and input use.

The four wheat farmers in Punjab with different production costs:



Farmer	Cost per Quintal (₹)	MSP Price Received (₹)	Producer Surplus (₹)
Harpreet	1,200	2,015	815
Baldev	1,400	2,015	615
Gurmeet	1,600	2,015	415
Amrit	1,800	2,015	215

All four farmers sell wheat at the MSP of ₹2,015, set by the government. Despite differing production costs, they each receive a surplus above their costs — this surplus is their producer surplus.

You’re planning to sell artisan chocolate-dipped banana pops at a weekend Gen-Z food festival. You want to outsource the production to one of four culinary entrepreneurs: Mary, Frida, Georgia, and Grandma. Each of them is willing to make the banana pops if the price they get covers their opportunity cost (cost of ingredients, time, and other commitments they’d give up).

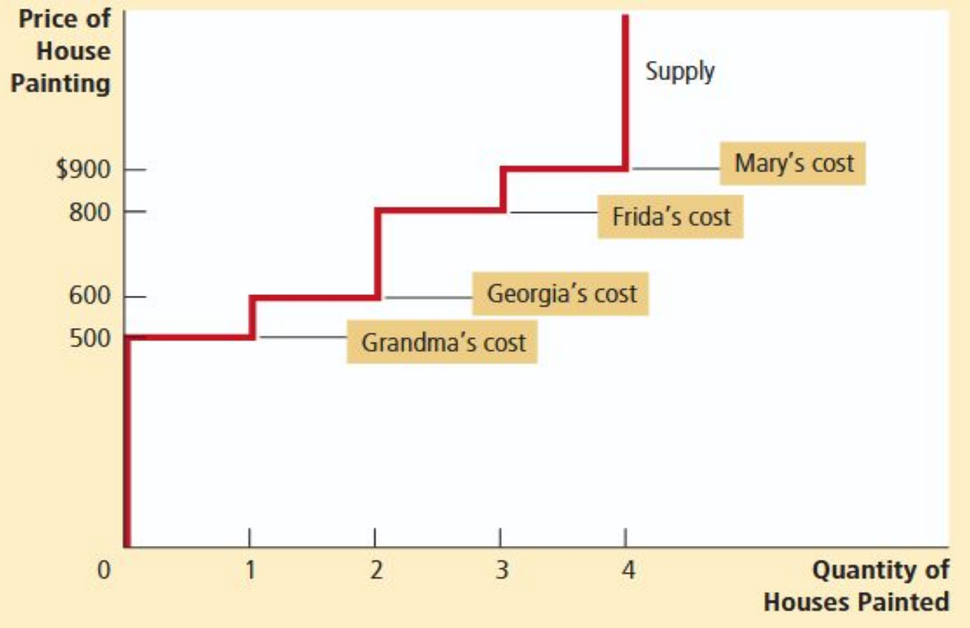
You decide to take bids from them and give the contract to the one who agrees to produce the pops for the lowest price.

Below is a table showing each entrepreneur’s cost to produce one batch of Artisan Chocolate Dipped Banana Pops:

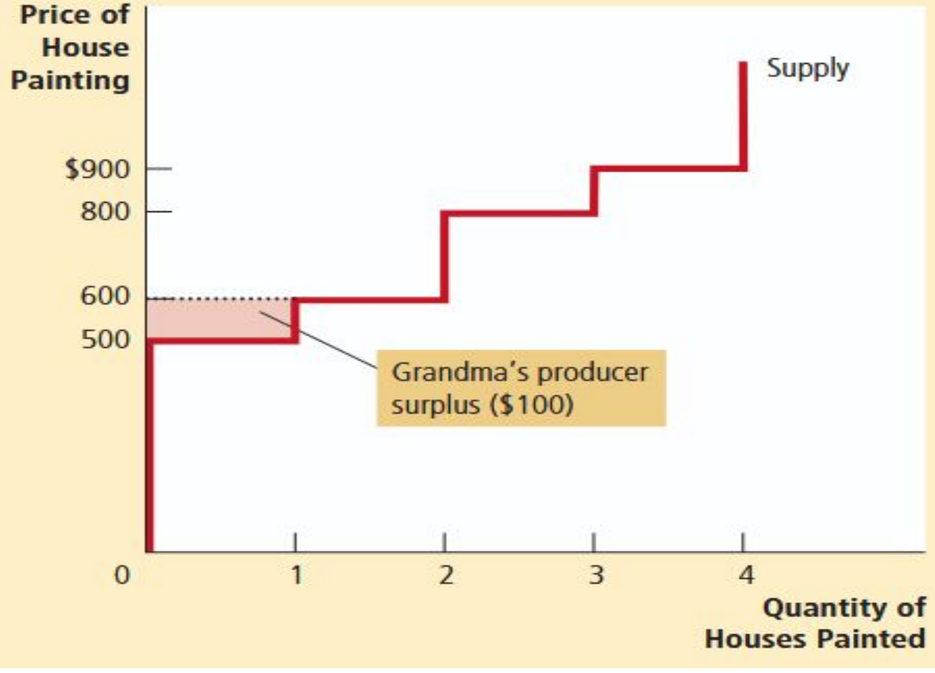


Seller	Cost
Mary	$900
Frida	800
Georgia	600
Grandma	500

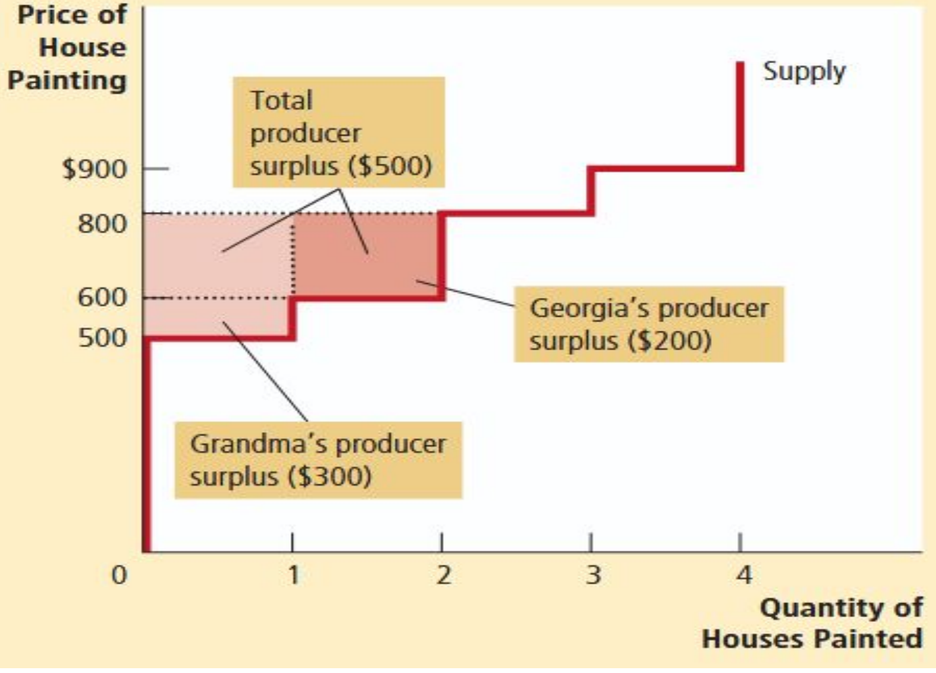
Price	Sellers	Quantity Supplied
$900 or more	Mary, Frida, Georgia, Grandma	4
$800 to $900	Frida, Georgia, Grandma	3
$600 to $800	Georgia, Grandma	2
$500 to $600	Grandma	1
Less than $500	None	0

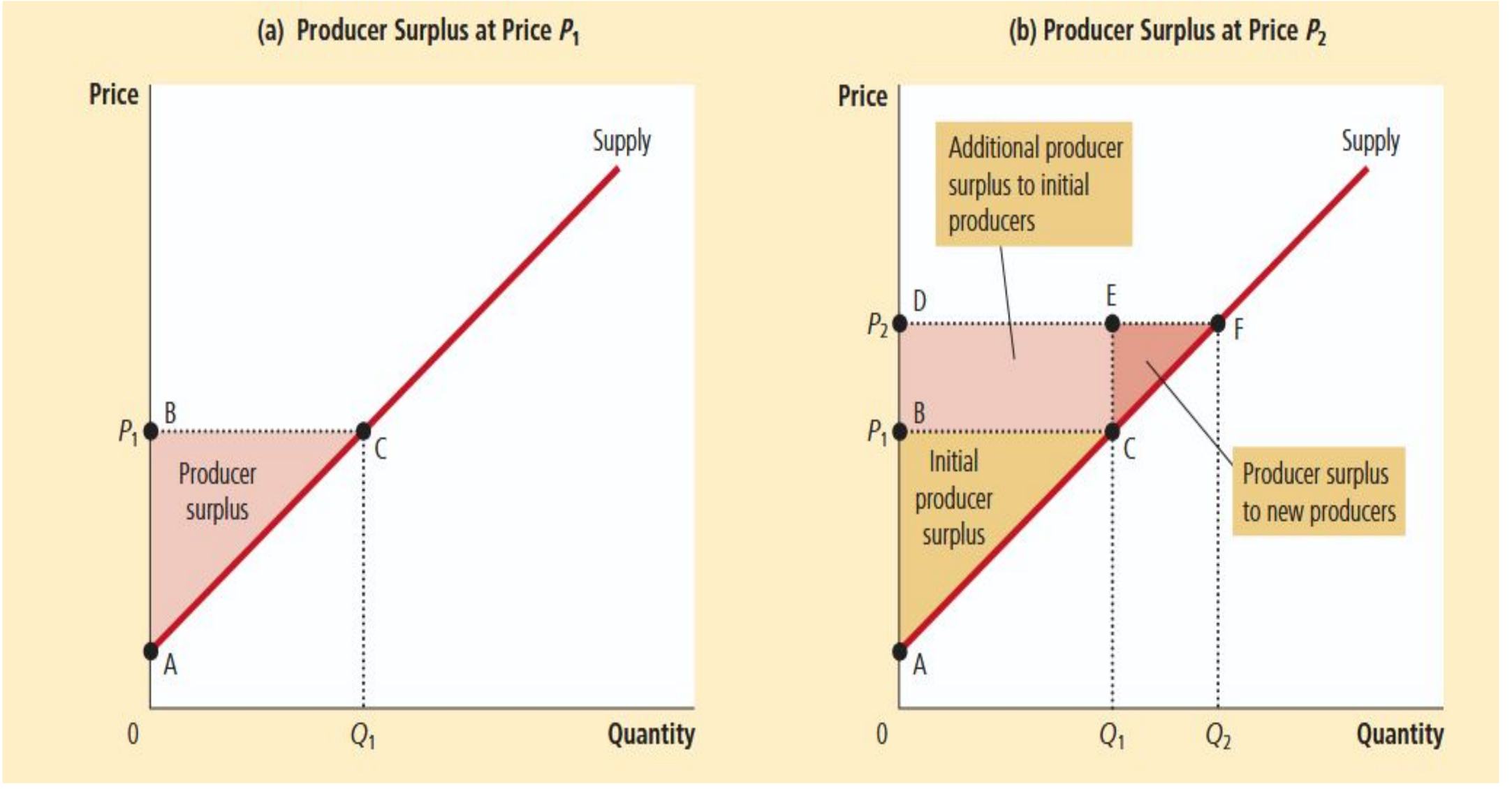


(a) Price = $600



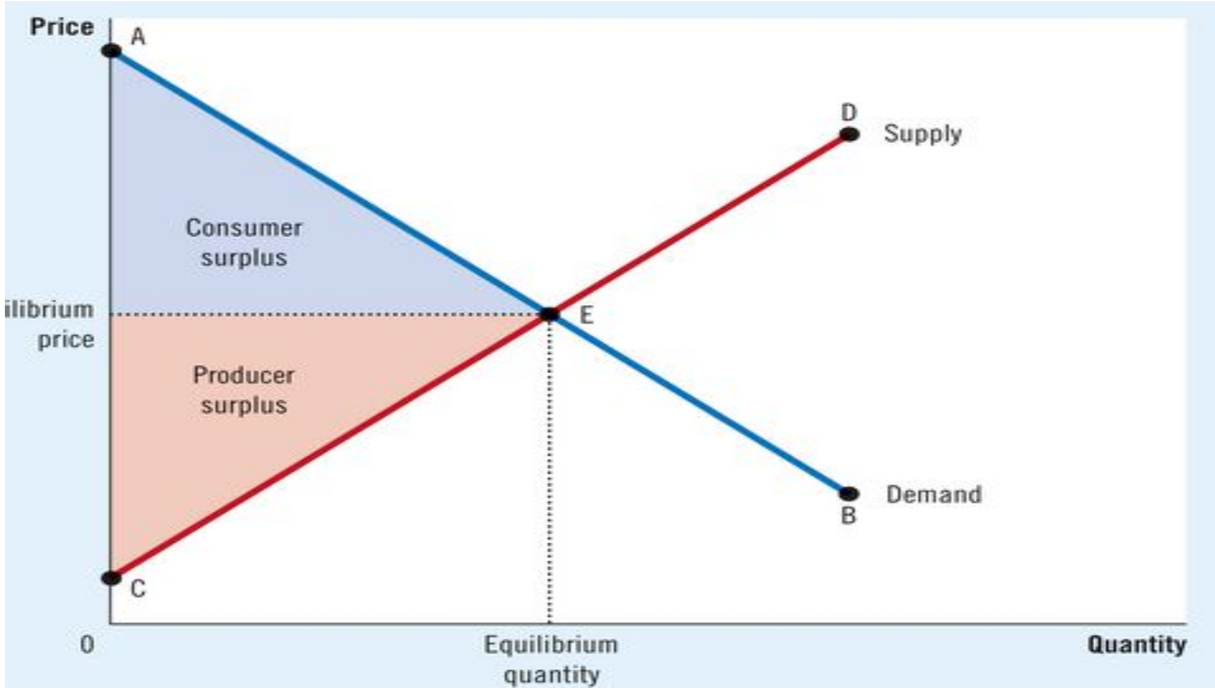
(b) Price = $800




(a) Producer Surplus at Price P_1
Price
Supply
P_1
B
C
Producer surplus
A
0
Q_1
Quantity
(b) Producer Surplus at Price P_2
Price
Additional producer surplus to initial producers
Supply
D
E
F
P_2
P_1
B
C
Initial producer surplus
Producer surplus to new producers
A
0
Q_1
Q_2
Quantity

Market Efficiency

A market is said to be efficient when resources are allocated in the best possible way, so that total welfare (consumer surplus + producer surplus) is maximized.



Formula

$$CS = \frac{1}{2} \times (Base) \times (Height)$$

$$PS = \frac{1}{2} \times (Base) \times (Height)$$

$$CS = \tfrac{1}{2} \times Q^* \times (P_{max} - P^*)$$

$$PS = \tfrac{1}{2} \times Q^* \times (P^* - P_{min})$$

In a local market, the demand and supply functions are:

$$Q_d = 180 - 3P$$
$$Q_s = -10 + 2P$$

where $Q$is quantity (units) and P is price (₹ per unit).

1. Find the equilibrium price and equilibrium quantity.
2. Find the maximum price (P_{max}) consumers are willing to pay,
3. Find the price (P_{min}) at which $Q_s = 0$
4. Calculate Consumer Surplus (CS), Producer Surplus (PS) and Total Surplus (TS) at equilibrium.
5. Draw a labelled demand–supply diagram showing equilibrium, CS and PS

Production

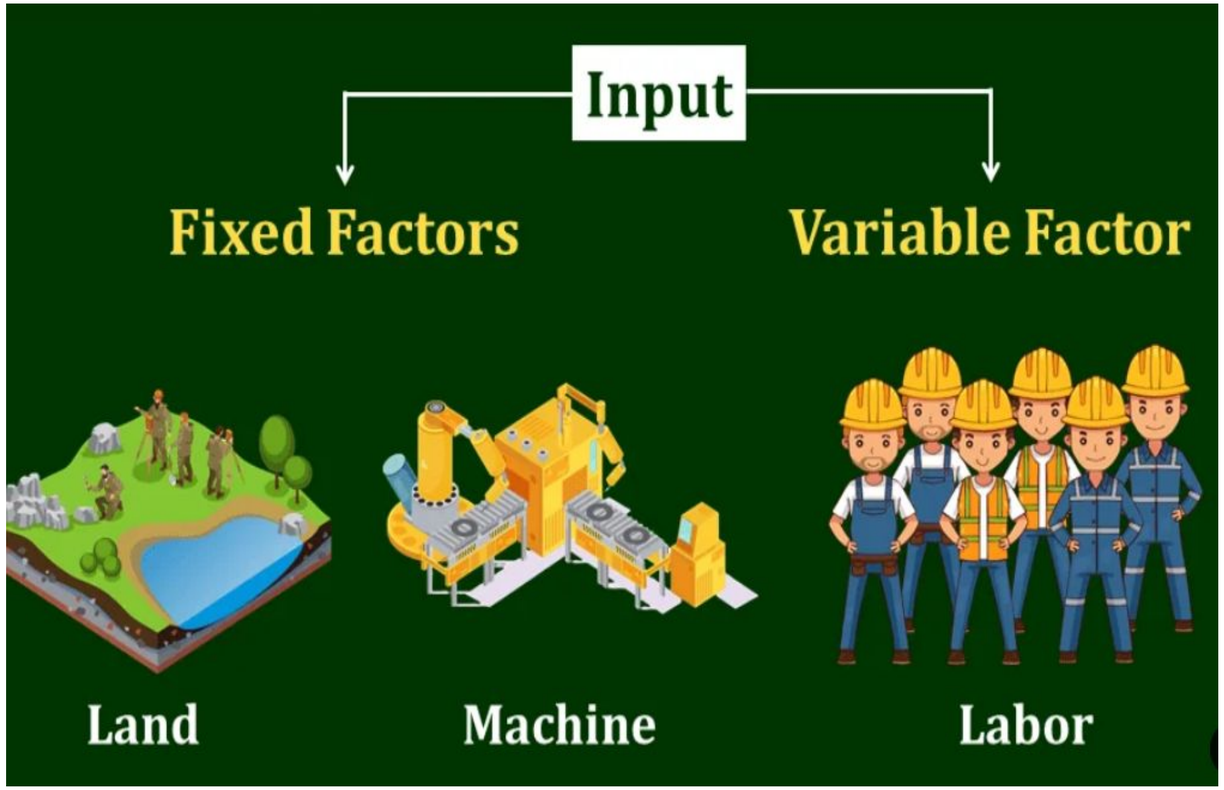


Production is the process of transformation of inputs into goods and services of utility to consumers and/or producers.

Types of Inputs

Technology

Fixed and Variable inputs



Ramesh decides to start an organic tomato farm, and in doing so, he brings together all four factors of production. He first rents 2 acres of land at a cost of ₹20,000 per year, which provides the natural resource base for farming. To cultivate the crops, he hires three farm workers, paying them ₹10,000 per month each. For capital, Ramesh invests in a tractor worth ₹3,00,000, an irrigation pump costing ₹50,000, and seeds and fertilizers for ₹1,00,000. Finally, using his entrepreneurial skills, he organizes these resources, takes the risk of focusing on organic tomatoes, and expects to sell 50,000 kg at ₹20 per kg, generating revenue.





Land (Rent) = ₹20,000

Labour = ₹3,60,000

Capital = ₹3,00,000 + ₹50,000 + ₹1,00,000 = ₹4,50,000

Total Cost = ₹20,000 + ₹3,60,000 + ₹4,50,000 = ₹8,30,000

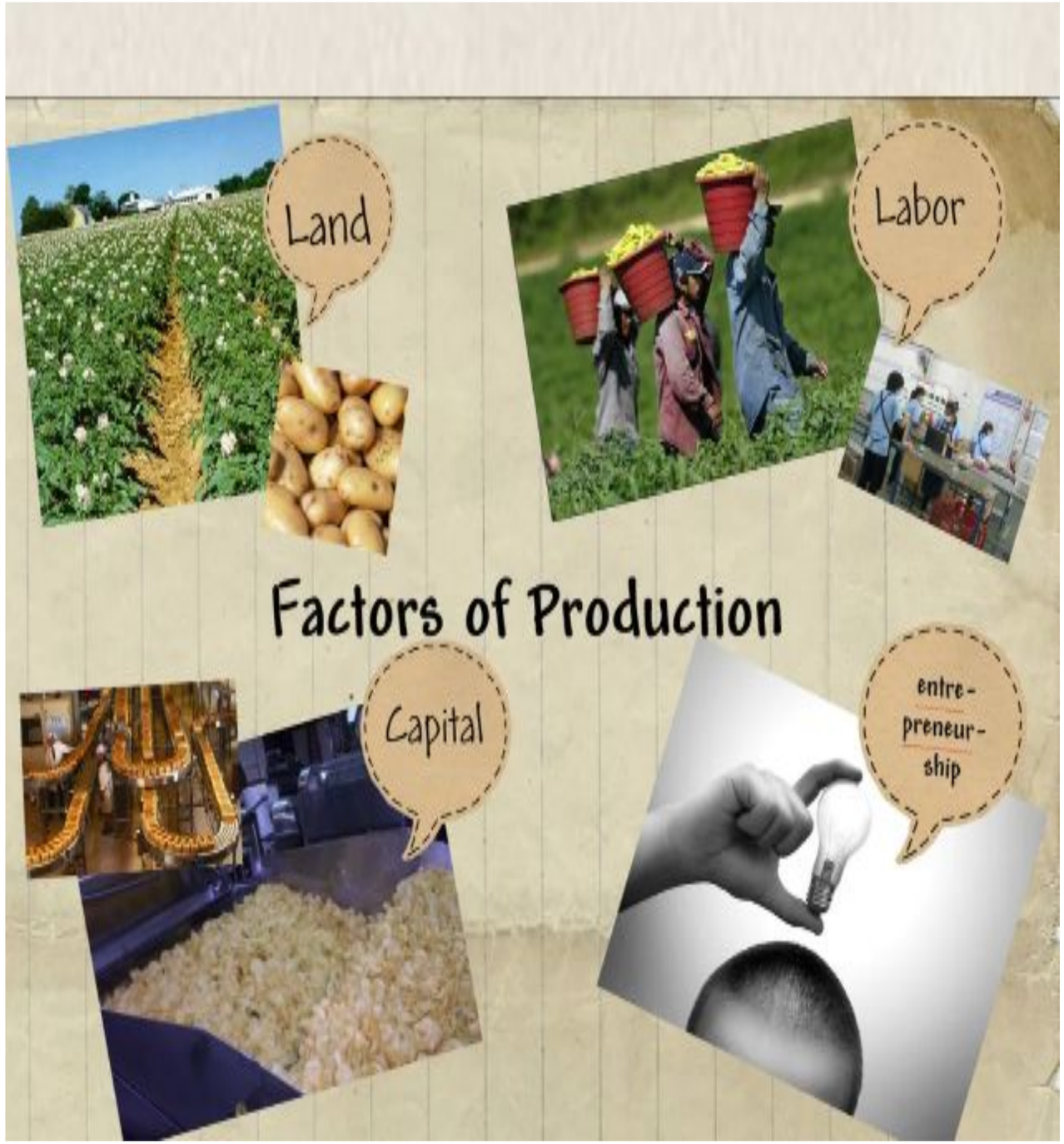
Revenue = ₹10,00,000

Profit = ₹10,00,000 – ₹8,30,000 = ₹1,70,000

Factors of Production

- Land (La)
- Labour (L)
- Capital (K)
- Enterprise (E)

$$Q = f(La, L, K, E)$$



The Factors of Production

Land



Land includes the "gifts of nature," or natural resources not created by human effort.

Rent

Capital



Capital includes the tools, equipment, and factories used in production.

Interest

Labor



Labor includes people with all their efforts and abilities.

Wages

Entrepreneurs



Entrepreneurs are individuals who start a new business or bring a product to market.

Profit

Income

Production Function With One Variable Input

Short Run

TOTAL PRODUCT / unit time (with only L variable)

$$TP_L = Q$$

- AVERAGE PRODUCT

$$AP_L = \frac{TP_L}{L} = \frac{Q}{L}$$

- MARGINAL PRODUCT

$$MP_L = \frac{dQ}{dL} = \frac{Q_2 - Q_1}{L_2 - L_1}$$

Suppose a **bakery** produces bread.

Workers	Bread Produced (TP)
1	10
2	25
3	45
4	60
5	70

If **3 workers produce 45 breads**, then:
Total Product = 45 breads

If:
Total product = 45 breads
Workers = 3
$AP = 45/3 = 15$

Workers	Total Product	Marginal Product
1	10	10
2	25	15
3	45	20
4	60	15
5	70	10

When workers increase from **2 to 3**:

$$MP = 45 - 25 = 20$$

So the **third worker adds 20 breads**.

Anita starts an organic corn farm and hires workers to cultivate and harvest corn. In the first year, she hires 1 worker, who produces 20,000 kg of corn. In the second year, she hires an additional worker (a total of 2 workers), and the total output rises to 45,000 kg.

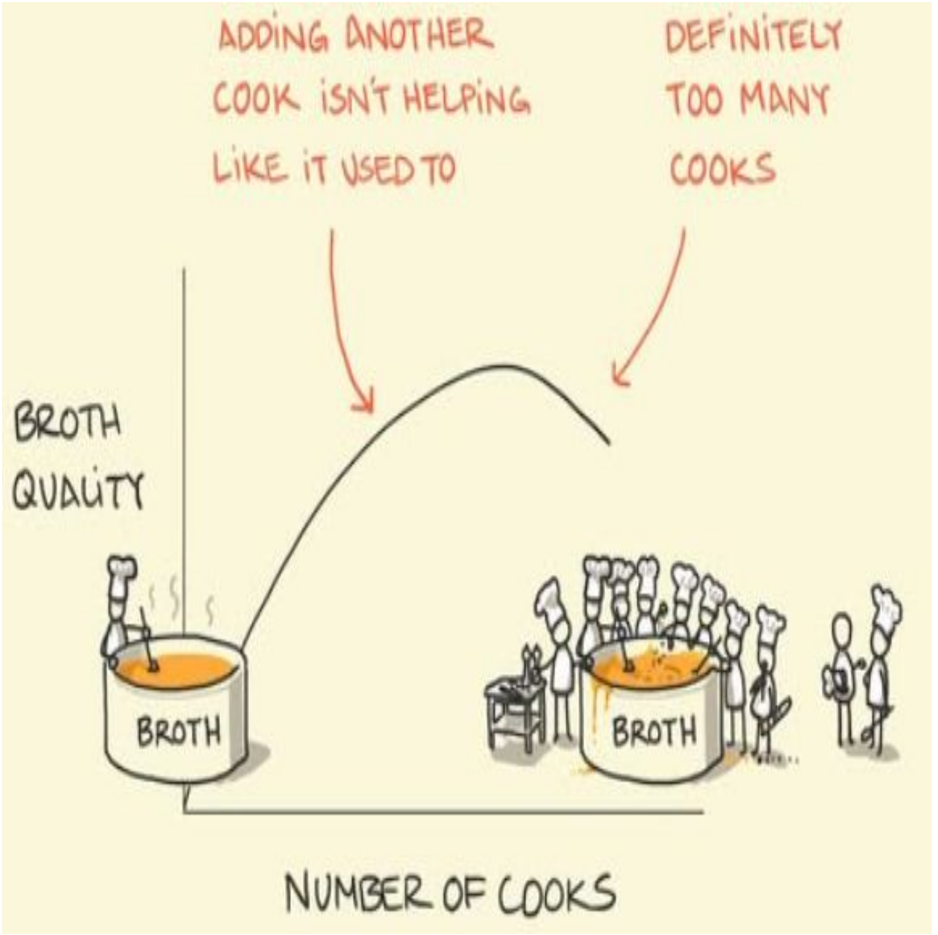


Workers (L)	TP (kg/year)	AP = TP ÷ L (kg/worker)	MP = ΔTP/ΔL (kg)
1	20,000	20,000	(20,000 − 0)/(1 − 0) = 20,000
2	45,000	22,500	(45,000 − 20,000)/(2 − 1) = 25,000

Production Function With One Variable Input

Short Run

Law of Variable Proportions, or the Law of Diminishing Marginal Returns, states that with an increase in the quantity of the variable factor, its marginal and average products will eventually decline, with other inputs remaining unchanged.



Stage 1: Increasing Returns	Each extra cook adds more; specialization boosts productivity.
Stage 2: Diminishing Returns	Output still rises, but each additional cook contributes less.
Stage 3: Negative Returns	Too many cooks spoil the broth; total output falls.

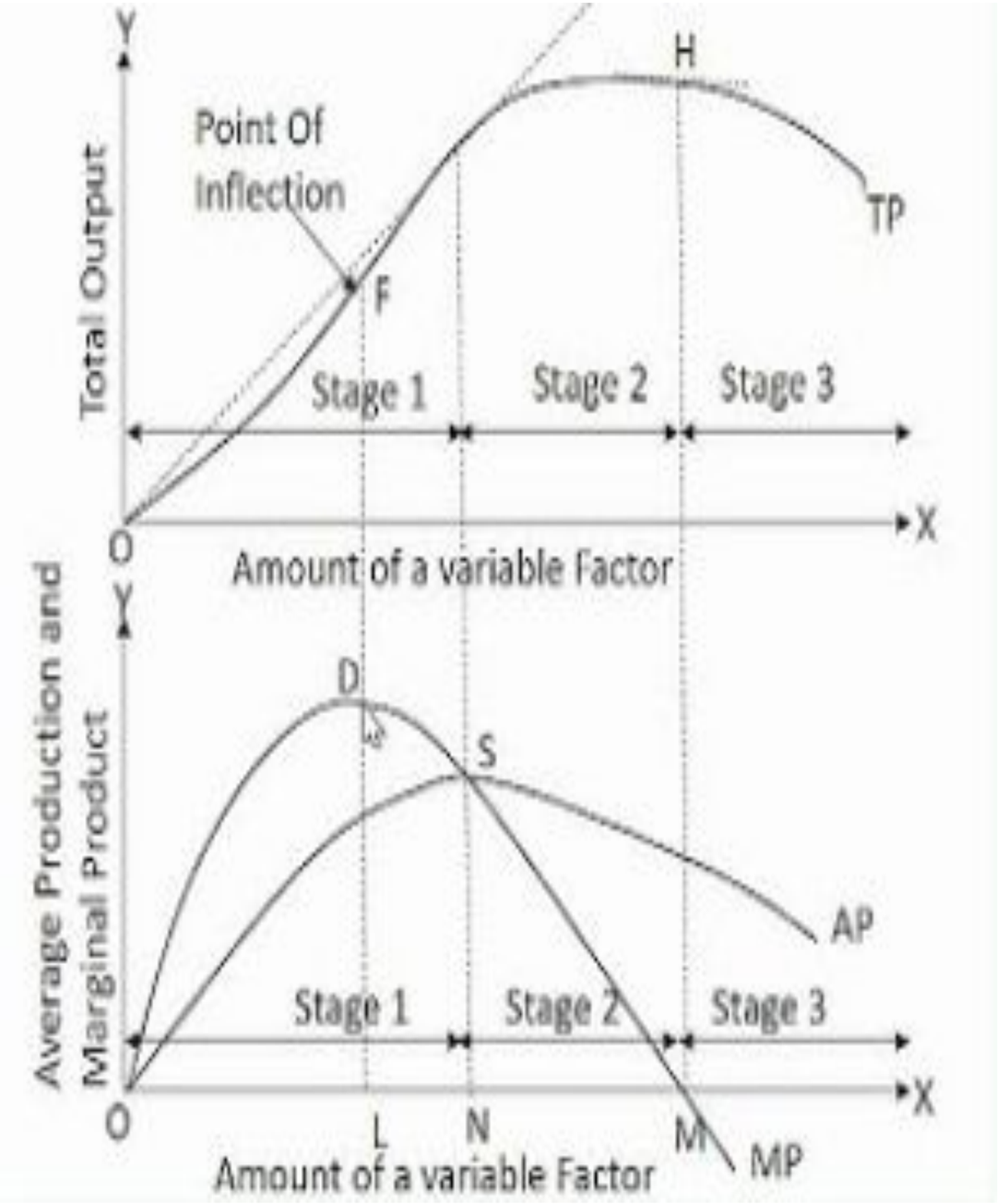






Stage of Production	Agriculture (Fertilizer on Crops)	Factory (Workers & Machines)	Office (Team Size)
Stage 1: Increasing Returns to scale/ Economies of scale	Small amount of fertilizer → big jump in crop yield (plants get nutrients).	Few workers + machines → division of labor increases efficiency.	Adding first employees → tasks divided, work gets done faster.
Stage 2: Diminishing Returns	More fertilizer still raises yield, but each extra dose helps less.	More workers increase output, but machines get overused → less gain per worker.	Adding more employees still increases output, but meetings/coordination slow things down.
Stage 3: Negative Returns to scale/ Diseconomies of scale	Excess fertilizer burns crops → total yield falls.	Too many workers → overcrowding, mistakes, breakdowns → output falls.	Too many employees → endless meetings, miscommunication → productivity drops.

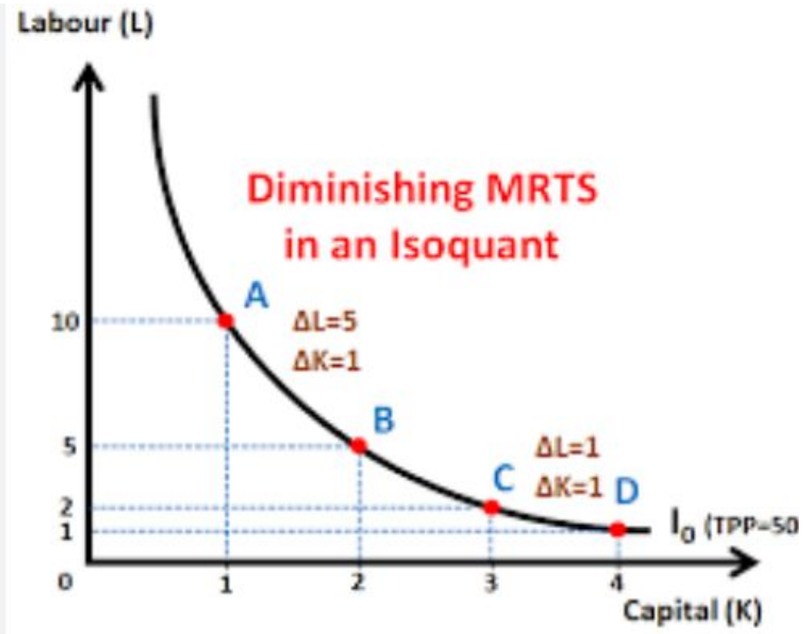
Stage	Total Product (TP)	Average Product (AP)	Marginal Product (MP)	Interpretation / Key Features
1 – Increasing Returns	Rises at increasing rate	Rises	Rises	Each additional worker increases efficiency; both AP and MP are rising but MP>AP
2 – Diminishing Returns	Rises at decreasing rate	Peaks then starts falling	Falls but positive	TP continues to rise but MP < AP; additional workers add less output
3 – Negative Returns	Falls	Falls	Negative	Too many workers cause overcrowding; TP decreases and MP becomes negative.



Production Function With Two Variable Input

Long Run

An **isoquant** is the locus of all technically efficient combinations for producing a given level of output.



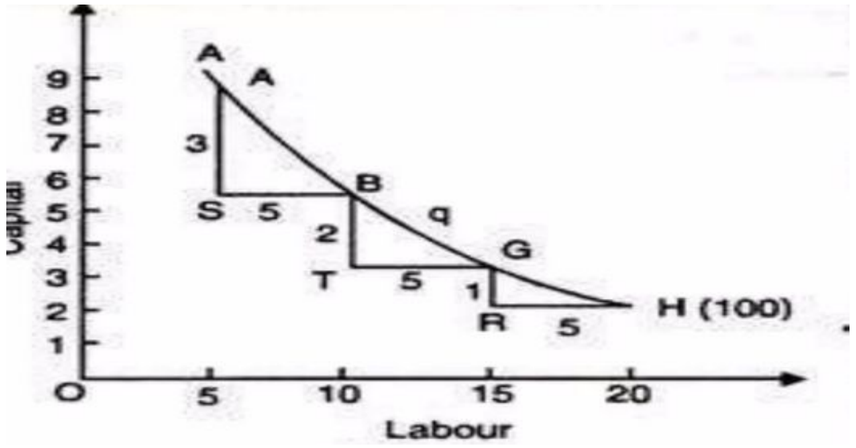
Marginal Rate of Technical Substitution

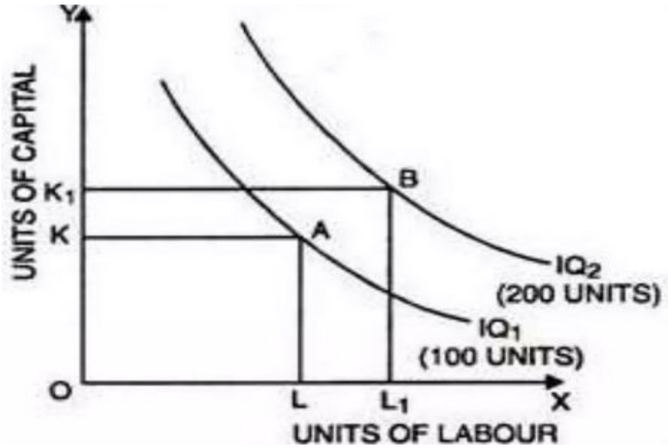
The MRTS is the Marginal Rate of (Technical) Substitution because it is substitution while maintaining constant output. For every production output quantity, there is a unique combination of inputs that minimizes cost called the Least-Cost Combination of Inputs.

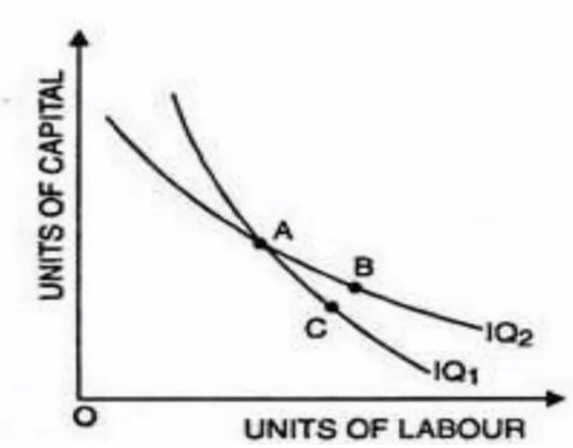
$$\text{MRTS}_{LK} = \frac{\Delta K}{\Delta L}$$

Properties of Iso-Quant

- Slopes downward to the right
- Convex to the origin.
- Iso-Quants cannot intersect with each other
- Higher Iso-Quant curve represents a higher level of output than a lower one.





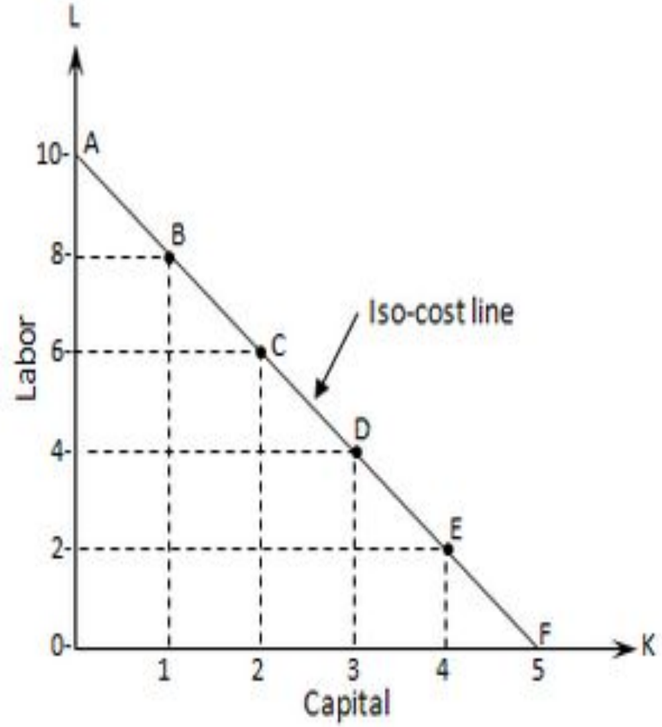


Iso Cost Line

The Iso-Cost line is the locus of points of all the different combinations of labor and capital that a firm can employ, given the total cost and prices of inputs.

Slope of the Iso-Cost line = - $\frac{P_L}{P_K}$

$$P_L Q_L + P_K Q_K = C$$



Let's say a firm has a budget of ₹10,000, and it needs to choose between hiring Labour (L) and renting Capital (K) (like machines).

Wage rate (w) = ₹500 per labourer

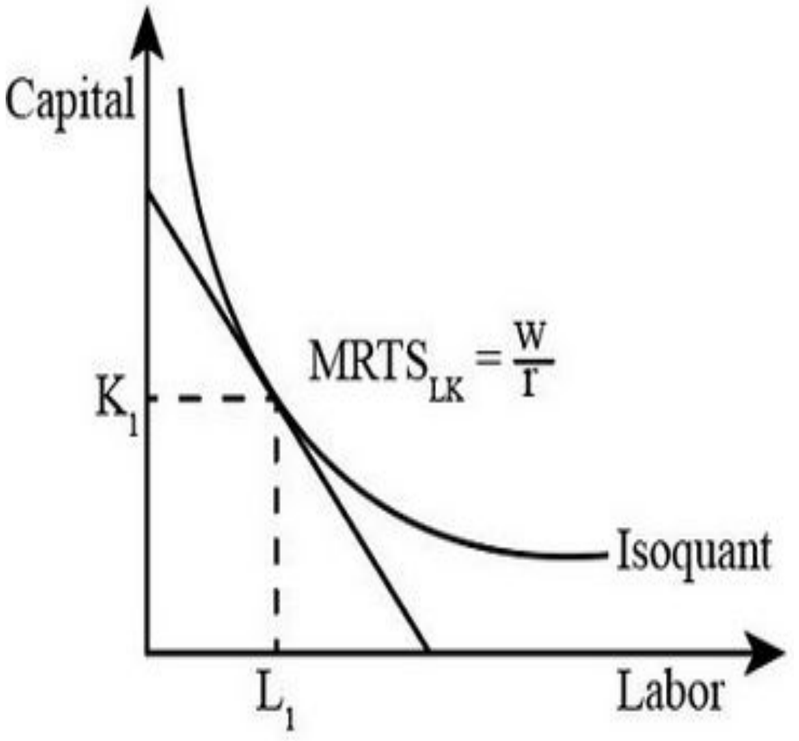
Rental rate of capital (r) = ₹1,000 per machine

Total budget = ₹10,000

$$C=wL+rK$$

$$\mathbf{C= P_L Q_L + P_K Q_K}$$

$$10{,}000 =500L+1000K$$



Labor (L)	Capital (K)	Check: 500L + 1000K = 10,000
0	10	0 + 10,000 = 10,000
2	9	1,000 + 9,000 = 10,000
4	8	2,000 + 8,000 = 10,000
6	7	3,000 + 7,000 = 10,000
8	6	4,000 + 6,000 = 10,000
10	5	5,000 + 5,000 = 10,000
12	4	6,000 + 4,000 = 10,000
14	3	7,000 + 3,000 = 10,000
16	2	8,000 + 2,000 = 10,000
18	1	9,000 + 1,000 = 10,000
20	0	10,000 + 0 = 10,000

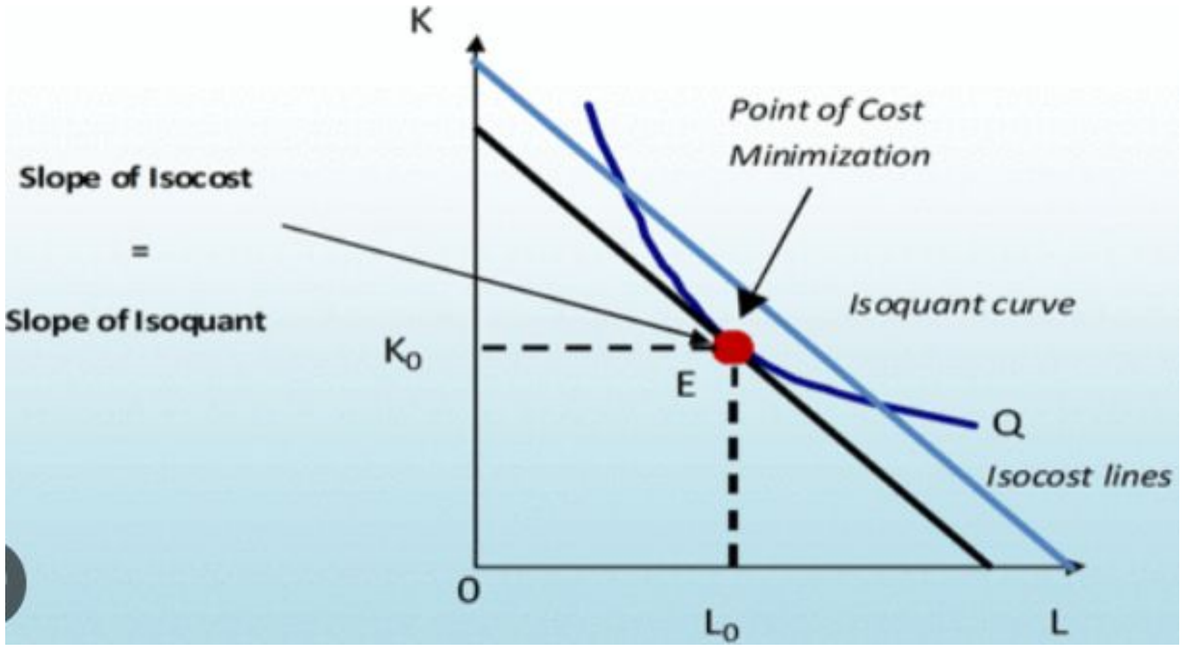
Test for Least-Cost Combination

The tangent point between an isocost and an isoquant represents the least cost.

$$\text{Slope of Isoquant} = -\frac{MP_L}{MP_K} = -\text{MRTS}_{L,K}$$

$$\text{Slope of Isocost} = -\frac{w}{r}$$

$$\boxed{\text{MRTS}_{L,K} = \frac{MP_L}{MP_K} = \frac{w}{r}} = \frac{\Delta K}{\Delta L}$$



Producer’s Equilibrium Through Iso-Quant and Iso-Cost Line

The necessary condition for producer equilibrium:

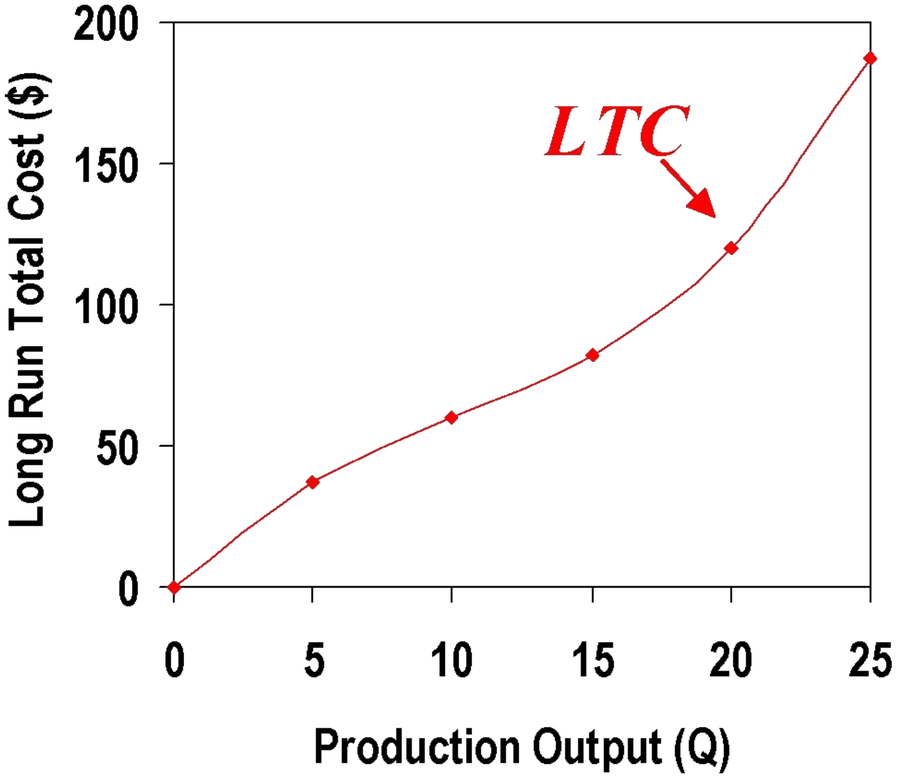
Slope of iso-quant = Slope of the iso-cost line

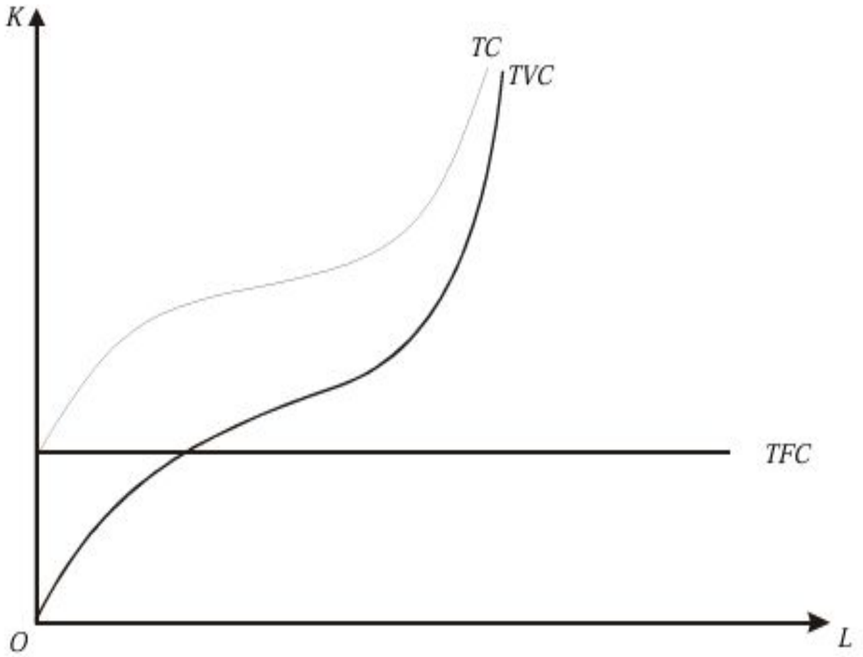
Types of Cost

- Opportunity Cost
- Implicit and explicit cost
- Economic Profit versus Accounting Profit
- Total cost (combination of fixed and variable cost)
- Average cost (combination of average fixed and variable cost)
- Marginal cost (combination of marginal fixed and variable cost)

Concept	Definition	Real-Life Example
Opportunity Cost	Value of the next best alternative foregone	Investing ₹5 lakhs in a tea café instead of an FD earning 6% interest. Opportunity cost = ₹30,000 (lost interest income).
Explicit Cost	Direct monetary outflows for business operations	₹20,000/month spent on rent and ingredients for a home bakery.
Implicit Cost	Opportunity cost of using owned resources	₹50,000/month salary foregone by quitting a job to run the bakery.
Accounting Profit	Total Revenue – Explicit Costs	Revenue = ₹80,000; Explicit costs = ₹20,000 ⇒ Accounting profit = ₹60,000.
Economic Profit	Total Revenue – (Explicit + Implicit Costs)	Revenue = ₹80,000; Total costs = ₹20,000 + ₹50,000 ⇒ Economic profit = ₹10,000.
Total Cost (TC)	Sum of Fixed Cost and Variable Cost	Tea packaging unit: FC = ₹1,00,000; VC = ₹1,00,000 ⇒ TC = ₹2,00,000 for 10,000 units.
Average Cost (AC)	Total Cost / Quantity or AC = AFC + AVC	TC = ₹2,00,000; Quantity = 10,000 units ⇒ AC = ₹20 per unit (AFC = ₹10, AVC = ₹10).
Marginal Cost (MC)	Cost of producing one additional unit	Producing 10,001 units costs ₹2,00,020; 10,000 units cost ₹2,00,000 ⇒ MC = ₹20.

Long-Run Total Cost (*LTC*)





Long-Run Total Cost is the least cost combination of inputs for each production quantity

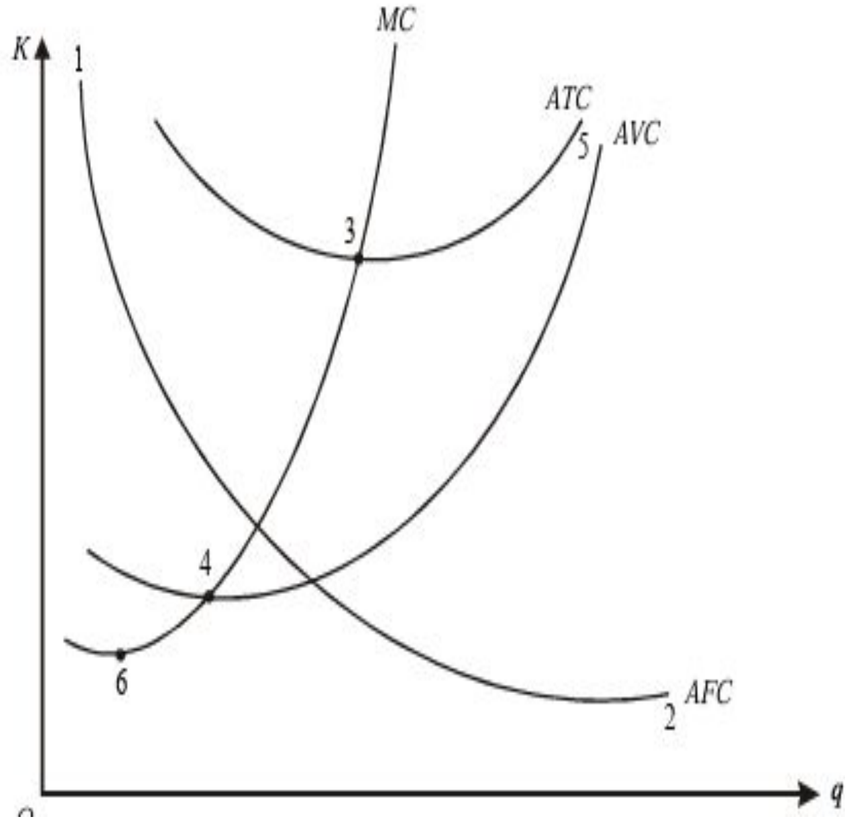
Long-Run Average & Marginal Cost (*LAC & LMC*)

Long-Run Average Cost

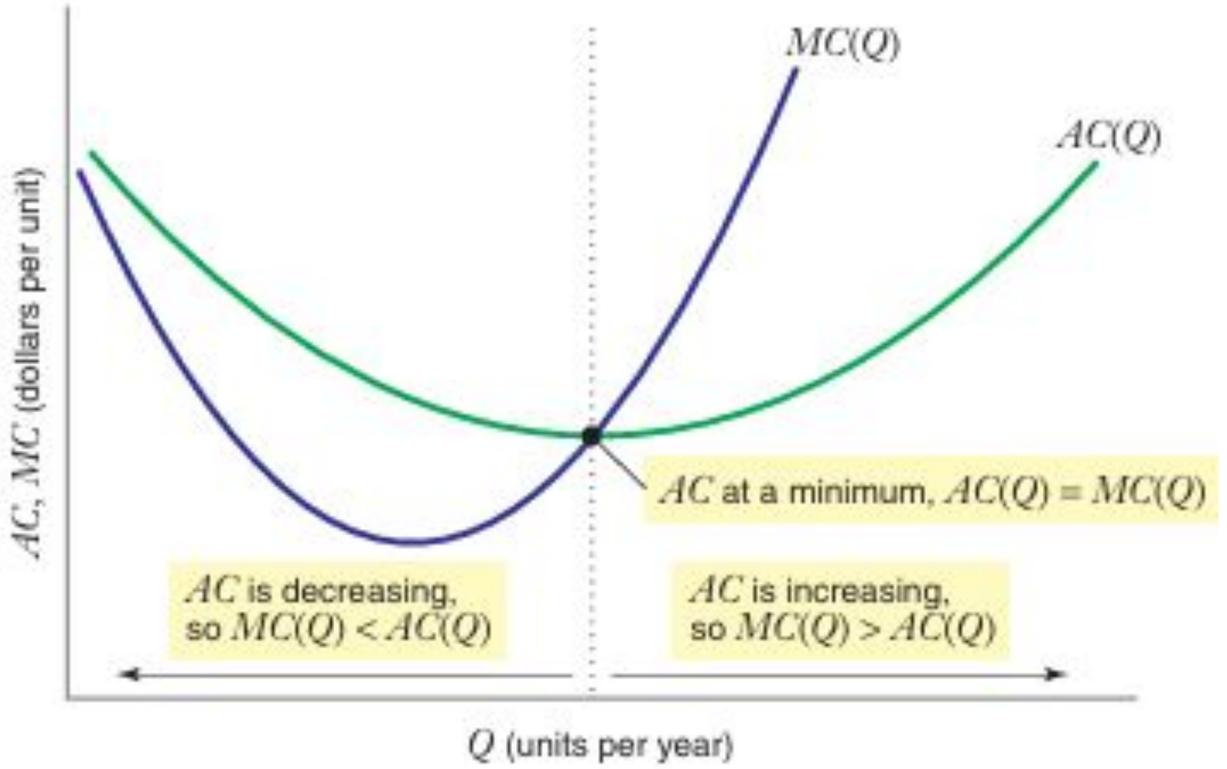
$$LAC = \frac{LTC}{Q}$$

Long-Run Marginal Cost

$$LMC = \frac{d(LTC)}{dQ}$$

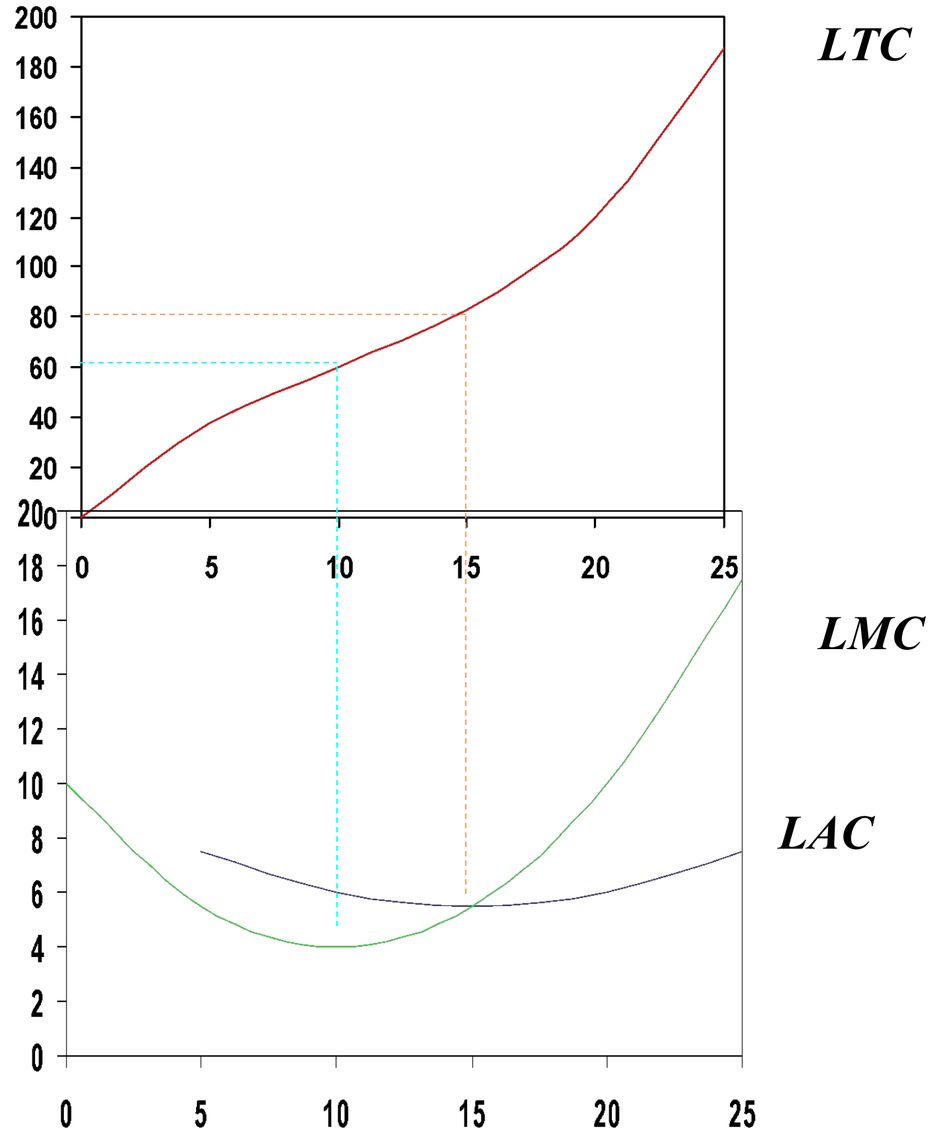


LAC	LMC
• In the beginning, average cost falls because larger scale lowers cost per unit (economies of scale). • At some point, the firm reaches the minimum efficient scale (MES) → lowest average cost. • Beyond this, average cost rises because of diseconomies of scale.	• When producing more units lowers average cost, marginal cost (MC)<AC. • When producing more units raises average cost, MC is>AC. • At the exact minimum of AC, MC = AC.


MC(Q)
AC(Q)
AC, MC (dollars per unit)
AC at a minimum, AC(Q) = MC(Q)
AC is decreasing, so MC(Q) < AC(Q)
AC is increasing, so MC(Q) > AC(Q)
Q (units per year)

Relationship of *LTC*, *LAC*, and *LMC*

Long-Run Marginal Cost (*LMC*) = Long-Run Average Cost (*LAC*) when **Long-Run Average Cost (*LAC*) is at its minimum**.

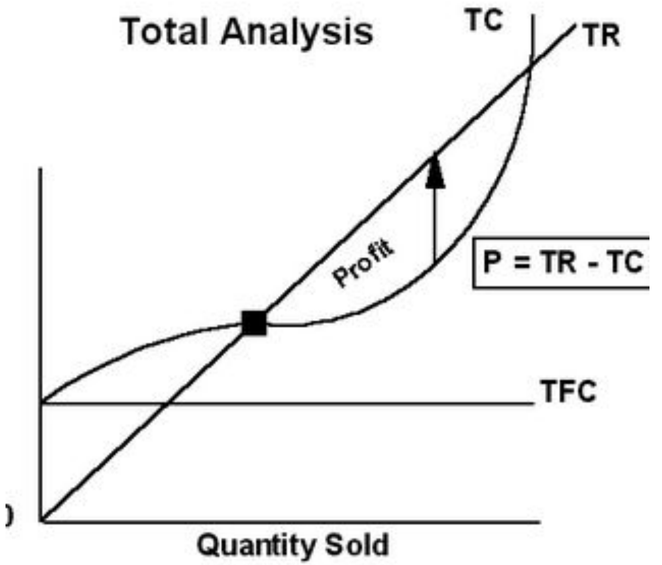


Stage	LTC	LAC	LMC	Relation
Stage 1	Increasing slowly	Falling	Below LAC	LMC < LAC
Stage 2	Linear (constant slope)	Minimum (flat)	=LAC	LMC = LAC
Stage 3	Increasing steeply	Rising	Above LAC	LMC > LAC

Total Revenue, Total Cost, and Profit

- The amount that the firm receives for the sale of its output is called its total revenue.
- The amount that the firm pays to buy inputs is called its total cost.
- Profit is a firm's total revenue minus its total cost:

$$\text{Profit} = \text{Total revenue} - \text{Total cost}$$







Market Structures and Decision Making

Type of market	*Number of firms*	*Nature of product*	*Number of buyers*	*Freedom of entry and exit*	*Examples*
Perfect competition	Very Large	Homogeneous (undifferentiated)	Very Large	Unrestricted	Agricultural commodities, shares unskilled labour
Monopolistic competition	Many	Differentiated	Many	Unrestricted	Retail stores, detergents
Oligopoly	Few	Undifferentiated or differentiated	Few	Restricted	Cars, computers, universities
Monopoly	Single	Unique	Many	Restricted	Indian Railways, Microsoft

	Market Structure	Competition	Nature of Product	Buyers/Sellers	Entry/Exit Conditions
Vegetable Mandi	Perfect Competition	Many small sellers and buyers	Homogeneous (identical vegetables)	No one controls price	Easy entry and exit
Indian Railways	Monopoly	Only one major provider	Unique service, no close substitutes	Single seller dominates	Very high barriers (infrastructure cost)
Telecom Industry (Jio, Airtel, Vodafone Idea, BSNL)	Oligopoly	Few large firms dominate	**Differentiated but similar services**	Each firm influences others' strategies	High barriers (licenses, tech costs)
Restaurants & Cafés	Monopolistic Competition	Many sellers	**Differentiated (taste, branding, ambience)**	No single firm controls market	Easy entry and exit

Oligopoly with Undifferentiated (Homogeneous) Products)

- **Cement industry** → ACC, UltraTech, Shree Cement (cement is largely the same product).
- **Steel industry** → Tata Steel, JSW Steel, SAIL (steel is a standard commodity).
- **Crude oil/petroleum** → (Saudi Arabia, UAE, etc. – oil is homogeneous).

Oligopoly with Differentiated Products

- **Automobiles** → Toyota, Ford, Hyundai, Maruti Suzuki (cars differ in design, brand image, technology).
- **Mobile phones** → Apple, Samsung, Xiaomi (different features, designs, ecosystems)

Monopolistic Competition

- Restaurants / Cafes – Each serves food or drinks, but the menu, taste, and ambiance differ (e.g., Starbucks vs local cafes).
- Clothing brands – Many brands sell shirts or jeans, but styles, designs, and quality differ (e.g., Zara vs H&M vs local boutiques).
- Toothpaste – Colgate, Pepsodent, Sensodyne (same basic product, differentiation through features like whitening, sensitivity, or herbal).
- Soap / Shampoo – Dove, Lifebuoy, Sunsilk (competing products with slight differences in formula, scent, or marketing).

Profit Maximization

First-order condition of profit maximization:

MR=MC

Second-order condition of profit maximization:

Slope of MR<Slope of MC



Perfect Competition

Features of Perfect Competition

- ✔ There are many buyers and many sellers in the market.
- ✔ The goods offered by the various sellers are largely the same (Homogeneous Product).
- ✔ Firms can freely enter or exit the market.
- ✔ Perfect Knowledge
- ✔ **Perfectly Elastic Demand Curve**
- ✔ Firm is a Price Taker

Selling Price (₹/kg) **10kg**	Total Revenue (₹)	Total Cost (₹)	Profit (₹)
19	190	180	10
20 (MP)	200	180	20
21	210	180	30

21

20

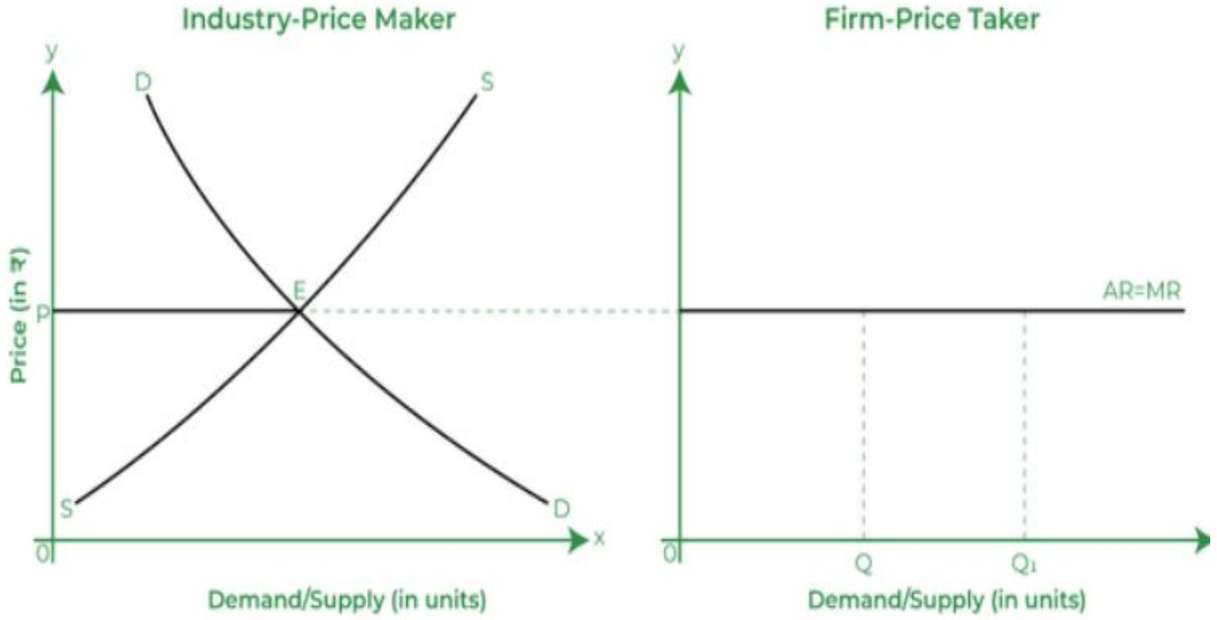
19

In the town of AgriVille, hundreds of small farmers sell identical fresh tomatoes in the local market every morning. Since all tomatoes are of similar size, color, and quality, buyers are indifferent to who they purchase from and only care about the price. Last week, the market price for 1 kg of tomatoes was ₹20. One farmer, Rani, tried selling at ₹22 hoping to earn more, while another farmer, Suresh, lowered his price to ₹18 to attract more buyers. However, Rani sold nothing because buyers chose cheaper sellers, and Suresh sold quickly but earned less than others. By the end of the day, all farmers returned to selling at ₹20 — the equilibrium price. This shows that in a perfectly competitive market, no individual seller has the power to influence the price; everyone is a price taker, and market forces of demand and supply determine the price.

Why did all farmers end up selling at ₹20 again?

Market Demand Curve And Firm's Demand Curve

- The demand curve for an individual firm is a horizontal straight line
- The market demand curve for the industry is a standard downward-sloping curve.



Price (₹/kg)	Total Market Demand (kg)
25	1000
20	2000
15	3000

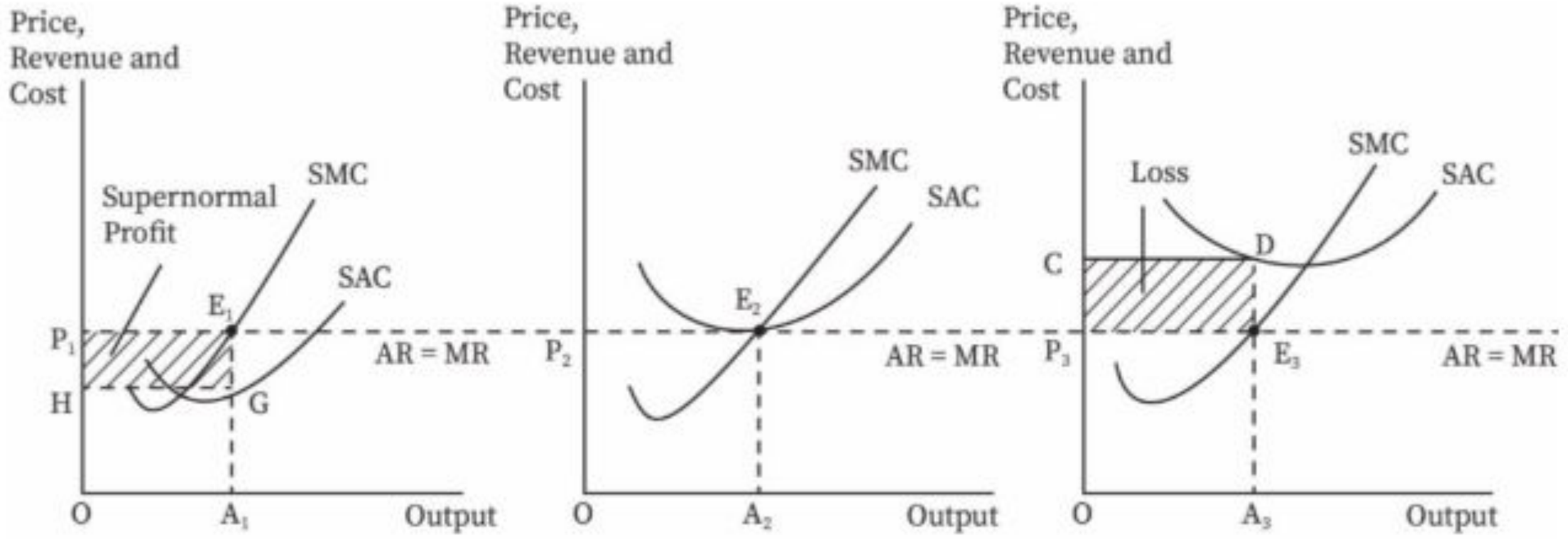
Demand And Revenue of a Firm

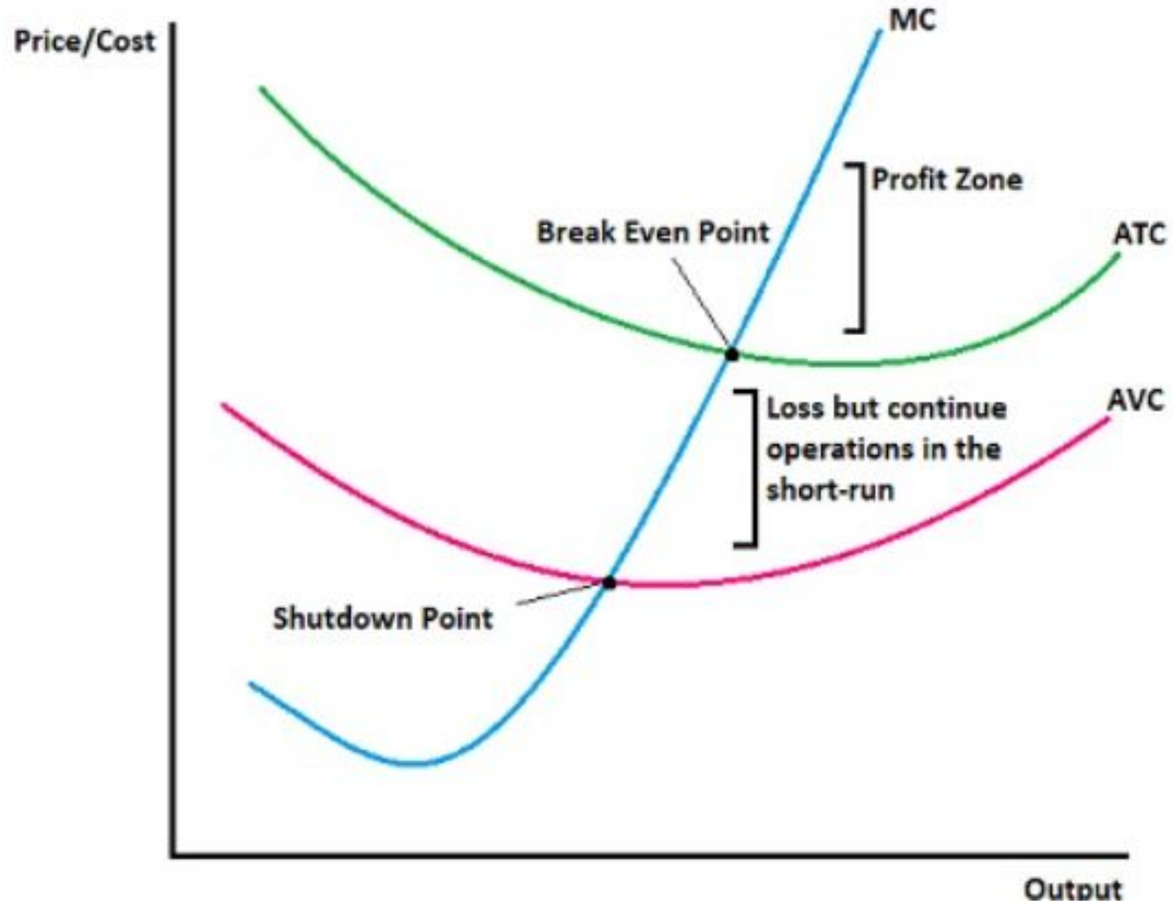
- Total revenue
- Marginal revenue
- Average revenue

Profit, Revenue, and Cost of a Firm

Short Run Equilibrium

- Case of Supernormal Profit
- Case of Normal Profit
- Case of Loss
- Special Case: Exit or Shut Down Point


Price,
Revenue and
Cost
Supernormal
Profit
SMC
SAC
E1
P1
H
G
AR = MR
O
A1
Output
Price,
Revenue and
Cost
SMC
SAC
E2
P2
AR = MR
O
A2
Output
Price,
Revenue and
Cost
Loss
SMC
SAC
D
C
P3
E3
AR = MR
O
A3
Output



P<AVC Shut down
P>AVC Continue

A perfectly competitive firm has:
FC = ₹500
VC = ₹900
Output = 100 units
Market price = ₹8

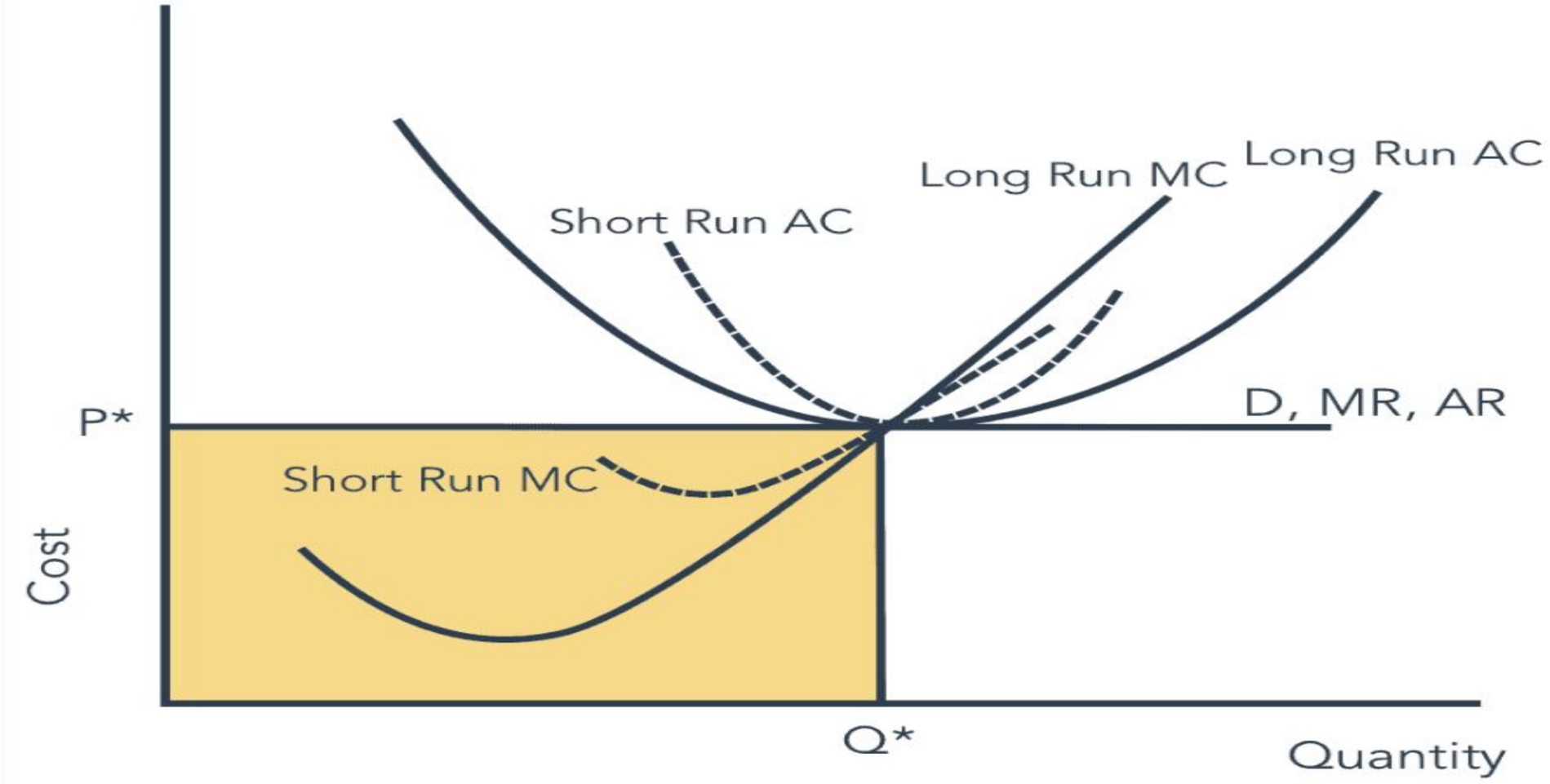
Market Supply Curve And Firm's Supply Curve

Portion of the MC curve above the AVC curve, as the MC curve (above AVC) shows how output changes with price, that's exactly what a supply curve does

Long Run Equilibrium

In the long run, perfectly competitive firms earn only normal profits.

The condition for long-run equilibrium in perfect competition is P = LAC = SAC = SMC = MR = AR


Long Run AC
Long Run MC
Short Run AC
D, MR, AR
P*
Short Run MC
Cost
Q*
Quantity

Thousands of small farmers across states like Karnataka, Maharashtra, and Andhra Pradesh produce and sell tomatoes in local mandis. Since the product is homogeneous and there are many sellers, no single farmer can influence the market price, making them price takers.

Prices in this market are determined purely by supply and demand. On days with high supply, prices fall, and when supply is limited, prices rise. Farmers cannot charge higher prices individually because buyers would simply purchase from other sellers.

Entry and exit in the market are relatively easy, and information about mandi prices is widely available. This ensures transparency and keeps the market competitive.

1. Why is the Indian tomato market considered a perfectly competitive market?

2. Who determines the price of tomatoes in this market?

3. What does it mean when we say farmers are “price takers”?

4. How does free entry and exit of farmers affect the market?

5. How do consumers benefit from perfect competition in the tomato market?

Monopoly

A monopoly is a market in which a single seller sells a product (or service) that has no substitute.

Organization	Type of Monopoly	Control Mechanism	Agricultural Sector
Monsanto	Private Monopoly	Patent rights over GM seeds (e.g., Bt cotton)	Seed Industry
ITC (Indian Tobacco Company)	Private Monopoly	Contract farming and market dominance	Tobacco Procurement
NDDB (Operation Flood, early phase)	Public Monopoly	Centralized procurement and distribution	Dairy Sector
IFFCO / NFL (before liberalization)	Public Monopoly	Government control over production and distribution	Fertilizer
Sugar Cooperatives (Maharashtra)	Local Monopoly	Single buyer of sugarcane in local region	Sugarcane

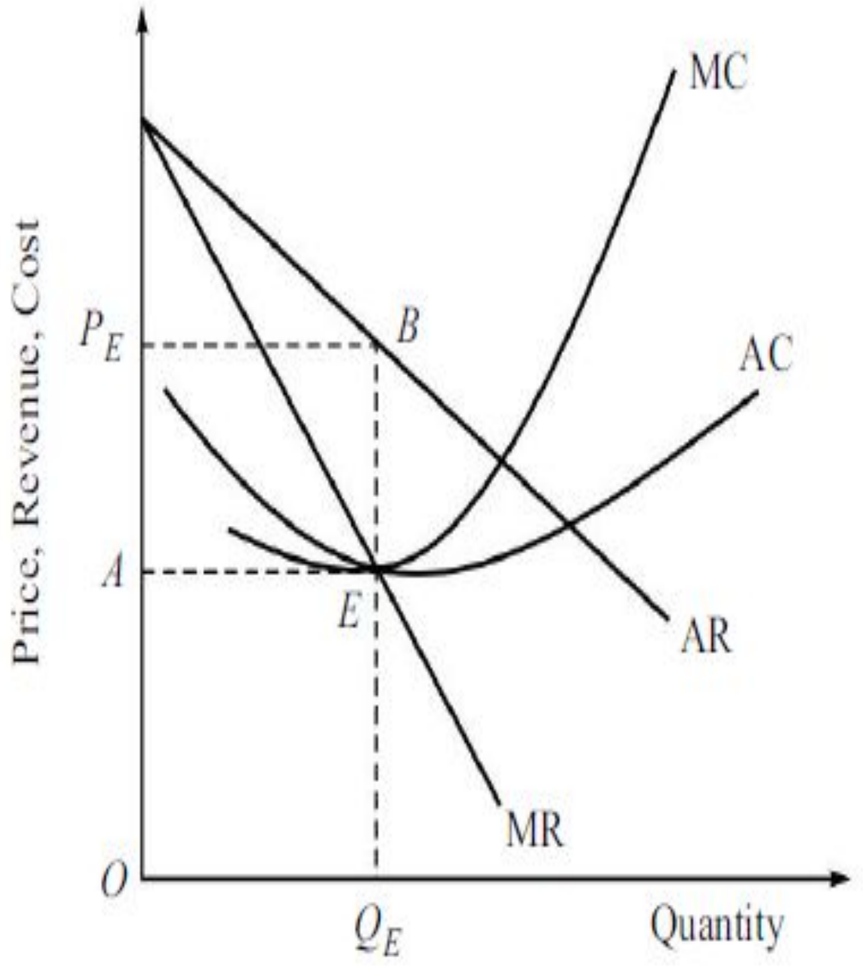
Features of Monopoly

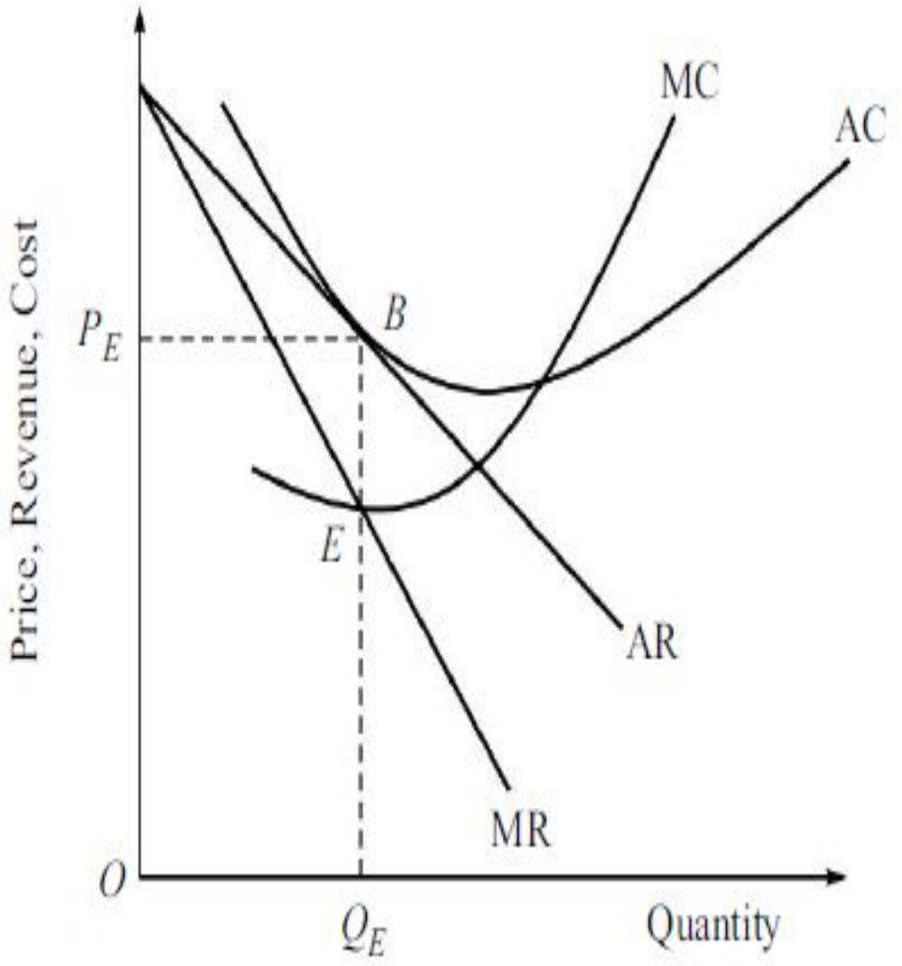
- ✔ One seller
- ✔ Single Product
- ✔ No Difference between Firm and Industry
- ✔ Independent Decision-Making
- ✔ Restricted Entry
- ✔ **Highly inelastic demand curve**
  - ✔ Essential product
  - ✔ No or very little close substitute
  - ✔ No option to change the price

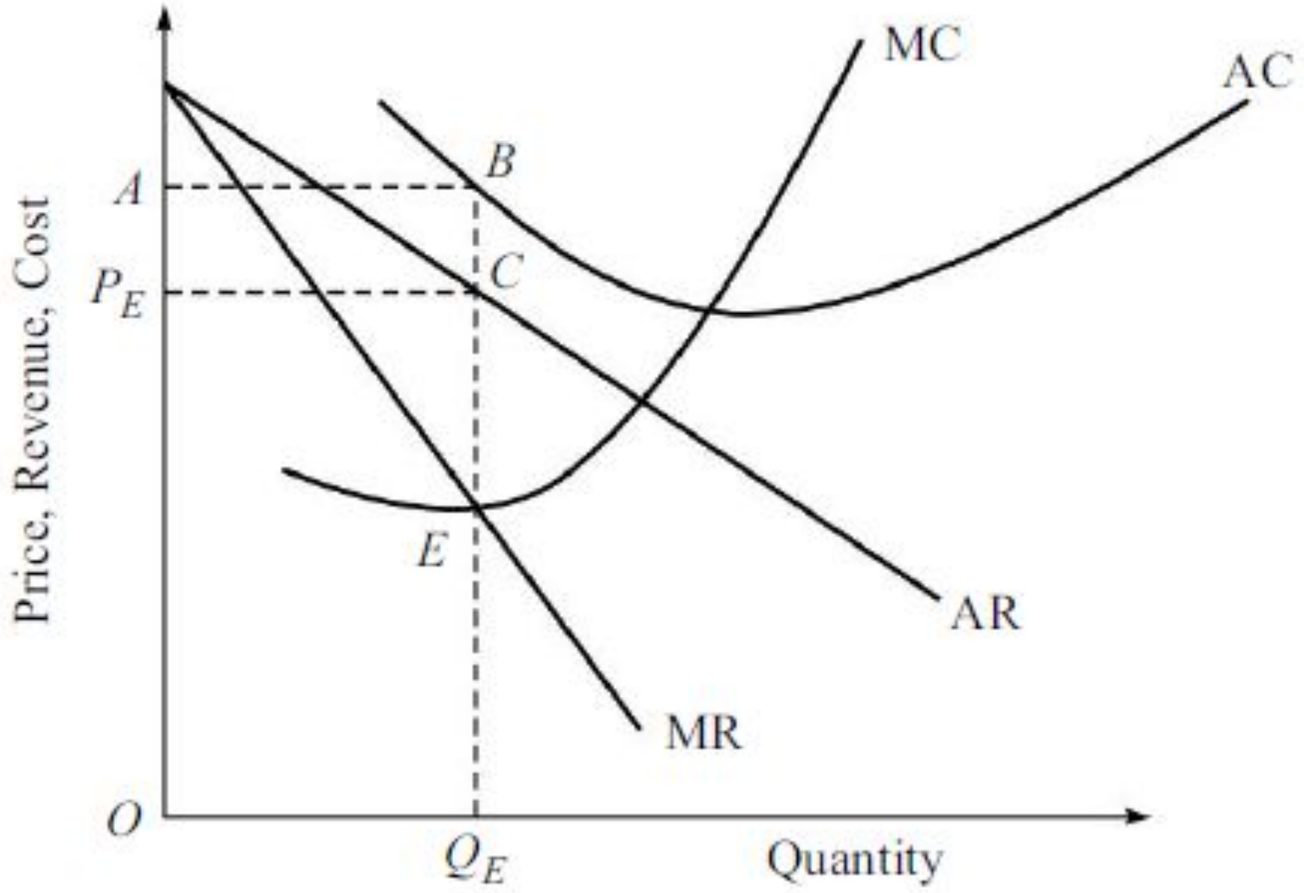
Demand And Marginal Revenue Curves For A Monopoly Firm

Price And Output Decisions In the Short Run

- Case of Supernormal Profit
- Case of Normal Profit
- Case of Loss

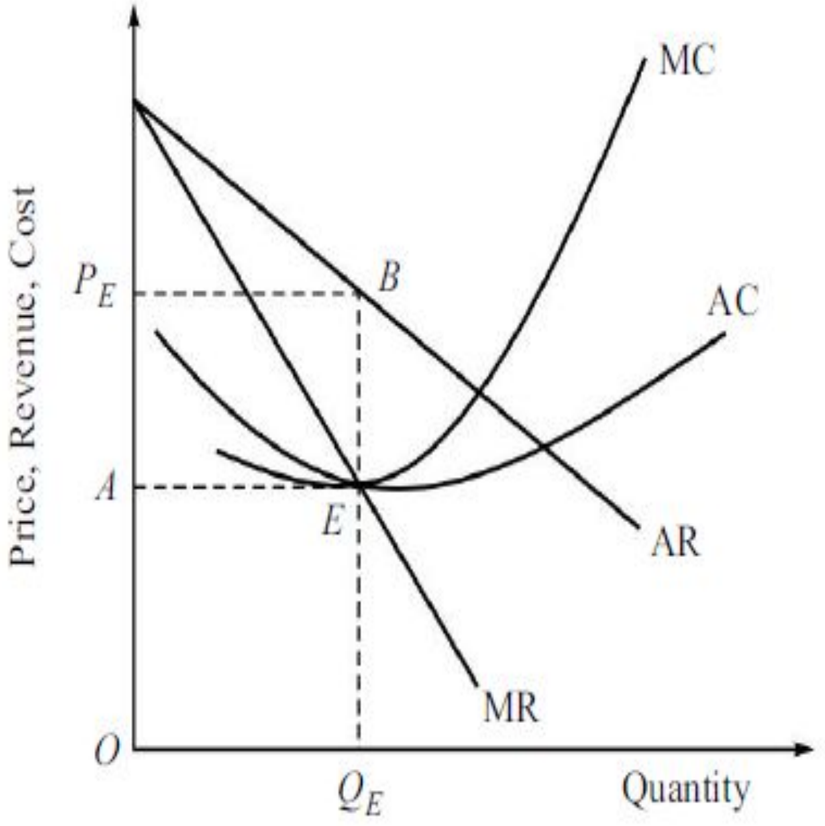

Price, Revenue, Cost
MC
AC
AR
MR
P_E
A
B
E
O
Q_E
Quantity


Price, Revenue, Cost
MC
AC
AR
MR
P_E
B
E
O
Q_E
Quantity


MC
AC
B
A
C
P_E
E
AR
MR
O
Q_E
Quantity
Price, Revenue, Cost

Price And Output Decisions In the Long Run

- Case of Supernormal Profit
- Case of Normal Profit


Price, Revenue, Cost
MC
AC
AR
MR
P_E
A
B
E
O
Q_E
Quantity

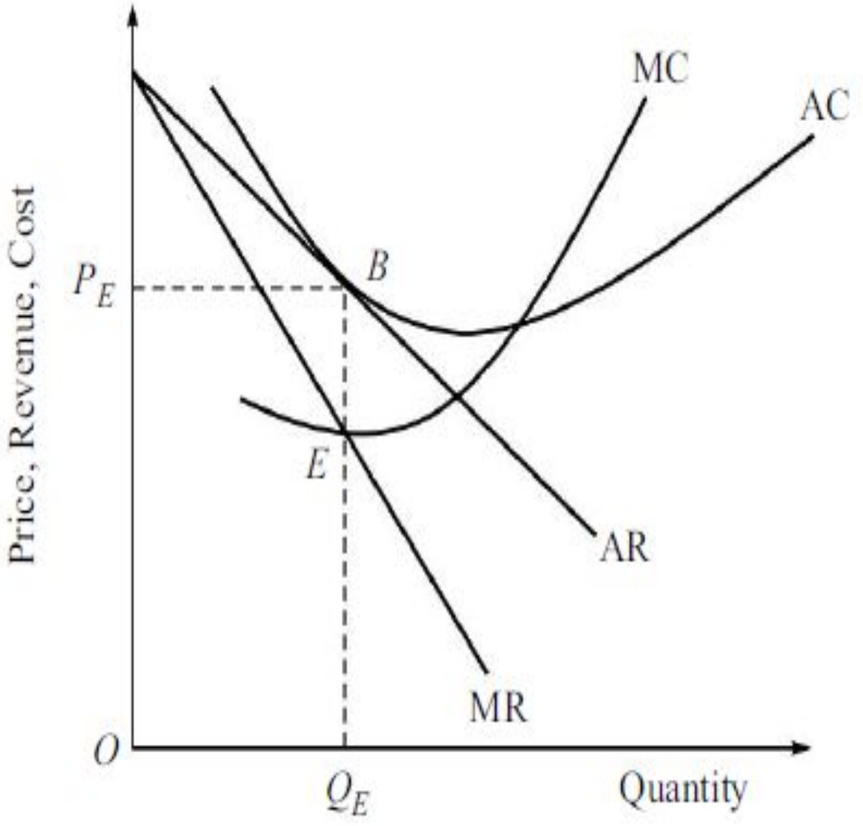
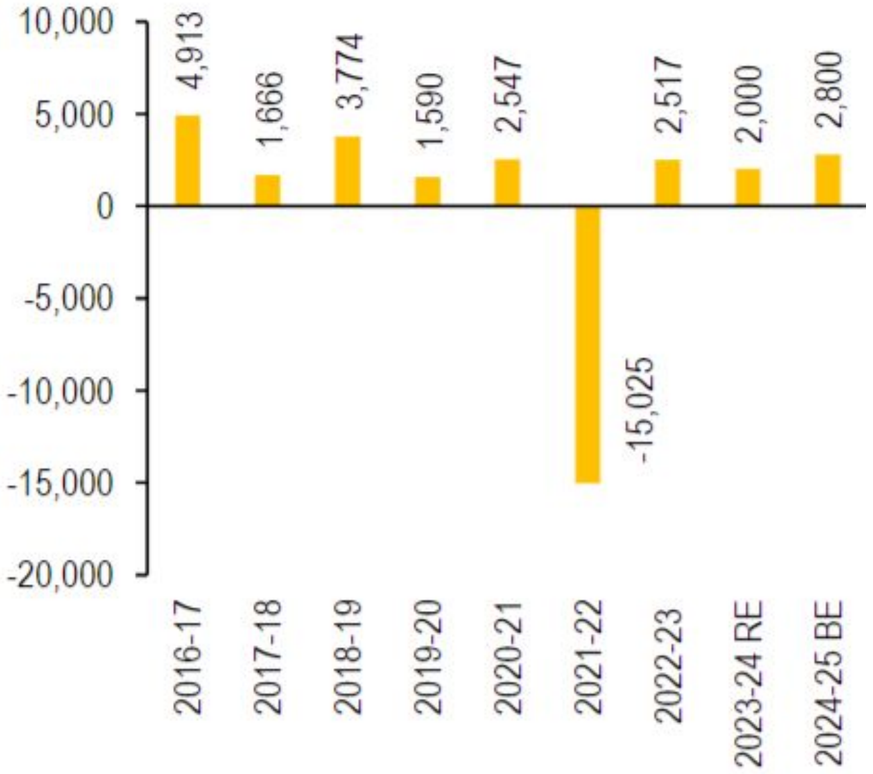

Price, Revenue, Cost
MC
AC
AR
MR
P_E
B
E
O
Q_E
Quantity

Figure 6: Revenue Surplus (in Rs crore)



Source: Expenditure Profile, Railway Statements, Union Budget Documents, 2018-19 to 2024-25; PRS.

The total revenue increased by nearly ₹49,000 crore, marking a 25% growth compared to the previous year. Freight revenue saw a strong performance, reaching ₹1.62 lakh crore, a 15% growth. Passenger revenue registered a remarkable 61% growth, reaching an all-time high of ₹63,300 crore in 2022-23.

Home / Companies / Results / Indian Railways earns record revenue of Rs 2.40 trillion in 2022-23

Indian Railways earns record revenue of Rs 2.40 trillion in 2022-23

Indian Railways earned Rs 63,300 Cr as passenger revenue in 2022-23 as compared to Rs 39,214 crore in 2021-22 which is 61% higher than the last financial year

Supply curve in Monopoly

- ✔ A monopolist will produce a different quantity at a different price.
- ✔ Thus, there is no unique, one-to-one relation between price and output.
- ✔ No Supply curve is present in a Monopoly market.

Indian Railways — A Monopoly in Public Transport

Indian Railways (IR) is one of the largest railway networks in the world and operates under the Government of India. It runs trains across the country and carries millions of passengers and tons of goods every day. Because only one company — Indian Railways — provides railway services across India, it is considered a monopoly. In this market, there are no close substitutes for railways because other transport options like buses or flights cannot carry as many people or goods at such a low cost.

Indian Railways sets its own prices with government approval, which means it is a price maker. It charges very low fares for passengers to make travel affordable for everyone. However, it charges higher rates for freight (goods transport) to cover its costs. This method of using profits from one group (freight) to support another (passengers) is called cross-subsidization. It helps poor passengers travel at a low price, but it also causes losses because the fares are often below the actual cost of running trains.

The government keeps control of Indian Railways because building a separate railway network is very expensive and requires large investments in land, tracks, and trains. If many companies tried to build their own networks, it would lead to duplication and waste of resources. That is why Indian Railways is called a natural monopoly — one company can serve the whole country more efficiently than several competing firms.

Although Indian Railways helps the public by providing cheap and reliable transport, it faces some problems. Trains are often crowded, delayed, and underfunded because the low fares reduce the profit needed for maintenance and upgrades. To improve efficiency, the government has started allowing private trains on some routes and has also promoted digital ticketing through IRCTC.

1. Who owns and manages Indian Railways?

2. Why is Indian Railways called a monopoly?

3. What is meant by cross-subsidization?

4. Why can’t private companies easily compete with Indian Railways?

5. How does Indian Railways help people through low passenger fares?

Monopolistic Competition

Monopolistic competition is a market situation in which a relatively large number of producers offer similar but not identical products.

Bakery Products





Bread



Cookies



Pastry

Hairdressing Industry





Hair Color



Hair Cuts



Facial

Restaurants





Burger



Pizza



Samosa

Smart Phone





iPhone



Samsung

Soft Drink





Coca-Cola



Pepsi

Clothing





ZARA



H & M

Taxi





Ola



Uber

Features of Monopolistic Competition

✔ Large number of buyers and sellers
✔ Differentiated Products but has a close substitute.
✔ Independent Decision-Making
✔ Imperfect Knowledge
✔ Unrestricted Entry and Exit
✔ **Highly elastic demand curve**

Monopoly VS Monopolistic Competition

Market	Demand Curve Shape	Reason	Example
Monopoly	Steep (Inelastic)	Only seller, no close substitutes → buyers less responsive to price.	One farmer has a rare variety of tomato (say, blue tomatoes). Buyers have no substitute → he can charge a high price → quantity demanded is lower.
Monopolistic Competition	Flat (Elastic)	Many firms, close substitutes → buyers highly responsive to price.	Monopolistic competition: Many farmers sell common red tomatoes → buyers can choose → price drops → quantity demanded is higher.

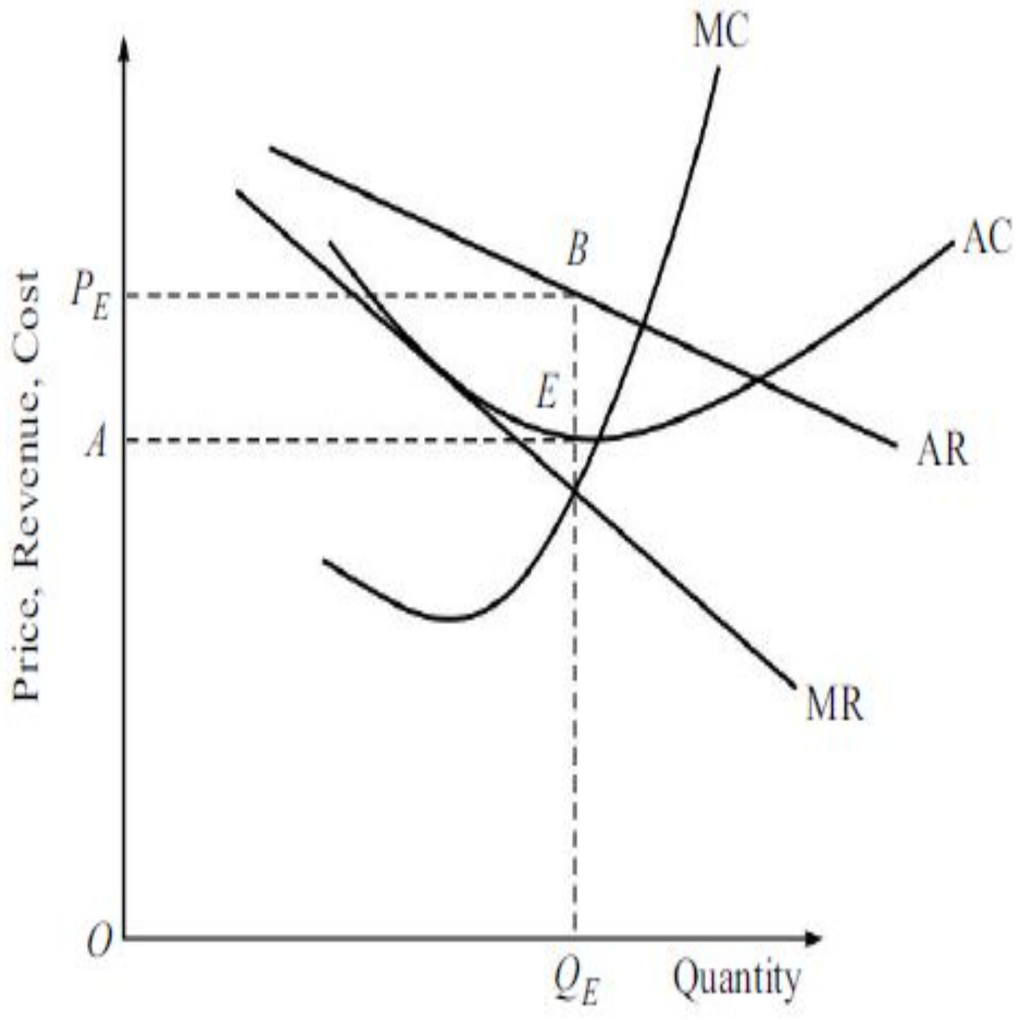


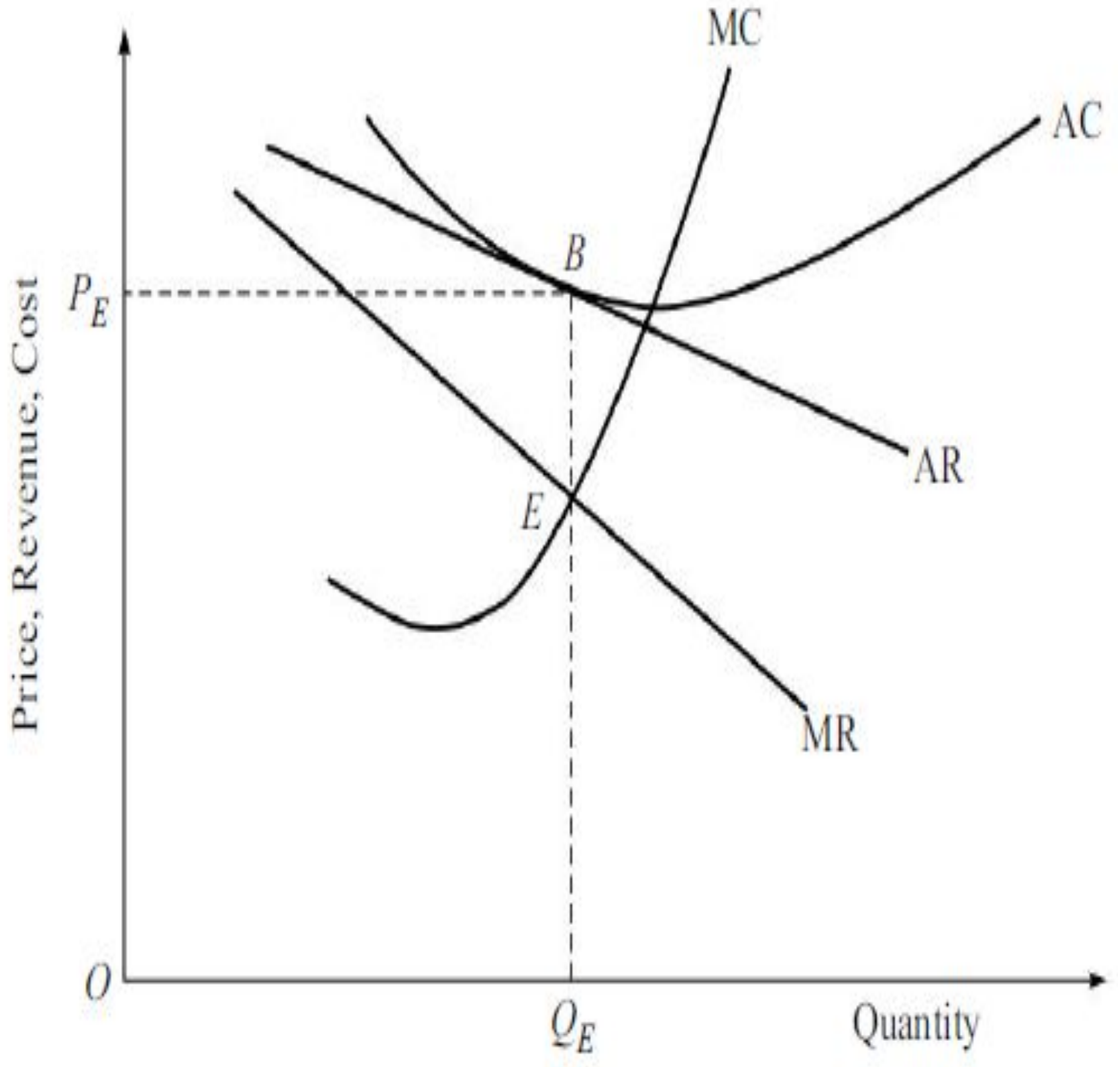


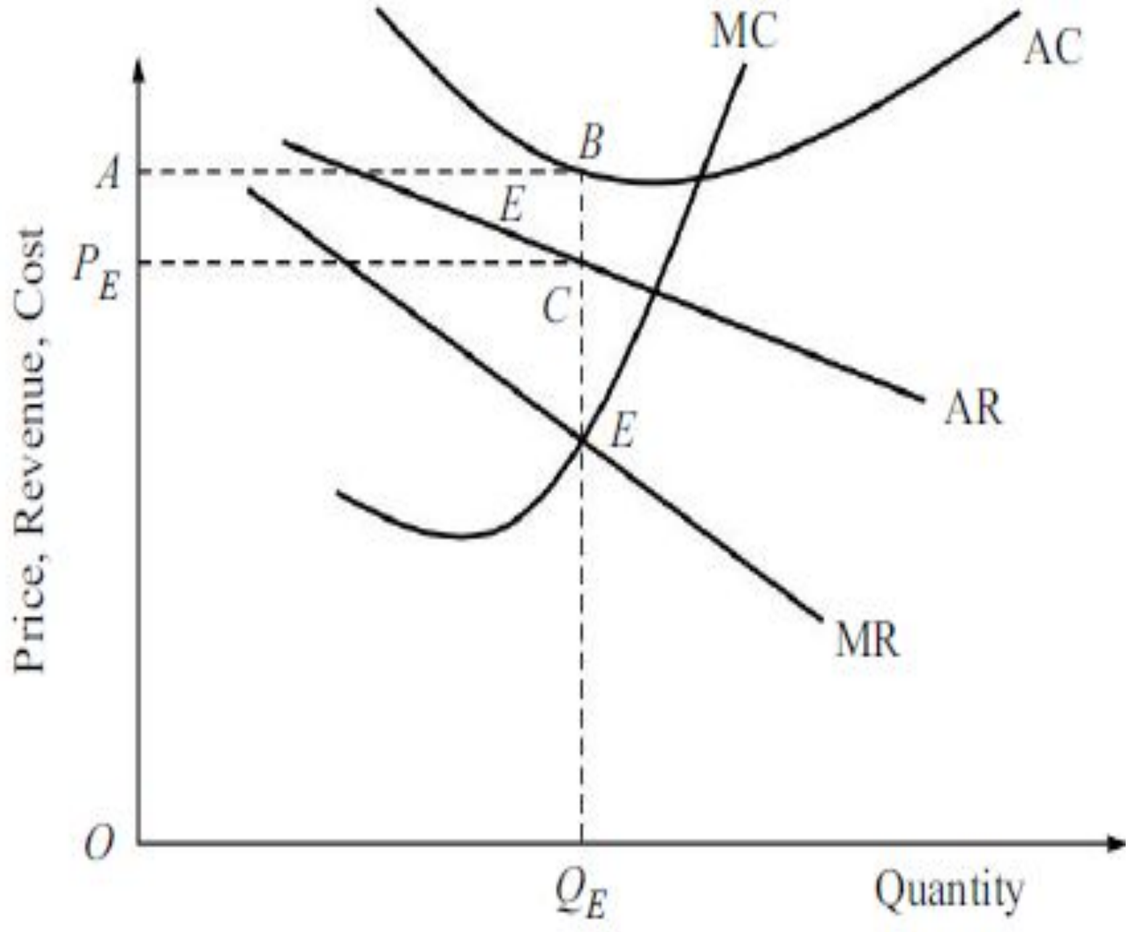
Demand And Marginal Revenue Curves For A Monopolistic Competition Firm

Price And Output Decisions In the Short Run

- Case of Supernormal Profit
- Case of Normal Profit
- Case of Loss

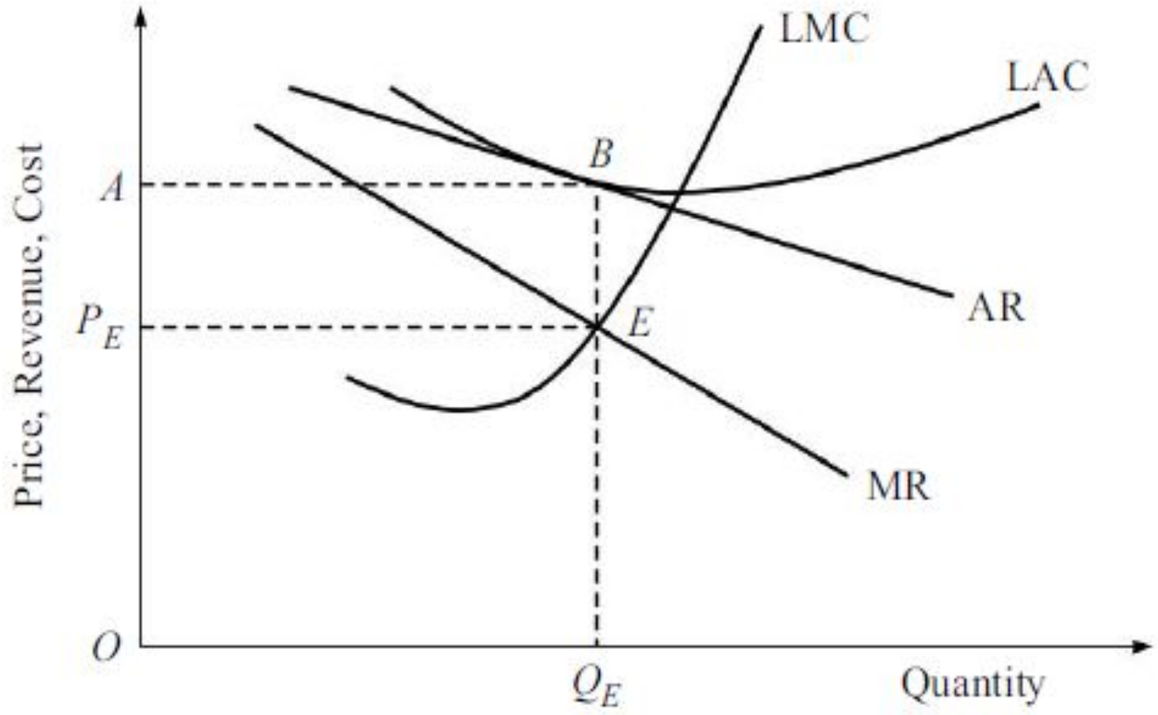

MC
AC
B
P_E
E
A
AR
MR
Price, Revenue, Cost
O
Q_E
Quantity


MC
AC
B
P_E
AR
E
MR
Price, Revenue, Cost
O
Q_E
Quantity


MC
AC
B
A
E
P_E
C
AR
E
MR
O
Q_E
Quantity
Price, Revenue, Cost

Price And Output Decisions In the Long Run

- Case of Normal Profit



Oligopoly

Oligopoly is a market where a few dominant sellers sell differentiated or homogenous products under the continuous consciousness of rivals' actions.

Features of Oligopoly

- ✔ Few sellers
- ✔ Differentiated (car) or Homogeneous (petrol) Products
- ✔ Independent Decision-Making
- ✔ <u>Restricted Entry and Exit</u>
  - ✔ Huge investment requirements
  - ✔ Strong consumer loyalty for existing brands
- ✔ Non-price competition: companies compete **not by lowering prices** but by improving other aspects like **quality, branding, features, customer service, or marketing**.
- ✔ **Indetermined Demand Curve (Highly elastic and highly inelastic)**

Price cuts = short-term revenue gain
Non-price competition = long-term profit, **brand loyalty, and sustainable market position**

Companies: Indian Oil

Price per litre: ₹100

Quantity sold: 1000 litres

If Indian Oil cuts price by 10% (₹90):

New quantity sold: 1050 litres (small increase)

Revenue before: 100 × 1000 = ₹100,000

Revenue after: 90 × 1050 = ₹94,500 → **Revenue falls**

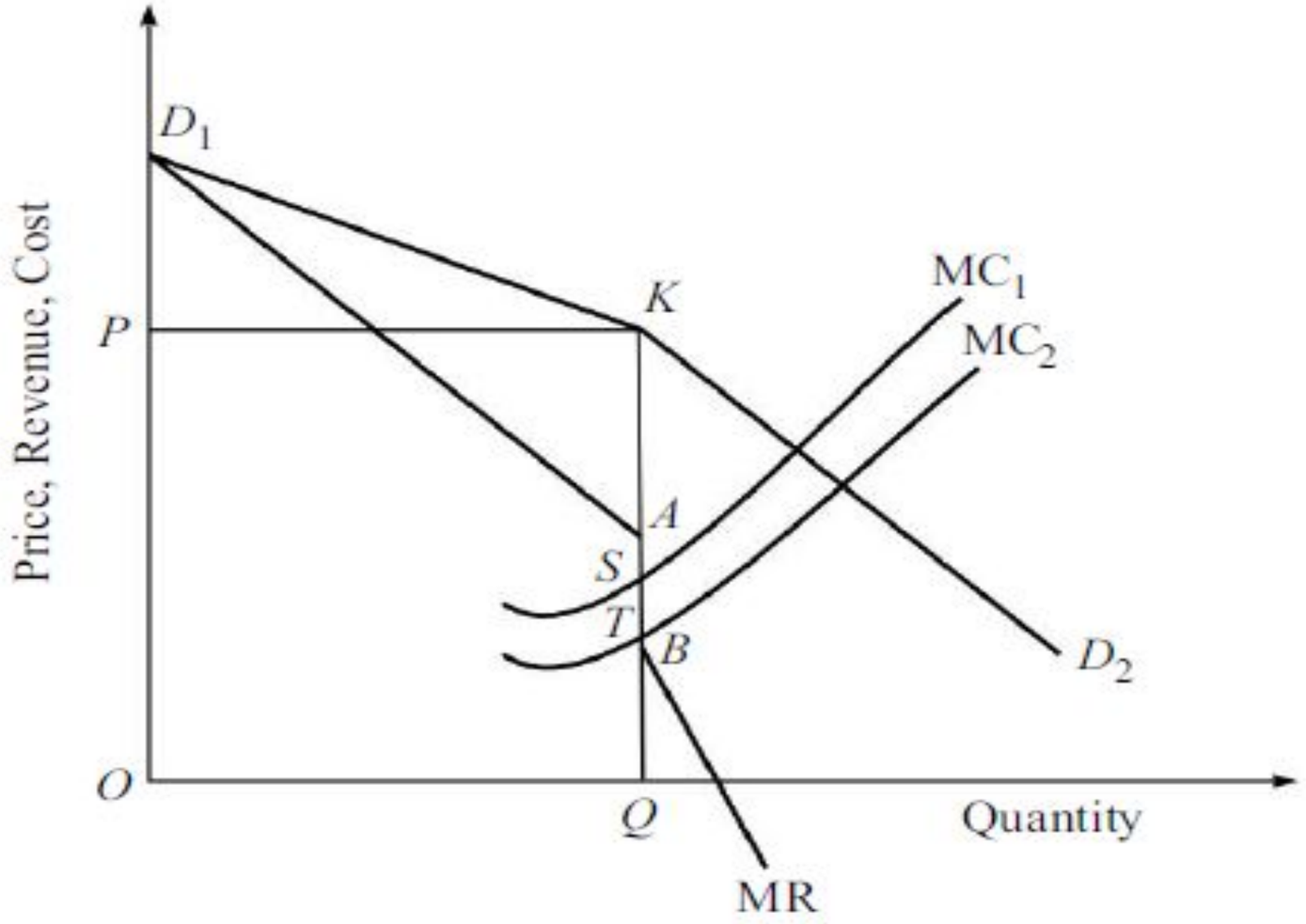
Companies: Hyundai
Price per car: ₹10,00,000
Quantity sold: 100 units
If Hyundai cuts price by 5% (₹9,50,000):
New quantity sold: 120 units
Revenue before: 10,00,000 × 100 = ₹1,00,00,000
Revenue after: 9,50,000 × 120 = ₹1,14,00,000 →
Revenue rises

Petrol (Undifferentiated, Inelastic)	Indian Oil	₹100 → ₹90	1000 → 1050 litres	₹1,00,000 → ₹94,500	Price cut reduces revenue → focus on service, loyalty, convenience
Cars (Differentiated, Elastic)	Hyundai	₹10,00,000 → ₹9,50,000	100 → 120 units	₹1,00,00,000 → ₹1,14,00,000	Price cut increases revenue short-term, but non-price (features, ads, design) ensures long-term profit, loyalty, brand value

Kinked Demand Curve

- This pioneering concept was invented by Paul Sweezy (1939).

- Two assumptions
  - If a firm increases its Price, others will not follow.
  - If a firm decreases its Price, others will follow.

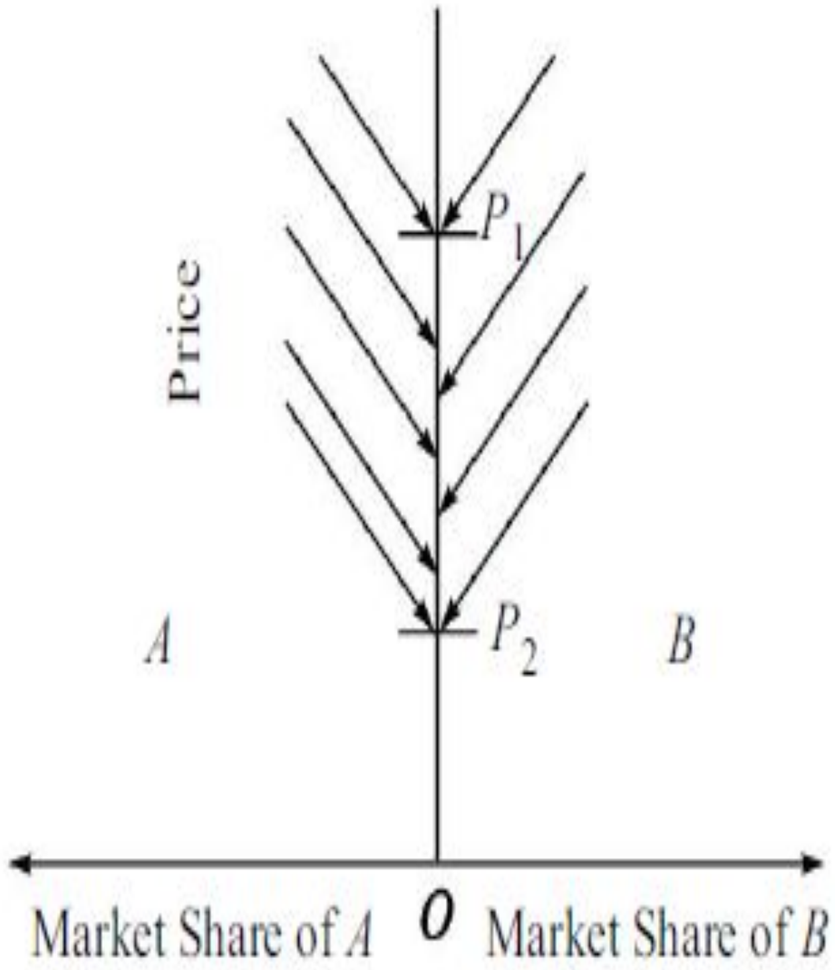

Price, Revenue, Cost
D_1
K
MC_1
MC_2
P
A
S
T
B
D_2
O
Q
Quantity
MR

Two companies operate in oligopolistic markets: Maruti Suzuki (Differentiated) and Indian Oil (Undifferentiated)

Product Type	Price Change	Competitors' Response	Demand Elasticity	Explanation
Undifferentiated (Petrol) Bharat Petroleum, Hindustan Petroleum, and Reliance	Price ↑	Competitors do not follow	Inelastic	Buyers switch immediately to cheaper brands → sharp drop in quantity demanded above price and vice versa
	Price ↓	Competitors follow fully	Inelastic	
Differentiated (Car) Hyundai, Tata Motors, Honda	Price ↑	Competitors partially follow	Elastic	Some loyal consumers stay → less drop in quantity demanded
	Price ↓	Competitors partially follow	Elastic	

Collusion Model – The Cartel

A cartel is a formal agreement between rival firms in an industry to collaborate instead of competing, usually to control prices, limit output, or divide the market.



- ✔ Firms A and B sell the same product (homogeneous).
- ✔ Initial price is P1.
- ✔ Firm A lowers price to attract more customers.
- ✔ Firm B fears losing customers → lowers price even more.
- ✔ Firm A retaliates by lowering price again → cycle continues.
- ✔ This keeps going until one firm gives up → price stabilizes at P2.

- ✔ Customers may benefit temporarily with lower prices.
- ✔ Firms' profits decline.
- ✔ Sometimes governments intervene to stop excessive price cuts, which can eliminate weak competitors unfairly.

OPEC (Organization of Petroleum Exporting Countries)

Saudi Arabia, UAE, Iraq, etc

Suppose global demand for oil is 100 million barrels/day.

If each OPEC country produced freely, total supply = 120 million barrels → price falls.

Cartel agreement: Produce only 90 million barrels → price rises → higher collective profits.

EXAMPLE 12.6 The Milk Cartel



In 1996, the federal government allowed milk producers in the six New England states to cartelize. The cartel—called the Northeast Interstate Dairy Compact—set minimum wholesale prices for milk, and was exempt from the antitrust laws. The result was that consumers in New England paid more for a gallon of milk than consumers elsewhere in the nation.

Studies have suggested that the cartel covering the New England states has caused retail prices of milk to rise by only a few cents a gallon. Why so little? The reason is that the New England cartel is surrounded by a fringe of noncartel producers—namely, dairy farmers in New York, New Jersey, and other states. Expanding the cartel, however, would have shrunk the competitive fringe, thereby giving the cartel a greater influence over milk prices.

Price Leadership

Aggressive: Implies leadership in which one organization establishes its supremacy by threatening the organizations to follow its leadership.

Barometric: refers to a leadership in which one organization declares the change in price at first and assumes that other organization would accept it. The organization does not dominate others and need not be a leader in the industry.

Dominant: refers to a type of leadership in which only one organization dominates the entire industry.

Aggressive	Forces price on others	Competitors follow because they must	Walmart (US retail market) Amazon Amul, India’s largest dairy cooperative
Barometric	Changes price first as a signal	Competitors follow if they want	Maruti Suzuki (India) American Airlines APMC (Agricultural Produce Market Committee)
Dominant	Largest firm sets price naturally	Competitors accept it	Indian Oil (India) Largest oil company IFFCO

Game Theory

Game theory is a mathematical tool that helps to study strategic situations in which players **optimize a variable** not only based on their own preferences but also on other players' decisions and reactions

Widely used in pricing, advertising, negotiation, and market entry.

Pricing strategies (e.g., airline fare wars)

Advertising competition (e.g., Coca-Cola vs Pepsi)

Negotiation tactics (e.g., buyer–supplier deals)

Market entry decisions (e.g., Jio entering telecom)

Types of Games

2 Person-Zero sum game: A game with two players, where one player's gain equals the other's loss, is known as a two-person zero-sum game. The sum of their net winning is zero. It is also a rectangular game because the payoff matrix is rectangular.

A **strategy** is a predetermined rule that specifies completely how one intends to respond to each possible circumstance at each stage of the game.

1. **Pure Strategy:** If the player **selects the same strategy each time**. In this case, **each player knows what other players would do** against his selected strategy.

2. **Mixed Strategy:** When the players use a combination of strategies and **each player leaves the other player guessing as to what he would do**, in case the other adopts a certain strategy.

Consider two neighboring farmers, **Farmer A and Farmer B**, who grow the same crop and must decide whether to **use chemical fertilizers** or **adopt organic methods**.

If both choose organic farming, they maintain soil health and can sell their produce at a premium price in niche markets.

If one farmer uses chemical fertilizers while the other stays organic, the one using chemicals may achieve a higher short-term yield and profit, while the organic farmer earns less.

However**, if both use chemicals**, soil fertility deteriorates over time, and both face lower yields and profits in the long run.

Players: Farmer A and Farmer B

Strategies: Use Chemical Fertilizers or Adopt Organic Methods

If both choose organic methods, they earn moderate, stable profits and preserve soil health.

If one chooses chemicals while the other stays organic, the chemical user gains a short-term advantage, while the organic farmer loses.

If both use chemicals, profits fall in the long run due to soil degradation and higher input costs.

Their optimal choice depends on what the other does.

	Farmer B: Organic	Farmer B: Chemical
Farmer A: Organic	A = 50, B = 50	A = 20, **B = 80**
Farmer A: Chemical	**A = 80**, B = 20	**A = 40, B = 40**

The final payoff they both get here is ₹40 each.
Therefore, the value of the game = ₹40 for each farmer.

If A switches → A's profit drops

If B switches → B's profit drops

The Nash Equilibrium is the point in a game where no player can improve their own outcome (payoff) by unilaterally changing their strategy, assuming all other players maintain their strategies unchanged.

Minimax-Maximin Principle

The selection of an optimal strategy by each player without the knowledge of the competitor's strategy is the basic problem of playing games. The objective of the study is to know how these players select their respective strategies so that they may optimize their payoff. Such a decision-making criterion is referred to as the minimax-maximin principle.

Saddle Point

If the minimax value = maximin value, then the game is said to have a saddle (equilibrium) point.

Procedure to determine Saddle (Equilibrium) Point

- Select the minimum element in each row of the payoff matrix and write them under the “row minima” heading. Then select the largest element among these elements and enclose it in a rectangle.

- Select the maximum element in each column of the pay matrix and write them under the “column maxima” heading. Then select the lowest element among these elements and enclose it in a circle.

- Find out elements that are the same in the circle as well as the rectangle and mark the position of such elements on the matrix. This element represents the value of the game and is called a saddle point.

Pure Strategies: Minimax-Maximin Principle: Games with saddle point

Two companies, A and B, sell two brands of flu medicine. Company A advertises in radio(A1), television(A2), and newspapers(A3). Company B, in addition to using radio(B1), television(B2), and newspapers(B3), also mails brochures(B4). Depending on the effectiveness of each advertising campaign, one company can capture a portion of a market from the other. The following matrix summarizes the percentage of the market captured or lost by Company A. Determine the optimal strategy and the value of the game. Explain the results.

	B1	B2	B3	B4
A1	8	-2	9	-3
A2	6	5	6	8
A3	-2	4	-9	5

A farmer's cooperative and a local government agency are negotiating a new three-year agreement regarding water usage rights and subsidies. Each party has 4 strategies to choose from:

1. Hard and aggressive negotiation
2. Reasoning and logical approach
3. Legalistic strategy
4. Conciliatory approach

The costs (or benefits) to each party for every pair of strategy choices are given in the matrix below.

	Company Strategies			
Union Strategies	*I*	*II*	*III*	*IV*
I	20	15	12	35
II	25	14	8	10
III	40	2	10	5
IV	− 5	4	11	0

Two competitive agricultural cooperatives are growing a new variety of crop under a government grant. To meet the demand, they have the option to allocate their resources for 8, 16, or 24 hours of labor per day. As the labor hours increase, so do the costs. Cooperative A has set up the matrix below, estimating the percentage of the market share it could capture based on different labor schedules:

	Cooperative B		
Cooperative A	C1:8 Hrs	C2: 16 Hrs	C3: 24 Hrs
S1: 8 Hrs	60%	56%	34%
S2: 16 Hrs	63%	60%	55%
S3: 24 Hrs	83%	72%	60%

Rule of Dominance

- It can be applied in 2 person-zero sum game.
- Large number of strategies available for each player.
- When saddle point does not exist.

Rules

- Rule of rows: If all the elements in a row are less than or equal to the corresponding elements in another row, this row is said to be dominated by the other row and can be deleted from the matrix.

- Rule of column: If all the elements in a column are more than or equal to the corresponding elements in another column, this column is said to be dominated by the other column and can be deleted from the matrix.

Mixed Strategies: Games Without Saddle Point: Algebraic Method (2 X 2 Matrix)

Reduce the following payoff matrix using the principle of dominance

	B1	B2	B3
A1	9	8	-7
A2	3	-6	4
A3	6	7	7

Two breakfast food manufacturers, ABC and XYZ are competing for an increased market share. The payoff matrix, shown in the following table, describes the increase in the market share for ABC and the decrease in the market share of XYZ. Determine optimal strategies for both the manufacturers and the value of the game.

Player XYZ					
Player ABC		Give Coupons	Decrease Price	Maintain present strategy	Increase advertising
	Give Coupons	2	-2	4	1
	Decrease Price	6	1	12	3
	Maintain present strategy	-3	2	0	6
	Increase advertising	2	-3	7	1

THE THEORY OF CONSUMER CHOICES

WHAT THE CONSUMERS CAN AFFORD

Imagine you have ₹200 in your pocket. You have just finished a three-hour programming lab. You are hungry, thirsty, and your friends are going to the canteen. You want to buy everything, but you have only ₹200.

- Why did you choose it?
- Why did you sacrifice other items?
- Would you choose differently if your budget increased to ₹300?
- What if the price of the one you chose fell?



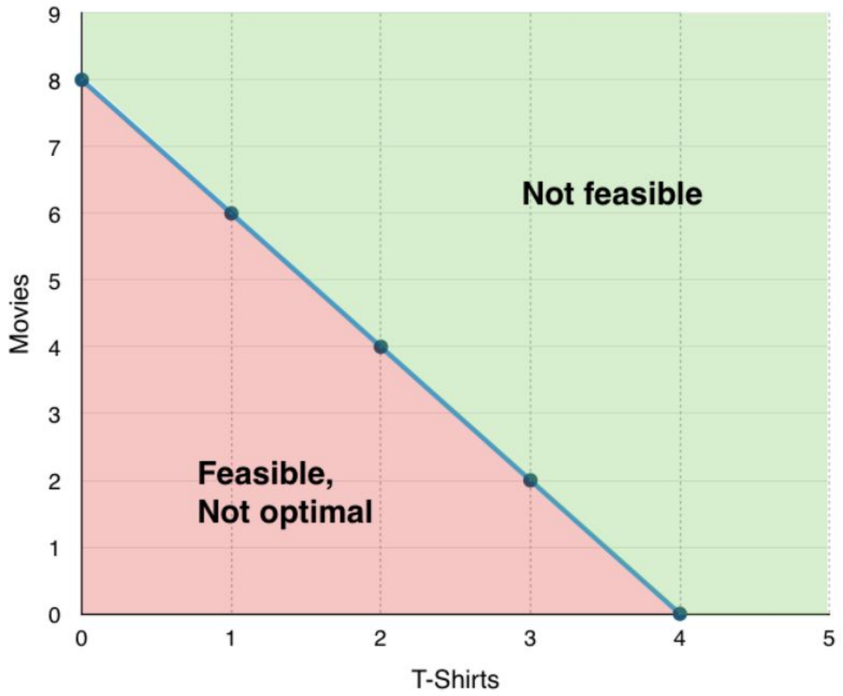
CONNECTING WITH ECONOMICS

Limited money	= Budget constraint
Personal likes	= Preferences
Best affordable combination	= Trade-off
Giving up one item for another	= Consumer optimum

BUDGET CONSTRAINTS

$$P_M \cdot Q_M + P_N \cdot Q_N = I$$

$$\text{Slope of the budget line} = \frac{P_M}{P_N}$$



BUDGET CONSTRAINTS

Imagine you have ₹1,000 to spend during a weekend on pizza and Pepsi. A pizza costs ₹100, while a Pepsi costs ₹20. You would like to enjoy more of both, but your income is limited, so you have to make choices.

1. Write the Budget Constraints equation.
2. Prepare a table of all combinations
3. Calculate the slope of it.
4. Draw a graph.
5. Imagine you have ₹2,000 to spend then write the Budget Constraints equation. with slope. Also draw the graph. (Income)
6. Imagine Pepsi cost is now ₹40. What will be the slope? Draw the graph. (Price)

WHAT THE CONSUMERS

WANT

Imagine I offer you a few combinations.

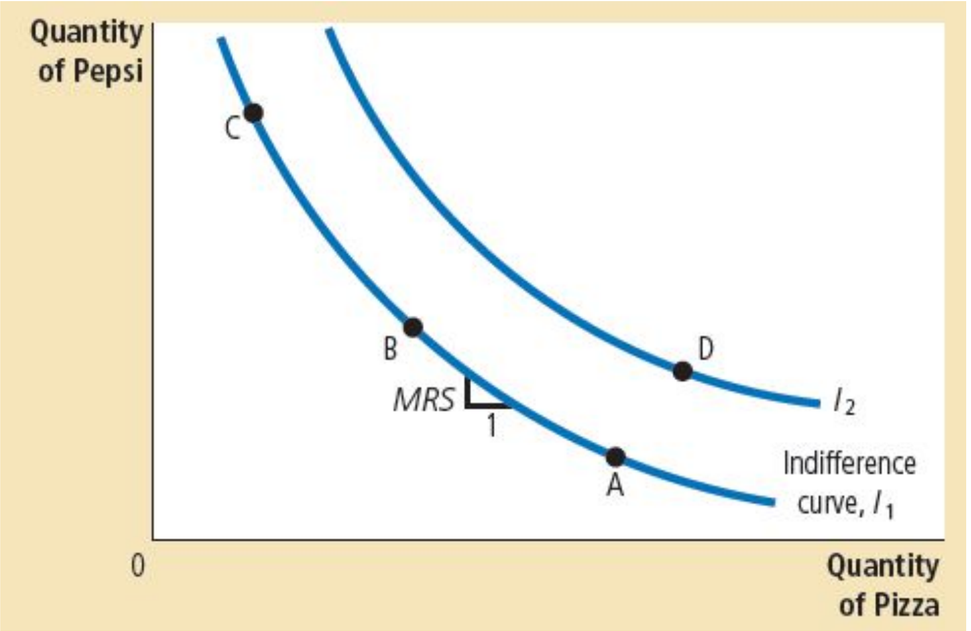
- Option A: 2 pizzas + 40 Pepsi
- Option B: 3 pizza + 35 Pepsi
- Option C: 4 pizza + 30 Pepsi
- Option D: 5 pizza + 25 Pepsi

Which would you choose?

INDIFFERENCE CURVE

A curve that shows consumption bundles that give the consumer the same level of satisfaction

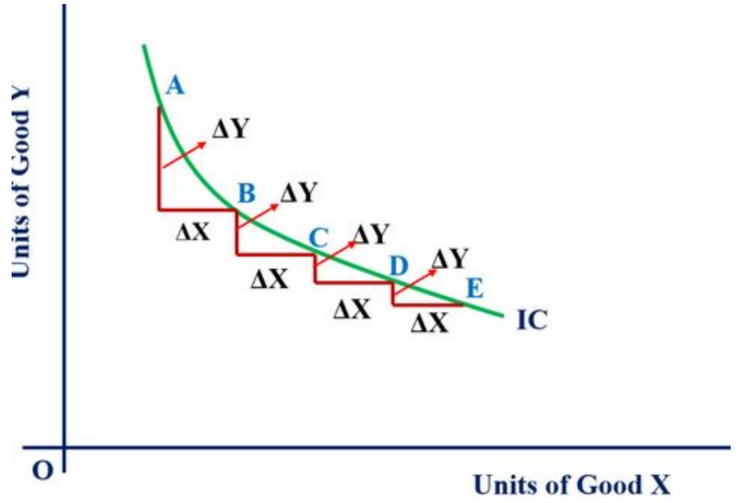
$$\text{Slope of IC} = \frac{\Delta \text{Pepsi}}{\Delta \text{Pizza}} = -MRS$$



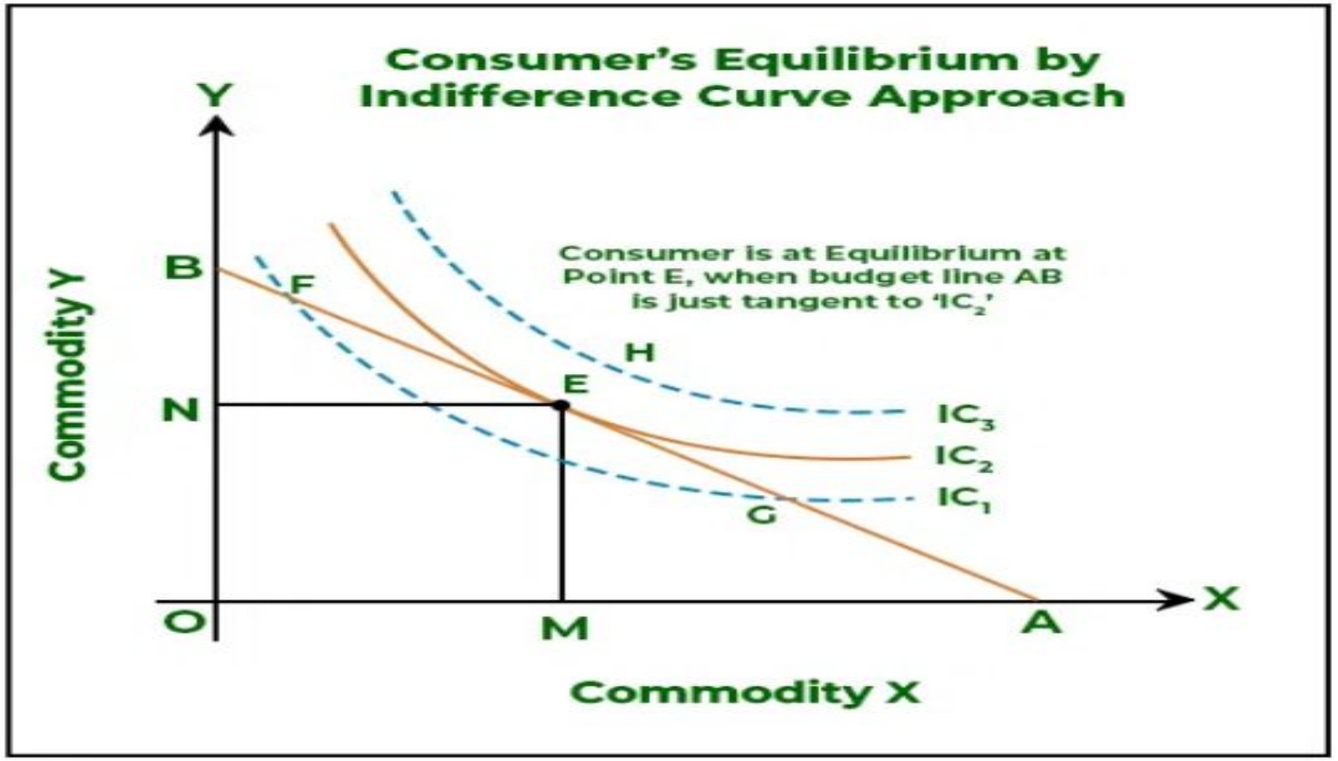
MARGINAL RATE OF SUBSTITUTIONS

the rate at which a consumer is willing to trade one good for another

$$\text{MRS} = {}^{-\Delta y}/_{\Delta x} = MU_y/MU_x$$



CONSUMER EQULIBRIUM



$$MRSx,y = MU_x/MU_y = P_x/P_y$$

PROPERTIES OF INDIFFERENCE CURVE

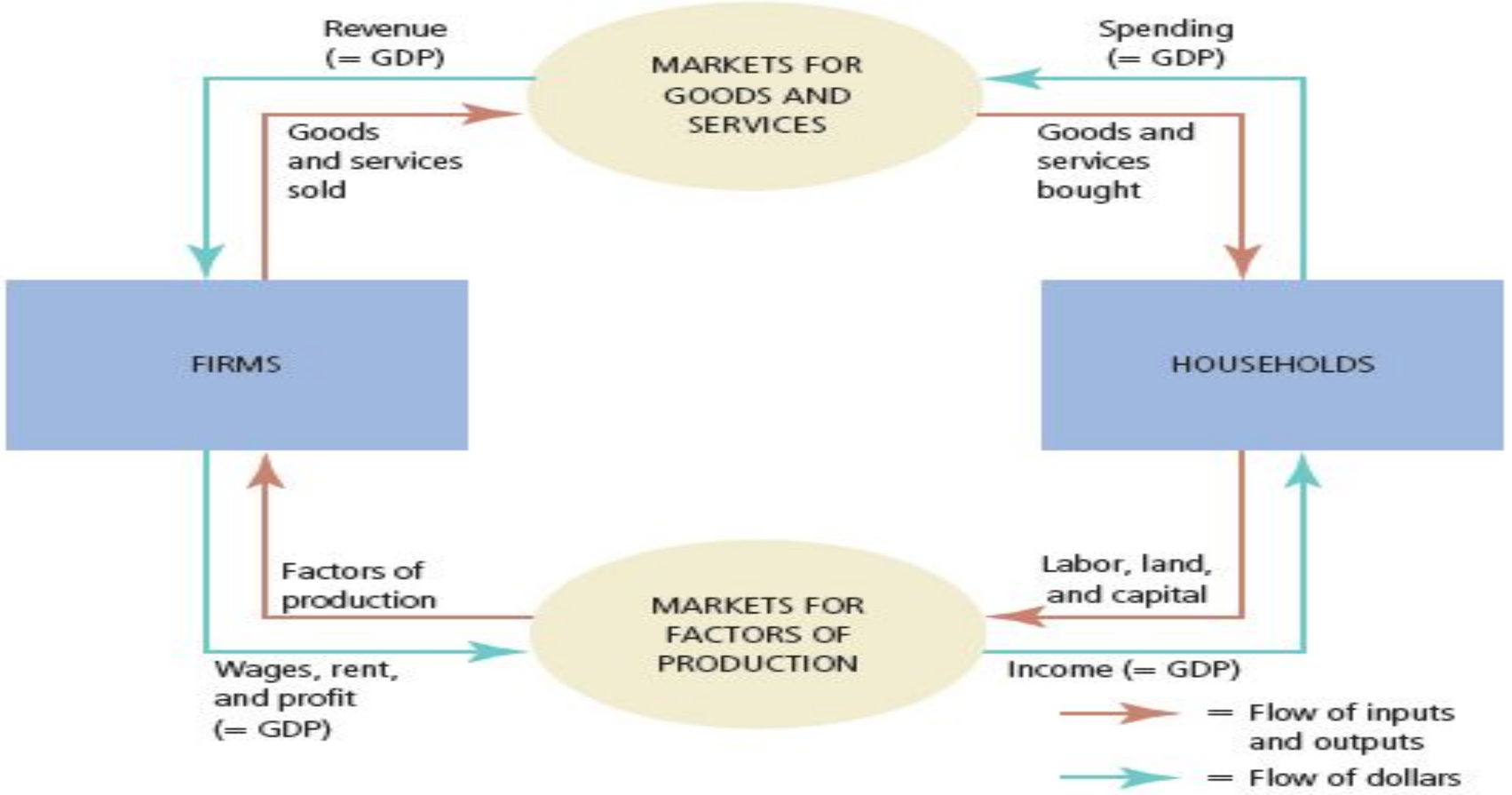
- IC’s slopes downward to the right
- IC is convex to the origin.
- IC cannot intersect to each other
- Higher IC represents a higher level of satisfaction than a lower one.

CONCEPT OF NATIONAL INCOME

Suppose you want to know whether India's economy is doing well.
What would you look at?

Can we combine all the economic activities of millions of people and firms into one number?

Income = Expenditure



One person's spending becomes another person's income.

GDP is the market value of all final goods and services produced within a country in a given period of time.

Market Value

Different products cannot simply be added together.
For example:
1 laptop = ₹50,000
1 notebook = ₹100
1 haircut = ₹500

Produced

GDP counts currently produced goods and services.
Suppose:

- Tata manufactures a new car this year → included.
- You sell your old car to another person → the value of the old car is not newly produced this year, so it is not counted again.

In a Given Period

- Quarterly
- Annually

Final

This is very important.
Suppose:
Steel → Computer → Final customer
If we count both the steel and the computer, we may count the same value twice.
Therefore:
GDP counts final goods, not intermediate goods

Within a Country

This is the meaning of Domestic in GDP.
It depends on where production takes place, not the nationality of the producer.
For example:
A Japanese company producing smartphones in India → production is part of India's GDP.
Because production occurred within India.

1. Tata Motors manufactures a new electric car in Karnataka. Does it count in India's GDP?
2. A student sells a second-hand motorcycle to a friend. Does its value count in current GDP?
3. Infosys develops a new software application for a Bengaluru client. Does it count in GDP?
4. An Indian student buys a newly imported iPhone. Does the entire purchase price add to India's GDP?
5. A factory produces new smartphones that remain unsold at the end of the year. Do they count in GDP?

COMPONENTS OF GROSS DOMESTIC PRODUCT

$$Y = C + I + G + NX$$

C – Consumption	A student buys a ₹6,00,000 laptop for personal use.
I – Investment	TCS purchases 500 new computers and servers to expand its IT operations. (Spending on things used to produce goods and services in the future.)
G – Government Purchases	The Government of India purchases computers, networking equipment and software for government offices.
NX – Net Exports	An Indian IT company provides software services to a US company (export), while an Indian company imports computer chips/semiconductors from Taiwan (import).

NOMINAL GDP

REAL GDP

Nominal GDP
The production of goods and services valued at current prices

Real GDP
The production of goods and services valued at constant prices

Year	Laptops Produced	Price per Laptop
Year 1 (Base Year)	100	₹50,000
Year 2	100	₹60,000

	Year 1	Year 2	Change
Production	100 laptops	100 laptops	0%
Nominal GDP	₹50 lakh	₹60 lakh	+20%
Real GDP	₹50 lakh	₹50 lakh	0%

	Nominal GDP	Real GDP
Meaning	GDP at current prices	GDP at constant prices
India, FY 2025–26	₹346.36 lakh crore	₹323.12 lakh crore
Growth Rate	8.9%	7.7%

GDP DEFLATOR

A measure of the price level calculated as the ratio of nominal GDP to real GDP times 100

GDP increased by 10%. Did India produce 10% more?

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

1. What is produced? → GDP
2. How much is produced? → Real GDP
3. At what price is it valued? → GDP Deflator

Prices and Quantities

Year	Price of Hot Dogs	Quantity of Hot Dogs	Price of Hamburgers	Quantity of Hamburgers
2010	$1	100	$2	50
2011	$2	150	$3	100
2012	$3	200	$4	150

Year	Calculation	Nominal GDP
2010	($1 × 100) + ($2 × 50)	$200
2011	($2 × 150) + ($3 × 100)	$600
2012	($3 × 200) + ($4 × 150)	$1,200

Year	Calculation using 2010 prices	Real GDP
2010	($1 × 100) + ($2 × 50)	$200
2011	($1 × 150) + ($2 × 100)	$350
2012	($1 × 200) + ($2 × 150)	$500

Year	Calculation	GDP Deflator
2010	($200 ÷ $200) × 100	100
2011	($600 ÷ $350) × 100	171.4
2012	($1,200 ÷ $500) × 100	240

Why should policymakers care about GDP?

- Measures economic growth
- Identifies economic slowdowns or recessions
- Helps formulate government policies
- Guides government spending and taxation decisions
- Helps plan infrastructure and development
- Shows the contribution of different sectors
- Helps assess investment and employment conditions
- Allows comparison of economic performance across years and countries
- Helps estimate average economic output through GDP per capita
- Provides a broad picture of the economy’s health

Year	Price of Milk ($)	Quantity of Milk (quarts)	Price of Honey ($)	Quantity of Honey (quarts)
2010	1	100	2	50
2011	1	200	2	100
2012	2	200	4	100

1. Compute nominal GDP, real GDP, and the
2. GDP deflator for each year, using 2010 as the base year.
3. Compute the percentage change in nominal GDP, real GDP, and the GDP deflator in 2011 and 2012 from the preceding year.

A Growing Economy, but Who Benefits?

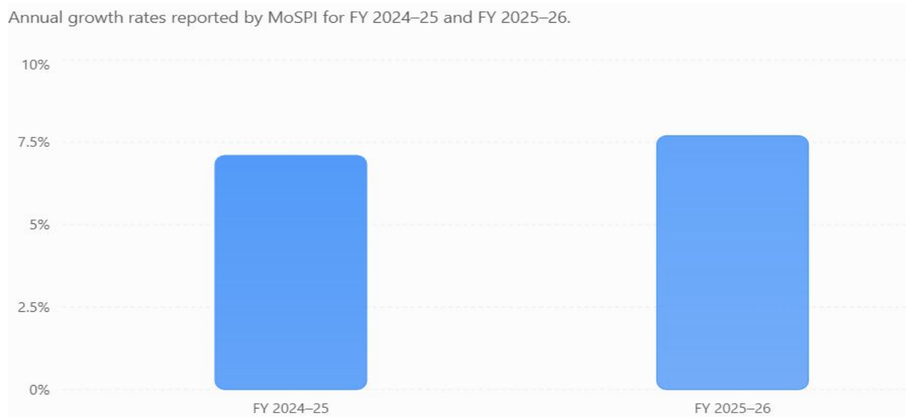
India's economy includes manufacturing, construction, agriculture, information technology, transport, banking and many other services. When these industries produce more goods and services, they contribute to the country’s GDP. A report on India's recent economic performance finds that India's real GDP growth increased from 7.1% in FY 2024–25 to 7.7% in FY 2025–26. Nominal GDP growth also increased, from approximately 8.9% to 8.9%?

Data below to compare the two measures accurately (MoSPI report, 2026) . The real GDP measures production using constant prices, while nominal GDP measures production using current prices. Stronger production may create jobs and business opportunities, but GDP growth alone cannot tell them whether every household's income has risen, whether prices have become more affordable, or whether the benefits of growth have been shared equally.

India's GDP performance

Indicator	FY 2024–25	FY 2025–26
Real GDP	₹299.89 lakh crore	₹323.12 lakh crore
Real GDP growth	7.1%	7.7%
Nominal GDP	₹318.07 lakh crore	₹346.36 lakh crore
Nominal GDP growth	—	8.9%

India's real GDP growth



Explain the difference between real GDP and nominal GDP, discuss how economic growth may affect employment and business opportunities, and critically examine whether the benefits of GDP growth are shared equally among all sections of society.

MEASURING THE COST OF LIVING

Imagine that your grandfather went to the market 20 years ago with ₹1,000. He bought rice, vegetables, milk, cooking oil, clothes, and other household necessities. Today, you also go to the market with ₹1,000. Can you buy the same quantity of goods?”

- Why can ₹1,000 buy fewer things today?
- Has the value of money changed, or have prices changed?
- If your salary increases by 10%, does that automatically mean you are better off?

In 2010, your salary = ₹20,000/month and In 2026, your salary = ₹50,000/month

Ask:

Who is richer?

A. 2010 person
B. 2026 person
C. Cannot say

Salary ↑ > Prices ↑ = Purchasing Power ↑

Salary ↑ < Prices ↑ = Purchasing Power ↓

CONSUMER PRICE INDEX

A measure of the overall cost of the goods and services bought by a typical consumer

$$\text{Consumer price index} = \frac{\text{Price of basket of goods and services in current year}}{\text{Price of basket in base year}} \times 100.$$

Step 1: Fix the basket
Step 2: Find the prices
Step 3: Calculate the basket's cost
Step 4: Select a base year
Step 5: Calculate inflation

A hypothetical economy in which consumers buy only hot dogs and hamburgers.

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

Basket = 4 hot dogs, 2 hamburgers

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2010	$1	$2
2011	2	3
2012	3	4

Step 3: Compute the Cost of the Basket of Goods in Each Year

2010	($1 per hot dog × 4 hot dogs) + ($2 per hamburger × 2 hamburgers) = $8 per basket
2011	($2 per hot dog × 4 hot dogs) + ($3 per hamburger × 2 hamburgers) = $14 per basket
2012	($3 per hot dog × 4 hot dogs) + ($4 per hamburger × 2 hamburgers) = $20 per basket

Step 4: Choose One Year as a Base Year (2010) and Compute the Consumer Price Index in Each Year

2010	($8 / $8) × 100 = 100
2011	($14 / $8) × 100 = 175
2012	($20 / $8) × 100 = 250

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year

2011	(175 − 100) / 100 × 100 = 75%
2012	(250 − 175) / 175 × 100 = 43%

CPI VS INFLATION

CPI = Price level
Inflation = Change in price level

CPI last year = 100
CPI this year = 120
Therefore:
Inflation = 20%

CPI tells us WHERE prices are.
Inflation tells us HOW FAST prices are rising.

WHY CPI IS IMPORTANT

- Measure inflation
- Track changes in the cost of living
- Compare purchasing power over time
- Adjust salaries and pensions
- Compare money values from different years
- Guide economic policy

PROBLEMS WITH CPI

Substitution Bias
2. New Goods
3. Quality Change

CPI VS GDP DEFLATOR

CPI	GDP Deflator
Measures price changes in **a representative basket** of consumer goods and services	Measures price changes in **domestically produced** final goods and services included in GDP
Includes imported consumer goods	Excludes imports (C+I+G+NX)
Uses a relatively fixed basket	Basket changes with production

A rise in the price of an imported phone can affect….

A rise in the price of a domestically produced machine can affect ……

WHY₹100 IN THE PAST IS NOT THE SAME AS TODAY

$$\text{Value in today's money} = \text{Past amount} \times \frac{\text{CPI today}}{\text{CPI in the past}}$$

Suppose a person earned ₹10,000 when the CPI was 100. Today, the CPI is 200.

NOMINAL INTEREST RATE

REAL INTEREST RATE

Real interest rate ≈ Nominal rate − Inflation rate

Riya deposits ₹10,000 in a bank at an annual interest rate of 10%. After one year, her savings increase to ₹11,000. She feels happy because she has earned ₹1,000 in interest. However, her friend Karan points out that prices have also increased by 10% during the same year. As a result, the goods that cost ₹10,000 last year now cost ₹11,000. Riya realises that although her money has increased, her purchasing power has remained almost unchanged.

1. What is Riya's nominal interest rate?
2. What is the inflation rate?
3. Calculate Riya's approximate real interest rate.
4. Has Riya's purchasing power increased? Explain briefly.

Suppose that people consume only three goods,as shown in this table:

	Tennis Balls	Golf Balls	Bottles of Gatorade
2011 Price	$2	$4	$1
2011 Quantity	100	100	200
2012 Price	$2	$6	$2
2012 Quantity	100	100	200

a. What is the percentage change in the price of each of the three goods?
b. Using a method similar to the consumer price index, compute the percentage change in the overall price level.

The residents of Vegopia spend all their income on three vegetables: cauliflower, broccoli, and carrots. In 2010, they purchased 100 heads of cauliflower for $200, 50 bunches of broccoli for $75, and 500 carrots for $50. In 2011, they purchased 75 heads of cauliflower for $225, 80 bunches of broccoli for $120, and 500 carrots for $100.

a. Calculate the price of each vegetable in 2010 and 2011.
b. Using 2010 as the base year, calculate the Consumer Price Index (CPI) for each year.
c. Calculate the inflation rate in 2011.

The residents of Energia spend their income on three energy sources: electricity, LPG cylinders, and petrol. In 2024, households purchased 500 units of electricity for ₹5,000, 20 LPG cylinders for ₹20,000, and 100 litres of petrol for ₹10,000. In 2025, households purchased 450 units of electricity for ₹5,400, 25 LPG cylinders for ₹30,000, and 120 litres of petrol for ₹14,400.

a. Calculate the price per unit of electricity, per LPG cylinder, and per litre of petrol in 2024 and 2025.
b. Using 2024 as the base year, calculate the Consumer Price Index (CPI) for 2024 and 2025. Use the quantities purchased in 2024 as the fixed basket.
c. Calculate the inflation rate in 2025.

PRODUCTION AND GROWTH

CENTRAL QUESTION

why did productive capacity expand so much?
AND
Why do some countries achieve this transformation while others struggle?

In the 1950s, …….. There was a poor country recovering from war. Today, it is globally known for semiconductors, automobiles, electronics and advanced technology.

How did this transformation happen?

Indicator	Earlier period	Recent data
GNI per person	$67, early 1950s	$36,624, 2024
Average annual real GDP growth	—	5.6%, 1980–2024
Life expectancy at birth	—	83.6 years, 2024
Internet use	—	98% of the population, 2024

Sources: World Bank Indicators

Country	GDP per capita	GDP growth	Life expectancy
United States	$84,534	2.8%	79 years
Germany	$56,104	About −0.2%	81 years
South Korea	$36,239	2.0%	84 years
China	$13,303	5.0%	78 years
India	$2,695	6.5%	72 years

Sources: World Bank Indicators, 2024

Across countries, average income and access to goods and services can differ greatly.

- Higher living standards are often associated with better nutrition, housing, healthcare and life expectancy.
- Living standards also change over time within the same country.
- Long-run economic growth is about sustained increases in an economy's capacity to produce goods and services.

What should India invest in to raise output per person—education, technology, infrastructure, healthcare, or all four?

WHAT IS PRODUCTIVITY?

Productivity is the quantity of goods and services produced per unit of labour input.

$$\text{Labour Productivity} = \frac{\text{Total Output}}{\text{Number of Workers}}$$

WHY IS PRODUCTIVITY IMPORTANT?

- Higher productivity can increase output per worker.
- Higher output can support higher real incomes.
- Businesses may expand and create employment opportunities.
- Society can consume more goods and services with available resources.
- Sustained productivity growth can improve living standards over time.

FOUR DETERMINANTS OF PRODUCTIVITY

1. Physical capital

Tools, equipment, machines and buildings used in production.

2. Human capital

Knowledge and skills gained through education, training and experience.

3. Natural resources

Inputs supplied by nature: land, rivers, minerals, sunlight and energy resources.

4. Technological knowledge

Society's understanding of better ways to produce goods and services.

CAN NATURAL RESOURCES ALONE MAKE A COUNTRY RICH?

Natural resources can create wealth, but a country's long-term prosperity also depends on how productively it uses its workers, technology, capital and resources.

WHY IS PRODUCTIVITY IMPORTANT?

Basis of Comparison	Saudi Arabia	Japan	India
Natural resources	Abundant oil reserves	Limited domestic supplies of many natural resources	Coal, iron ore, fertile land and renewable-energy potential
Main source of economic strength	Oil exports and investment in other sectors	Advanced manufacturing, technology and skilled workers	Agriculture, manufacturing, IT and services
Major challenge or strategy	Reduce dependence on oil by developing tourism, manufacturing and services	Import essential raw materials and energy resources	Improve skills, infrastructure, technology and industrial productivity
Economic lesson	Natural resources can generate substantial wealth, but dependence on one resource creates risks	A country can achieve high living standards despite limited natural resources	Natural resources create greater value when combined with investment, skills and technology

PRODUCTION FUNCTION

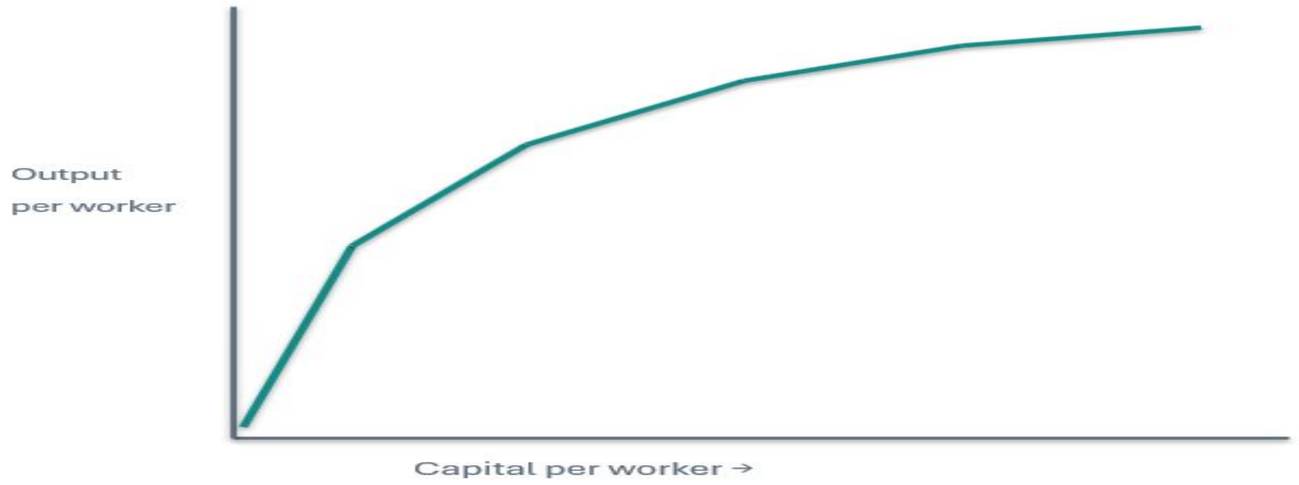
A production function shows the relationship between the inputs used in production and the total output produced.

$$Y = A\,F(L, K, H, N)$$

Improved technology can increase output even when other inputs remain unchanged.

DIMINISHING RETURNS TO CAPITAL

When more capital (machines, tools, equipment) is added while labour and other factors remain constant, total output increases, but the additional output generated by each extra unit of capital gradually decreases.



PRODUCTION FUNCTION: CRS

Suppose a factory initially produces 100 units.

Input/output	Initial situation	After doubling all inputs
Workers	10	20
Machines	5	10
Raw materials	100 kg	200 kg
Output	100 units	200 units

$$F(2L, 2K, 2H, 2N) = 2F(L, K, H, N)$$

PRODUCTION FUNCTION: IRS

Suppose a factory initially produces 100 units.

Input/output	Initial situation	After doubling all inputs
Workers	10	20
Machines	5	10
Raw materials	100 kg	200 kg
Output	100 units	250 units

$$\underbrace{F(2L, 2K, 2H, 2N)}_{\text{Output after doubling inputs}} > \underbrace{2F(L, K, H, N)}_{\text{Twice the original output}}$$

PRODUCTION FUNCTION: DRS

Suppose a factory initially produces 100 units.

Input/output	Initial situation	After doubling all inputs
Workers	10	20
Machines	5	10
Raw materials	100 kg	200 kg
Output	100 units	180 units

$$\underbrace{F(2L, 2K, 2H, 2N)}_{\text{Output after doubling inputs}} < \underbrace{2F(L, K, H, N)}_{\text{Twice the original output}}$$

ECONOMIC GROWTH AND PUBLIC POLICY

What can government policy do to raise productivity and living standards?

1. Saving and investment	Government investment in highways, railways, and industrial facilities.	Investment creates productive assets. Better transport allows businesses to move goods faster and reduce costs.
2. Physical capital	PM GatiShakti infrastructure planning in India.	Coordinating road, rail, port, and logistics projects can reduce delays and improve the movement of goods.
3. Human capital (education)	Public schools, technical colleges, and vocational training.	Students acquire skills that help them operate machinery, write software, solve engineering problems, and become more productive workers.
4. Health and nutrition	Public healthcare, vaccination, nutrition programmes, and sanitation.	Healthier workers can attend work more regularly and perform tasks more effectively.
5. Foreign direct investment (FDI)	Hyundai Motor Group's announced $15.5 billion investment plan in India through 2032, reported in October 2026.	Foreign investment can expand manufacturing capacity, support R&D, create jobs, and introduce new technologies. These are planned investments, not all completed expenditure. Reuters

ECONOMIC GROWTH AND PUBLIC POLICY

6. Property rights and political stability	Courts that enforce contracts and laws protecting business ownership.	Entrepreneurs are more willing to build factories and hire workers when they have confidence that their assets and contractual rights will be protected.
7. Free trade	Indian businesses exporting medicines, software, textiles, and engineering products.	Exporting gives firms access to more customers, while importing machinery and components can improve production efficiency.
8. Research and development	Government research grants and university-industry collaboration.	Researchers develop new products and methods; firms can adopt these innovations to improve output and quality.
9. Technological progress	Digital payments, industrial automation, and precision farming.	Technology can reduce transaction costs, improve accuracy, save time, and increase output from existing resources.

ECONOMIC GROWTH AND PUBLIC POLICY

10. Manufacturing incentives	India's Production Linked Incentive (PLI) schemes.	Eligible firms receive incentives linked to specified production-related criteria. By March 2026, official government reporting recorded over ₹2.40 lakh crore in actual investment under the schemes.
11. Natural-resource management	Efficient irrigation, watershed management, and sustainable forestry.	Better resource management can improve agricultural output while reducing waste and protecting future production capacity.
12. Population and employment	Skill-development programmes and opportunities for women to enter paid employment.	Workers gain skills and access to jobs, allowing the economy to use a larger share of its available talent.
13. Entrepreneurship and competition	Startup support, business registration reforms, and easier access to finance.	New firms can introduce products, create jobs, and pressure existing businesses to improve productivity.
14. Infrastructure and logistics	Integrated transport and logistics planning under PM GatiShakti.	Better connections between factories, suppliers, markets, and ports can reduce transport delays and make Indian products more competitive.

SAVING, INVESTMENT, AND THE FINANCIAL SYSTEM

Imagine you have completed your B.Tech and want to start an economic forecasting company. You need ₹10 lakh to buy computers, furniture, software, and office equipment. You have saved only ₹2 lakh. Where will you get the remaining ₹8 lakh?

Option A • Borrow

Get a loan from a bank or another lender; repay principal plus interest.

Option B • Sell ownership

Invite investors to provide funds in exchange for a share of future profits.

How does one person's saving, finance another person's investment?

Two broad types of financial institutions

1. Financial markets

Savers provide funds directly to borrowers.

Examples:
- Bond market
- Stock market

2. Financial intermediaries

Institutions stand between savers and borrowers.

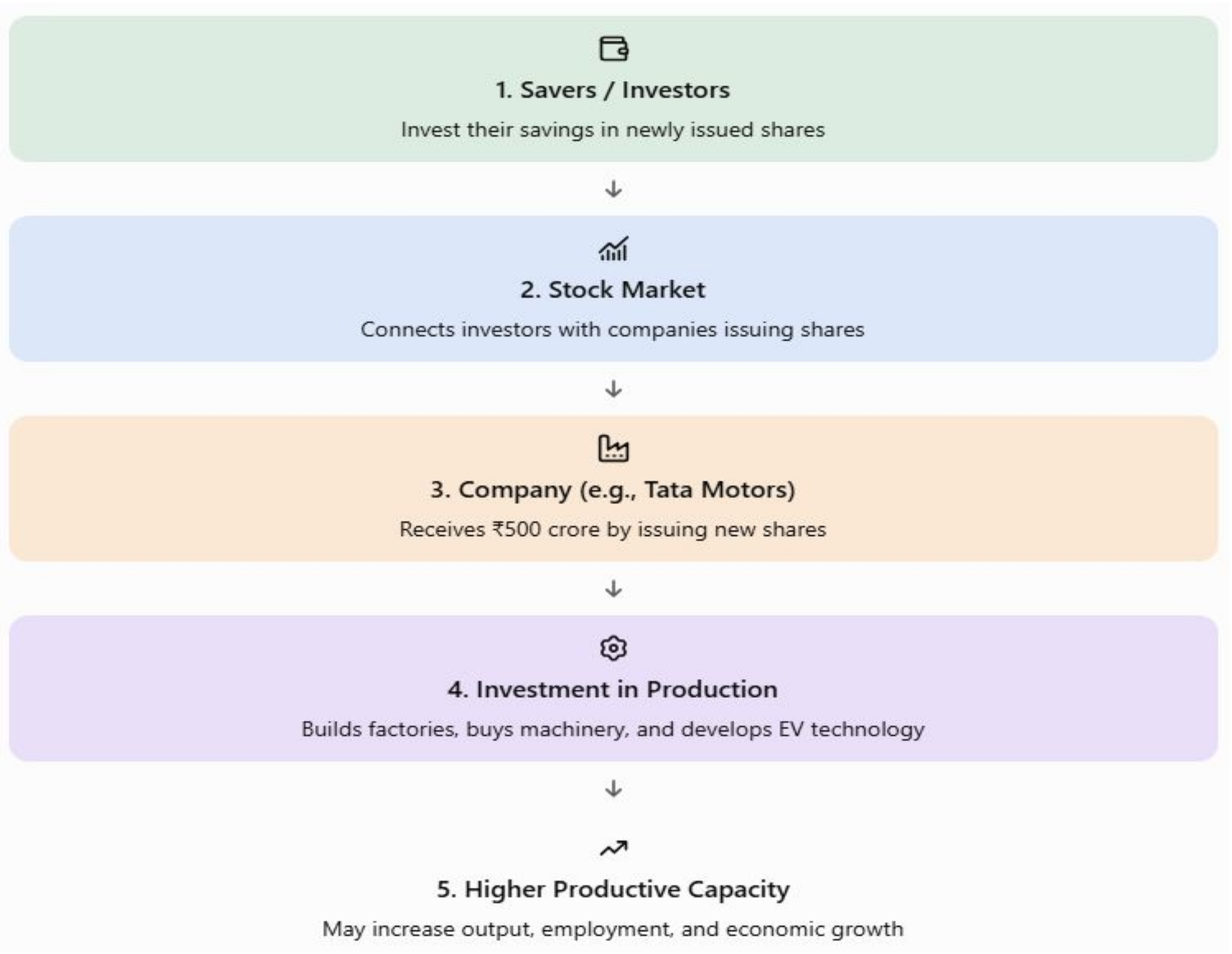
Examples:
- Banks
- Mutual funds

Both channels help move funds toward households and firms that need financing.

1. Financial markets (Bond Market)



1. Financial markets (Stock Market)



Bonds and stocks: debt versus ownership

BOND = DEBT FINANCE

1. Meaning: You lend money to a company.
2. Return: You usually receive interest.
3. Repayment: The company is expected to repay the borrowed money at maturity, subject to its ability to pay.
4. Risk: Generally less risky than shares, but you may lose money if the company fails to pay.
5. Your role: You are a lender (creditor), not an owner.
6. When investors trade: Money goes to the seller when an existing bond is sold to another investor.

STOCK = EQUITY FINANCE

Meaning: You become a part-owner of a company.
Return: You may receive dividends and earn a profit if the share price rises.
Repayment: The company does not normally repay your investment on a fixed date.
Risk: Generally riskier because the share price can rise or fall significantly.
You are an owner (shareholder) of a small part of the company.
Money goes to the seller when an existing share is sold to another investor—not to the company.

Bondholders lend, and shareholders own.

The financial system connects savers and borrowers

Its central job is to direct resources from people who spend less than they earn to people who want to borrow.

SAVERS

Households or institutions with funds available for the future.

Funds

FINANCIAL SYSTEM

Markets and intermediaries coordinate the flow of funds.

Loans/capital

BORROWERS

Households and firms seeking funds for spending or investment.

Why it matters: saving can support investment in capital, raise productivity and contribute to higher living standards over time.

The financial system connects savers and borrowers

Steps	Example
1. Saver	Priya earns ₹40,000 per month and saves ₹5,000.
2. Financial intermediary	The bank uses its funds to provide business loans while maintaining required reserves.
3. Borrower	The owner takes a ₹10 lakh loan to buy a new machine.
4. Investment	The new machine helps the factory produce more goods.

Economic benefits: Daily production rises from 100 to 150 units.

Living standards: Workers may earn more and afford better food, housing and education.

What makes bonds different?

TERM / MATURITY

How long until the principal is repaid? Longer-term bonds often pay more to compensate for added risk. (Longer time = higher interest.)

CREDIT RISK

How likely is the **borrower to miss interest or principal payments**? Higher perceived default risk generally requires a higher rate. (HIGH RISK= higher interest.)

TAX TREATMENT

How is interest taxed? Tax advantages can allow some bonds to offer lower interest rates. (HIGH TAX= higher interest.)

Job A: Pays $100,000, but after taxes, you only take home $70,000.
Job B: Pays $75,000, but because of a special tax rule, you get to keep $75,000 completely tax-free.

Banks and mutual funds

BANKS

1. Accept deposits from savers.
2. Use deposits to make loans.
3. Pay depositors interest and charge borrowers interest.
4. Also help people make payments through deposits, checks and debit cards.

MUTUAL FUNDS

1. Sell shares to investors.
2. Pool the money to buy a portfolio of stocks and/or bonds.
3. Allow small investors to diversify.
4. Investors bear the portfolio's gains and losses.

Diversification reduces dependence on the performance of a single company, but it does not eliminate risk.

GDP links income and expenditure

NATIONAL INCOME ACCOUNTING

$$Y = C + I + G + NX$$

Y

Total income / GDP

C

Consumption

I

Investment in new capital

G + NX

Government purchases + net exports

An accounting identity must hold by definition: every dollar of expenditure is counted in one of the components.

From GDP to national saving

Assume a closed economy: no international trade or international borrowing/lending, so NX = 0.

$$Y = C + I + G$$

$$Y - C - G = I$$

$$S = I$$

National saving is the income left after consumption and government purchases. In this model, it equals investment.

Saving is not the same as investment

SAVING

Income not spent on current consumption.

Example: Larry deposits unspent income in a bank or buys a bond.

INVESTMENT

Purchase of new capital goods or structures.

Example: a firm builds a factory, or a household buys a newly built home.

Buying an existing share or bond is a financial transaction—not, by itself, new investment in the national income accounts.

National saving has two parts

Let T = taxes minus transfer payments.

$$S = (Y - T - C) + (T - G)$$

PRIVATE SAVING

$Y - T - C$

Household and firm income remaining after taxes and consumption.

PUBLIC SAVING

$T - G$

Government tax revenue remaining after purchases.

If T > G: budget surplus → public saving is positive.

If G > T: budget deficit → public saving is negative.

The market for loanable funds

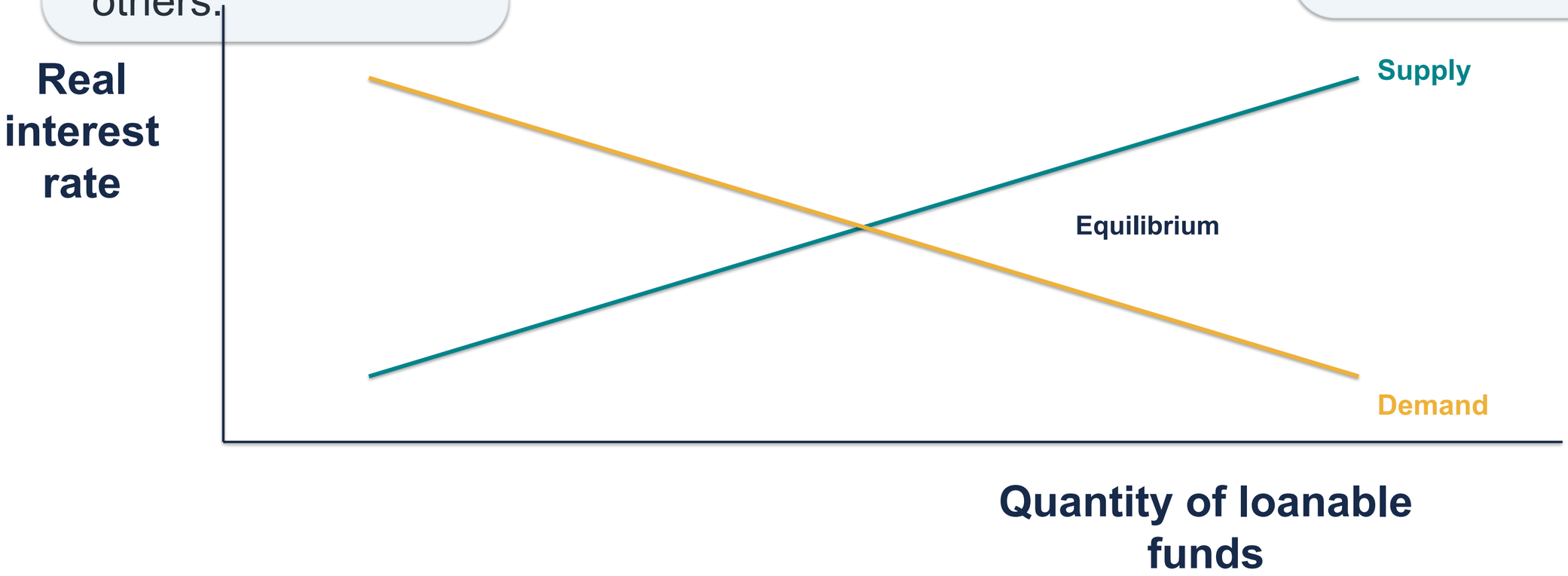
SUPPLY

Comes from saving by households and others.

Interest rate = the price of borrowing and return on saving

DEMAND

Comes from households and firms borrowing to invest.



The market for loanable funds

Person A: A salaried employee earns ₹60,000 per month and saves ₹10,000. She deposits the money in a bank.
Supplies loanable funds

Person B: An entrepreneur
A small factory owner wants to purchase a new machine worth ₹5 lakh but does not have enough money. He borrows from the bank.
Demands loanable funds

Person C: A household
A family takes a home loan to purchase a newly constructed house.
Demands loanable funds

The market for loanable funds

Feature	**Supply of loanable funds**	**Demand for loanable funds**
Comes from	People who save	Firms and households that borrow to invest
Main source	Saving	Investment
When interest rates rise	People generally want to save more	People generally want to borrow less
Curve	Upward-sloping	Downward-sloping

How the interest rate restores equilibrium

Interest rate below equilibrium

- Quantity supplied is less than quantity demanded.
- A shortage of loanable funds emerges.
- Lenders have an incentive to raise interest rates.

Interest rate above equilibrium

- Quantity supplied exceeds quantity demanded.
- Lenders compete to attract borrowers.
- Interest rates are pushed down.

Equilibrium: quantity of loanable funds supplied = quantity demanded.

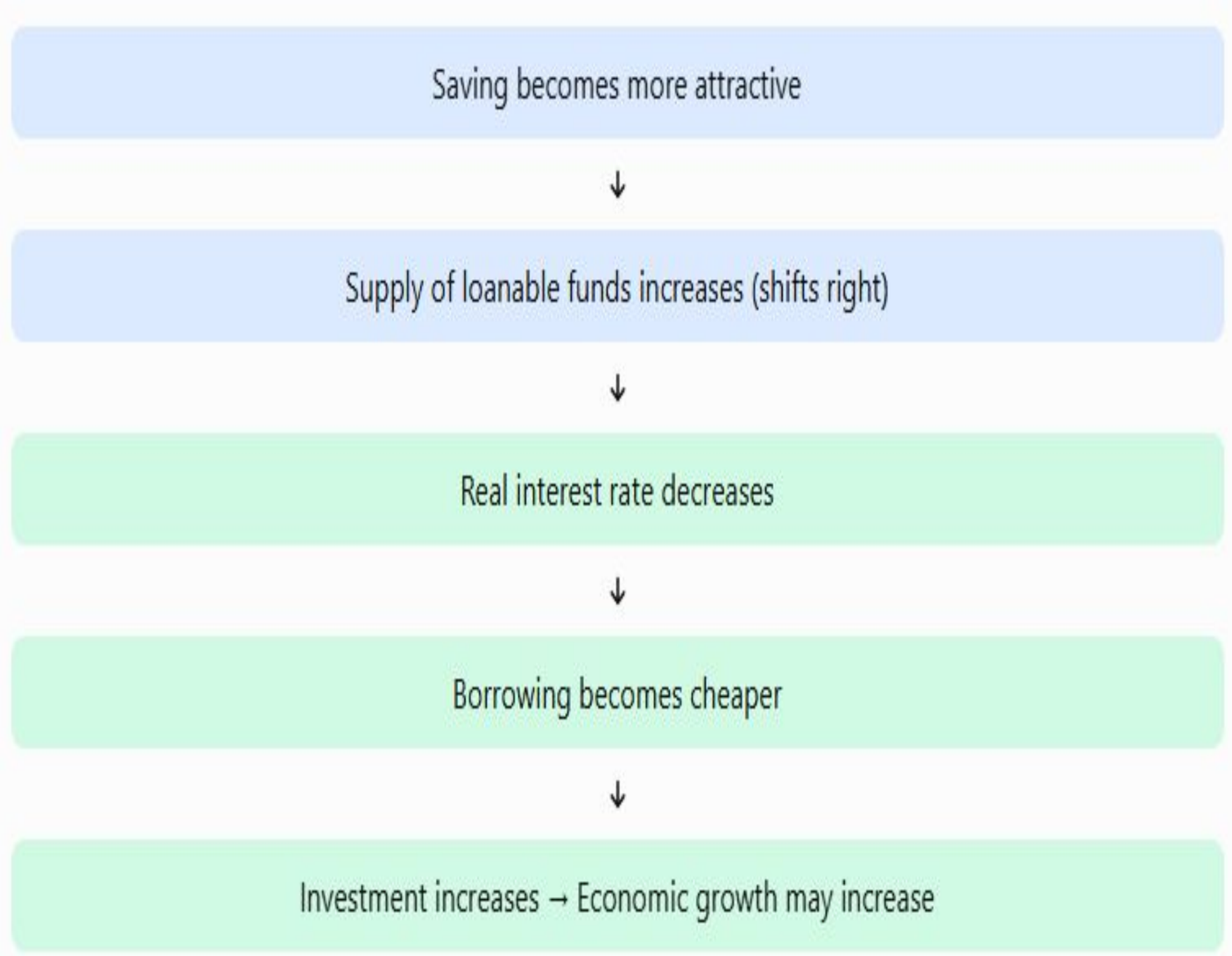
Policy 1 • Incentives to Save

The government provides tax benefits on eligible savings or retirement investments.

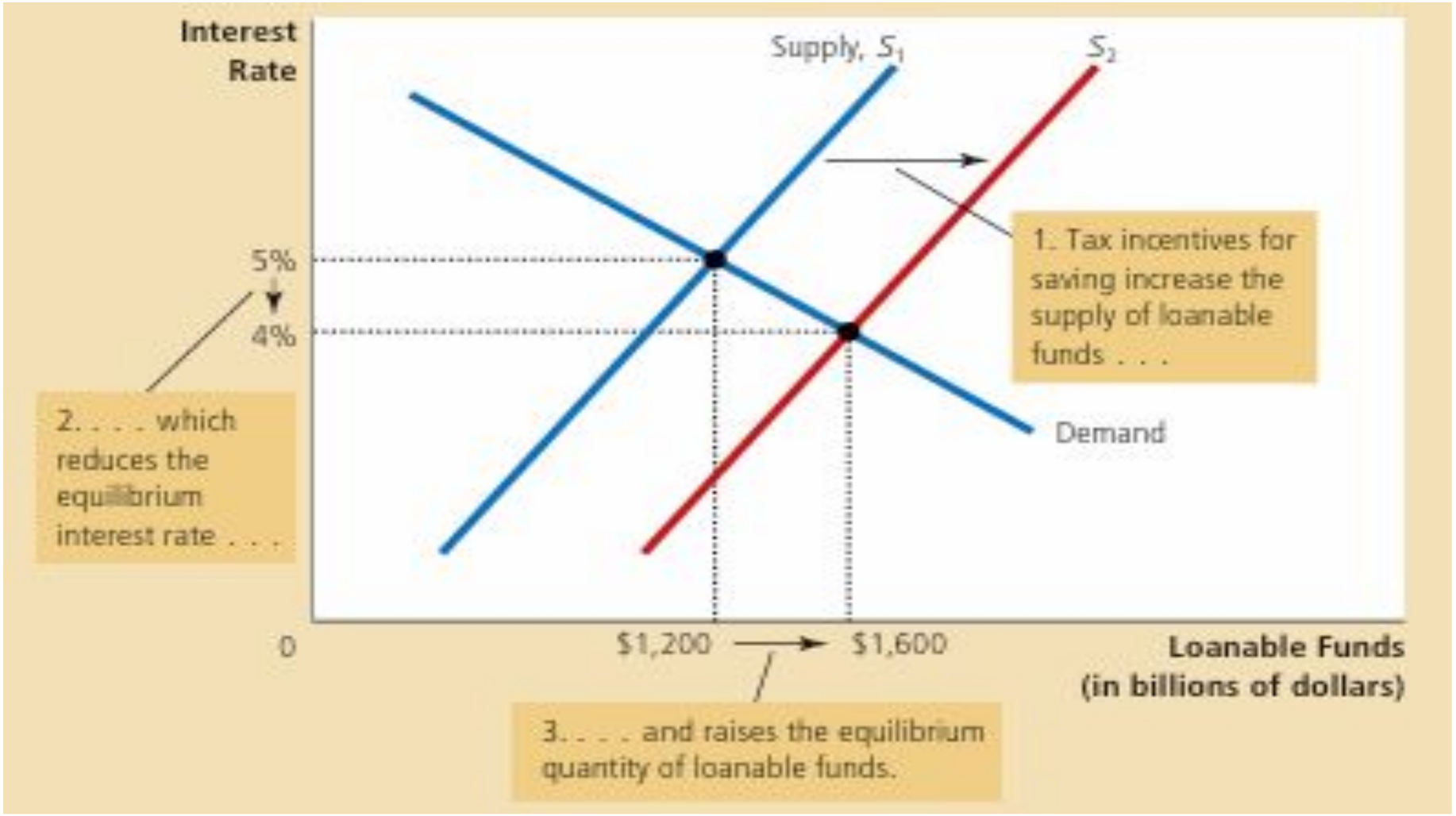
1. People find saving more attractive.
2. Supply of loanable funds increases (shifts right).
3. The equilibrium real interest rate falls.
4. Businesses can borrow more cheaply and investment increases.

Saving incentives → lower interest rate → higher investment.

Policy 1 • Incentives to Save



Policy 1 • Incentives to Save



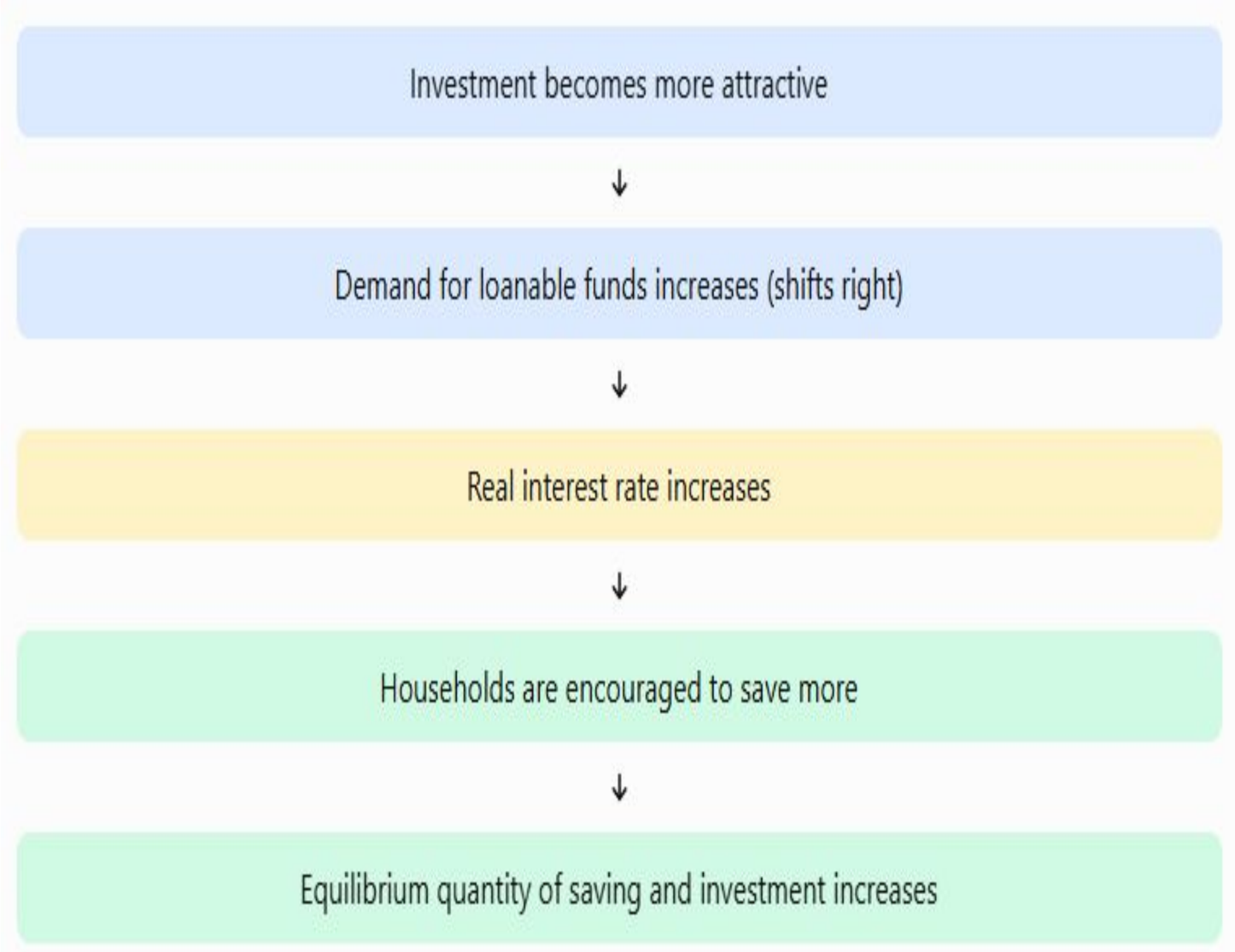
Policy 2 • Incentives to Invest

The government gives a tax credit to a company purchasing new manufacturing equipment.

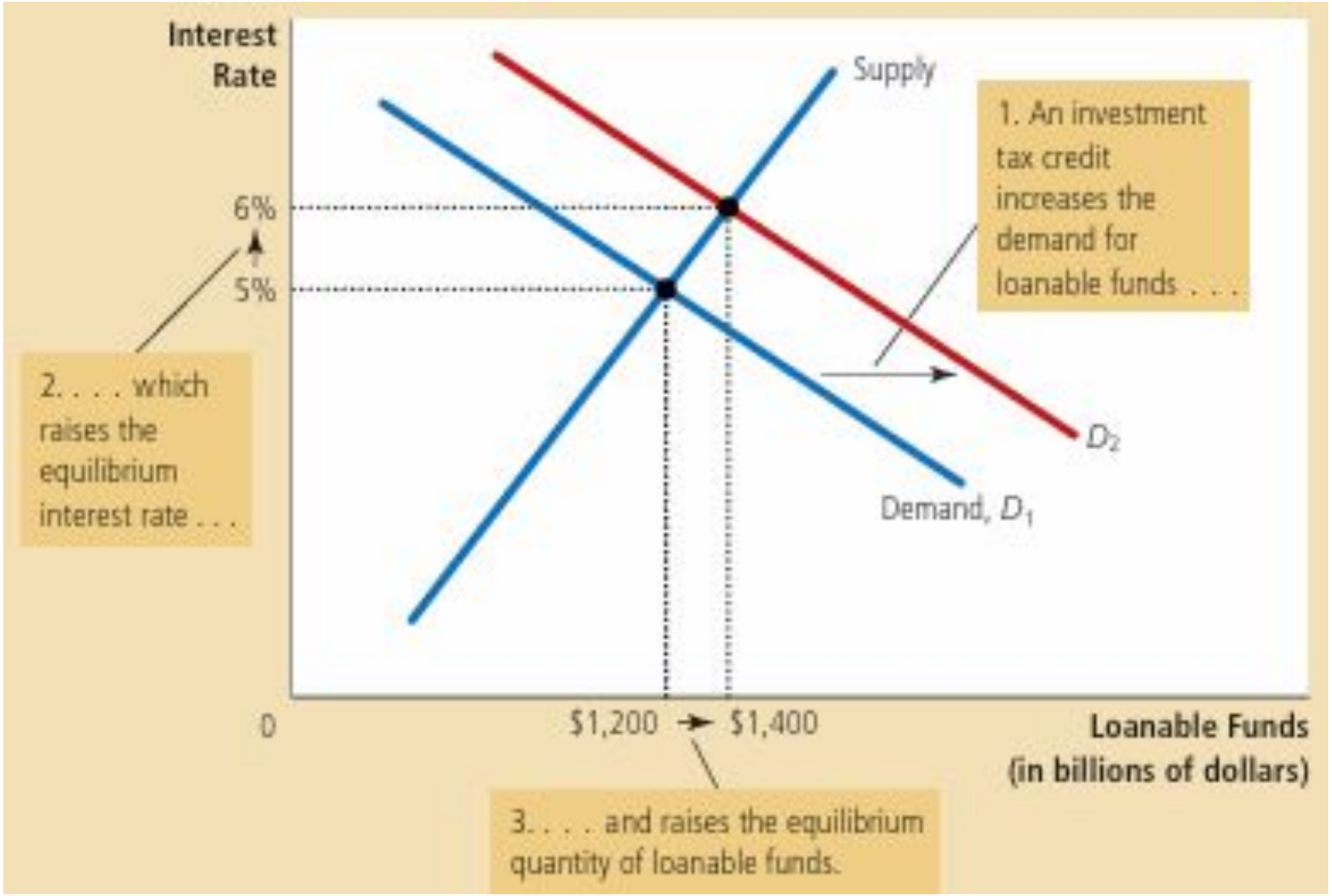
1. The company finds a new machine more financially attractive.
2. Demand for loanable funds increases (shifts right).
3. The equilibrium real interest rate rises.
4. Higher interest rates encourage households to save more, increasing the equilibrium quantity of funds.

Investment incentives → higher interest rate → higher equilibrium saving and investment.

Policy 2 • Incentives to Invest



Policy 2 • Incentives to invest



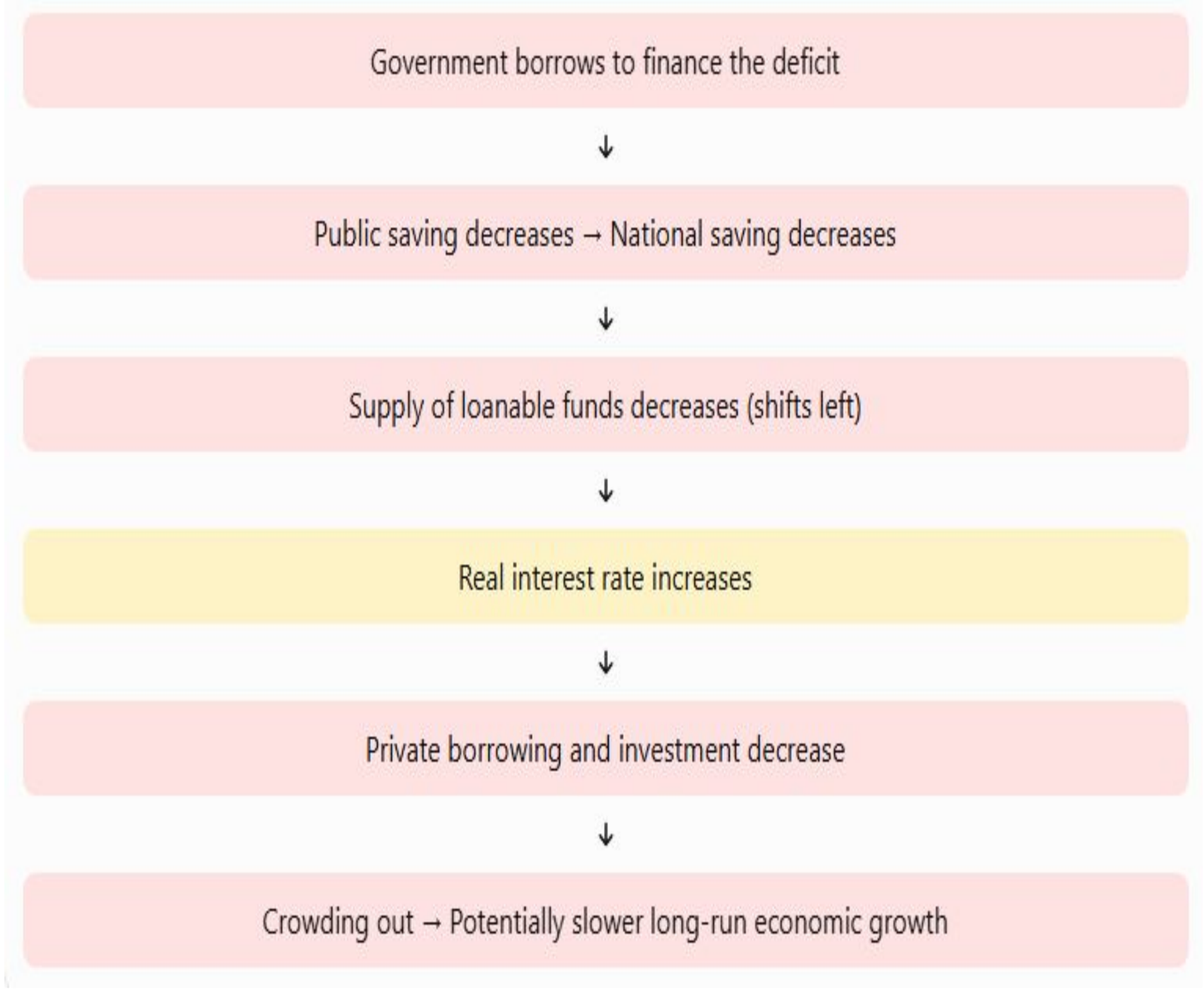
Policy 3 • Budget Deficits And Crowding Out

The government collects ₹100 crore in taxes but spends ₹130 crore. It has a deficit of ₹30 crore and borrows to finance it.

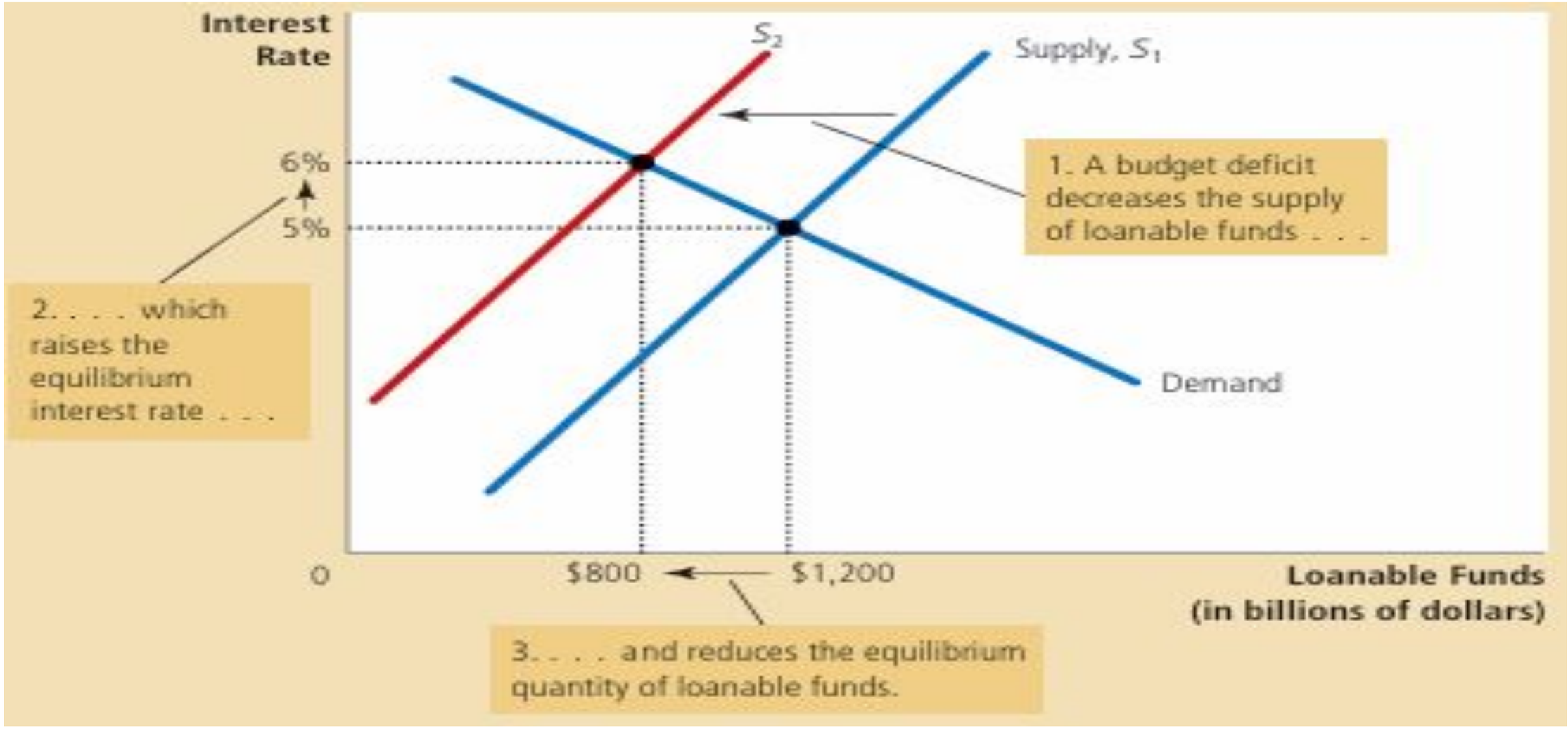
1. The government spends more money than it collects in taxes, so it borrows money.
2. This reduces the money available for lending in the economy.
3. Interest rates go up because money becomes harder to get.
4. Businesses may avoid taking loans because borrowing becomes expensive. As a result, they may invest less in new factories and machines.

Budget deficit → lower national saving → higher interest rate → lower private investment.

Policy 3 • Budget Deficits And Crowding Out



Policy 3 • Budget Deficits And Crowding Out



Policy 3 • Crowding Out

Crowding out occurs when government borrowing pushes up interest rates and reduces private investment.

• Find Out the Preferred Policy

Many families in Bengaluru decide to save more money in bank deposits instead of spending it on shopping and entertainment. Banks now have more funds available to lend to businesses and individuals.

The Indian government plans to build new highways but does not have sufficient tax revenue to finance the project. It borrows money from financial markets to cover the additional expenditure.

An Indian electric-vehicle company expects sales to increase. It decides to borrow money to build a new factory and purchase machinery. The company needs additional funds to expand production.

The government increases borrowing to finance public infrastructure because tax revenue is insufficient. As government demand for funds rises, interest rates increase. A private manufacturing company finds borrowing more expensive and postpones building a new factory.

An Indian software company experiences rising demand for cloud computing services. It borrows money to purchase new servers and expand its data centre. More businesses are now seeking loans to finance investment projects.

CONCLUSION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus vestibulum at risus non efficitur. Etiam at auctor felis. Nam finibus, orci in viverra varius, nibh metus interdum urna, id condimentum elit lectus eu leo. Maecenas nisl lectus, malesuada vitae blandit a, mollis ut urna. Aliquam scelerisque maximus velit nec faucibus.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus vestibulum at risus non efficitur. Etiam at auctor felis. Nam finibus, orci in viverra varius, nibh metus interdum urna, id condimentum elit lectus eu leo.

- Economic principles explain how limited resources are allocated efficiently.
- Scarcity forces individuals and organizations to make choices.
- Opportunity cost highlights the value of alternatives that are given up.
- Supply and demand determine market prices and production levels.
- Marginal thinking and incentives improve decision-making.
- Productivity and market efficiency support long-term economic growth.

HANOVER AND TYKE

Smarter Choices, Better Futures

THANK YOU FOR ATTENTION

Understanding economic principles helps people make smarter financial decisions, improve business strategies, and contribute to a stronger and more sustainable economy.